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(54) **INTERPRETING MEANING OF CONTENT**

(71) Applicant: **entigenlogic LLC**, Schaumburg, IL (US)

(72) Inventors: **Frank John Williams**, Rossmoor, CA (US); **David Ralph Lazzara**, Huntington Beach, CA (US); **Donald Joseph Wurzel**, Savannah, GA (US); **Stephen Emerson Sundberg**, Chicago, IL (US); **Ameeta Vasant Reed**, Deerfield, IL (US); **Dennis Arlen Roberson**, Chapin, SC (US); **Thomas James MacTavish**, Inverness, IL (US); **Karl Olaf Knutson**, Palatine, IL (US); **Jessy Thomas**, Palatine, IL (US); **Niklas Josiah MacTavish**, Inverness, IL (US); **David Michael Corns, II**, Elgin, IL (US); **Andrew Chu**, St. Charles, IL (US); **Theodore Mazurkiewicz**, Lake Zurich, IL (US); **Gary W. Grube**, Barrington Hills, IL (US)

(73) Assignee: **entigenlogic LLC**, Schaumburg, IL (US)

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continuation of application No. 18/103,772, filed on Jan. 31, 2023, now Pat. No. 11,880,656, which is a continuation of application No. 16/855,445, filed on Apr. 22, 2020, now Pat. No. 11,586,819.

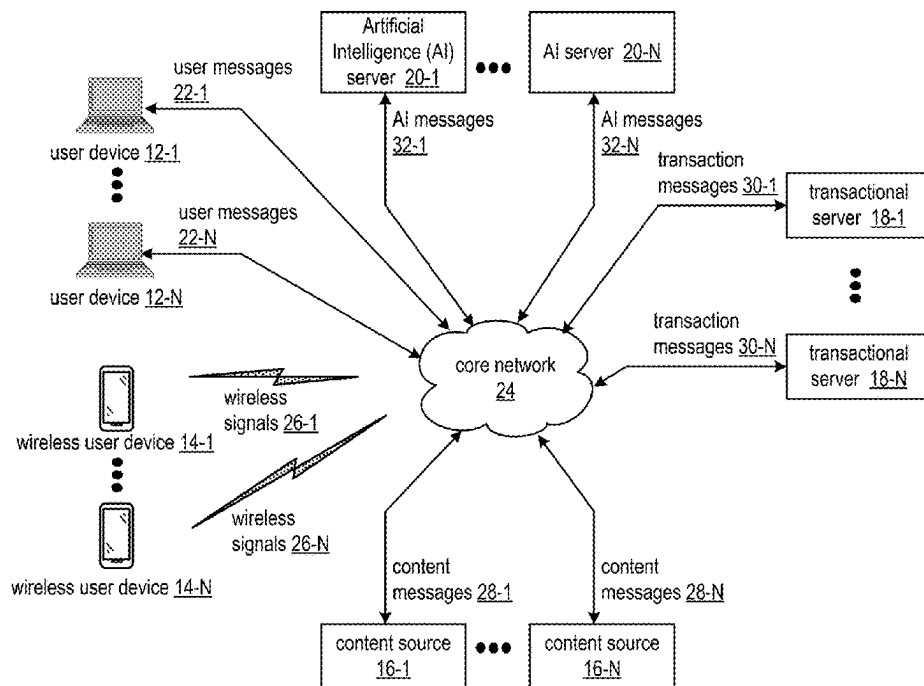
(60) Provisional application No. 62/839,892, filed on Apr. 29, 2019.

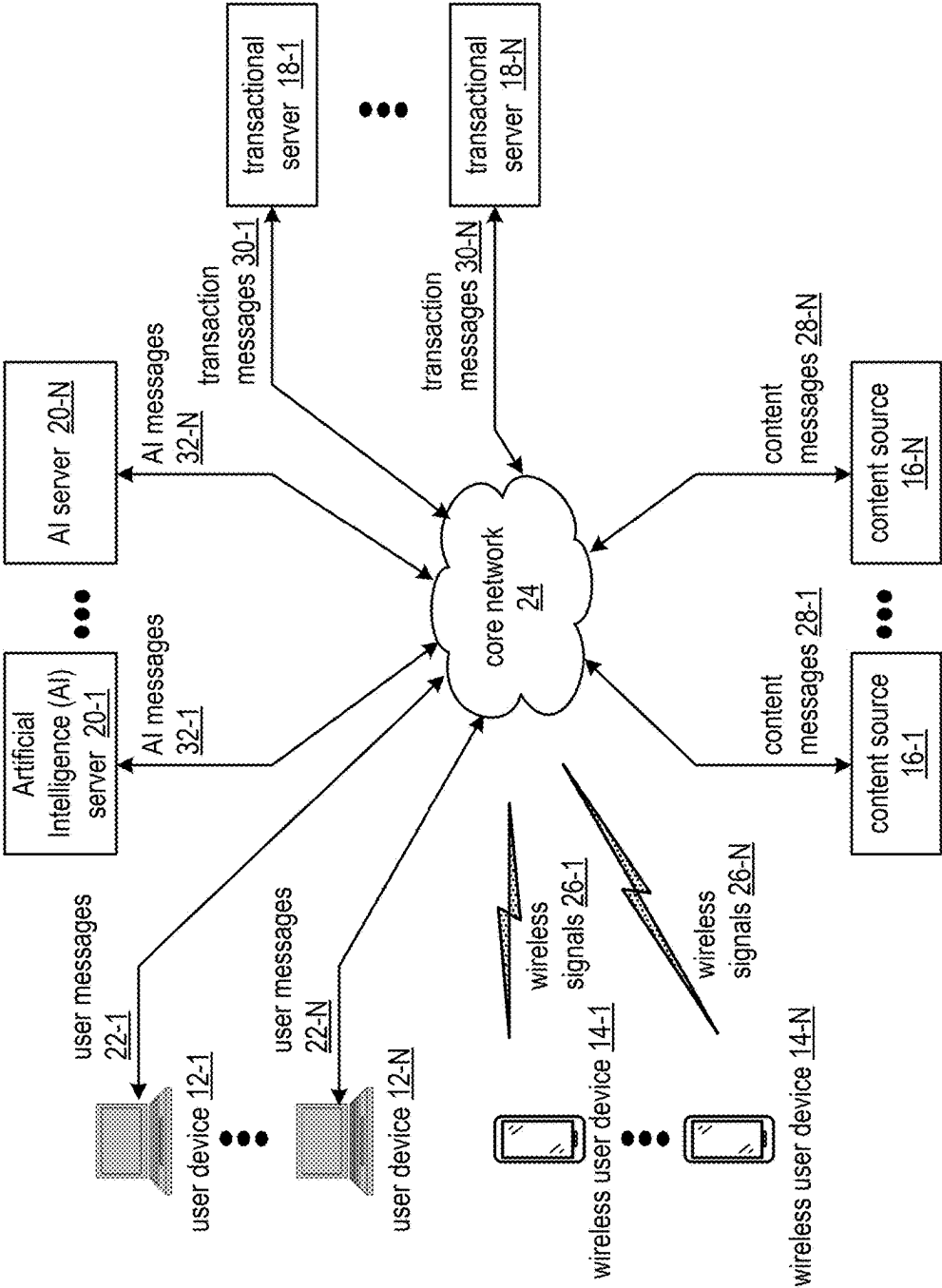
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(57) **ABSTRACT**

A method for execution by a computing device includes identifying sets of identigens for words of a phrase. The method further includes identifying levels of pairing of sequentially adjacent identigens between each adjacent set of identigens and interpreting every permutation of sequentially adjacent identigens to produce permutations of entigen groups. The method further includes determining permutation scores for the permutations of entigen groups and selecting an entigen group based on the permutation scores to produce a selected entigen group that represents a most likely meaning interpretation of the phrase.





computing system 10

FIG. 1

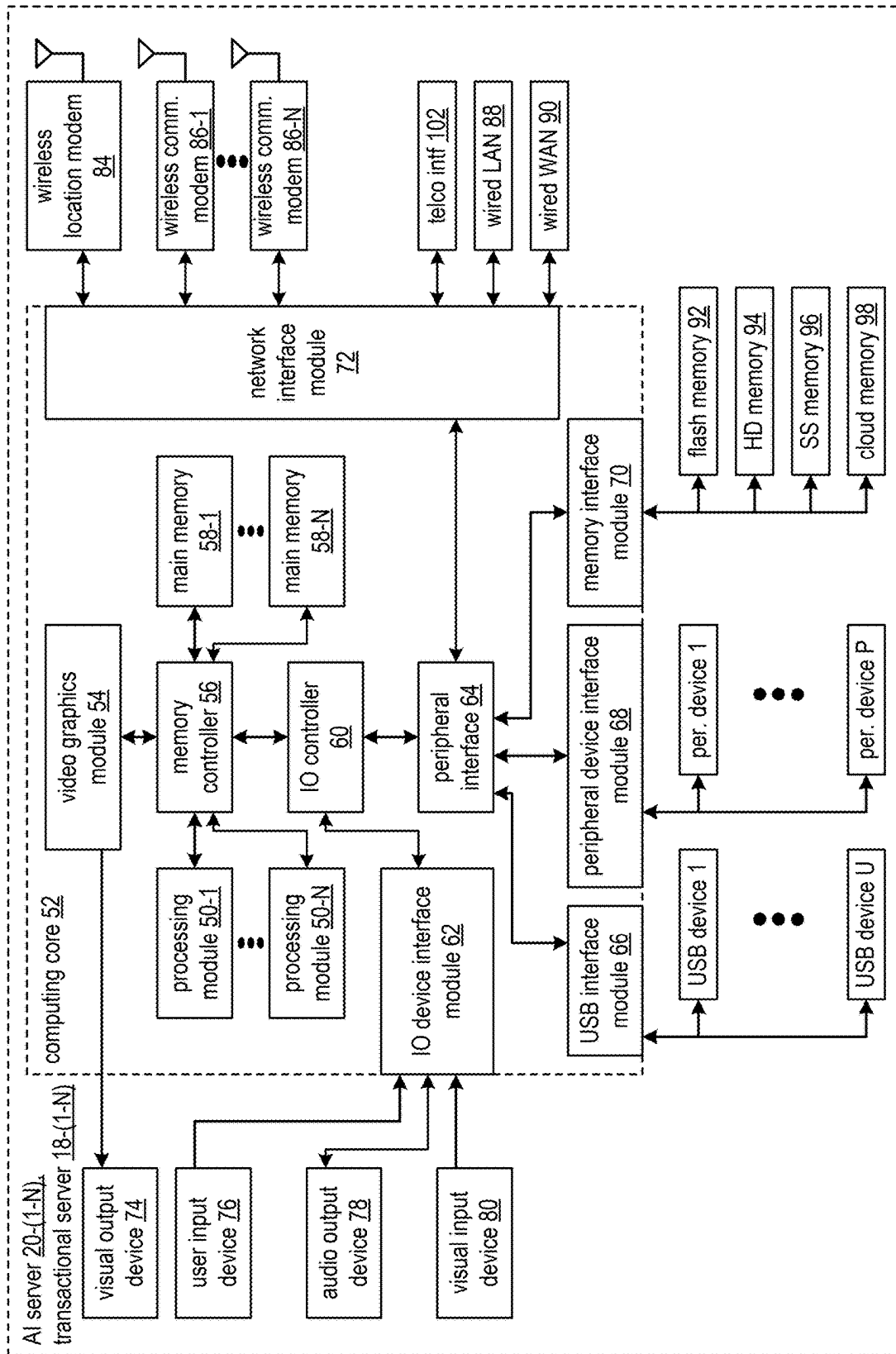


FIG. 2

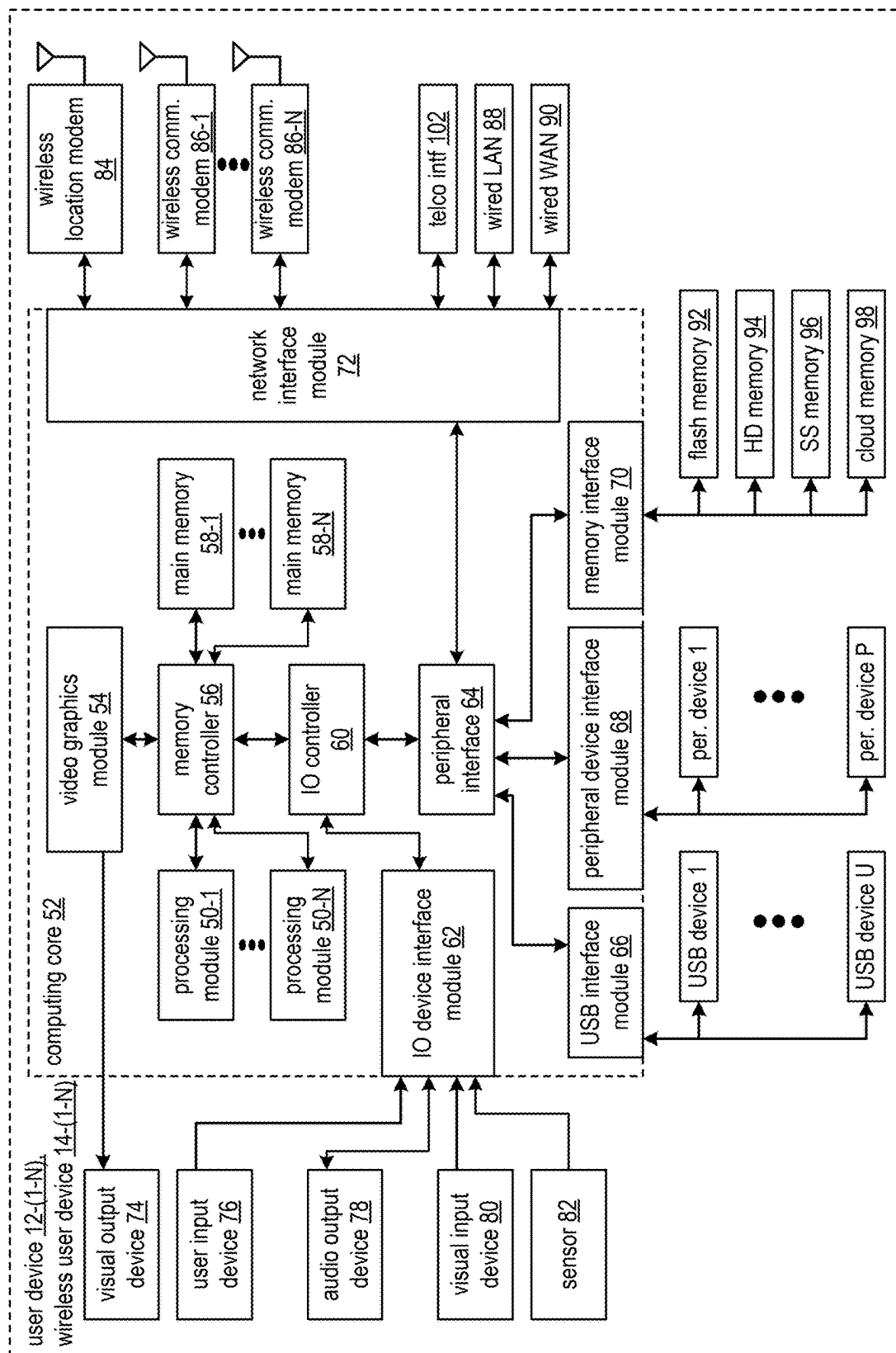


FIG. 3

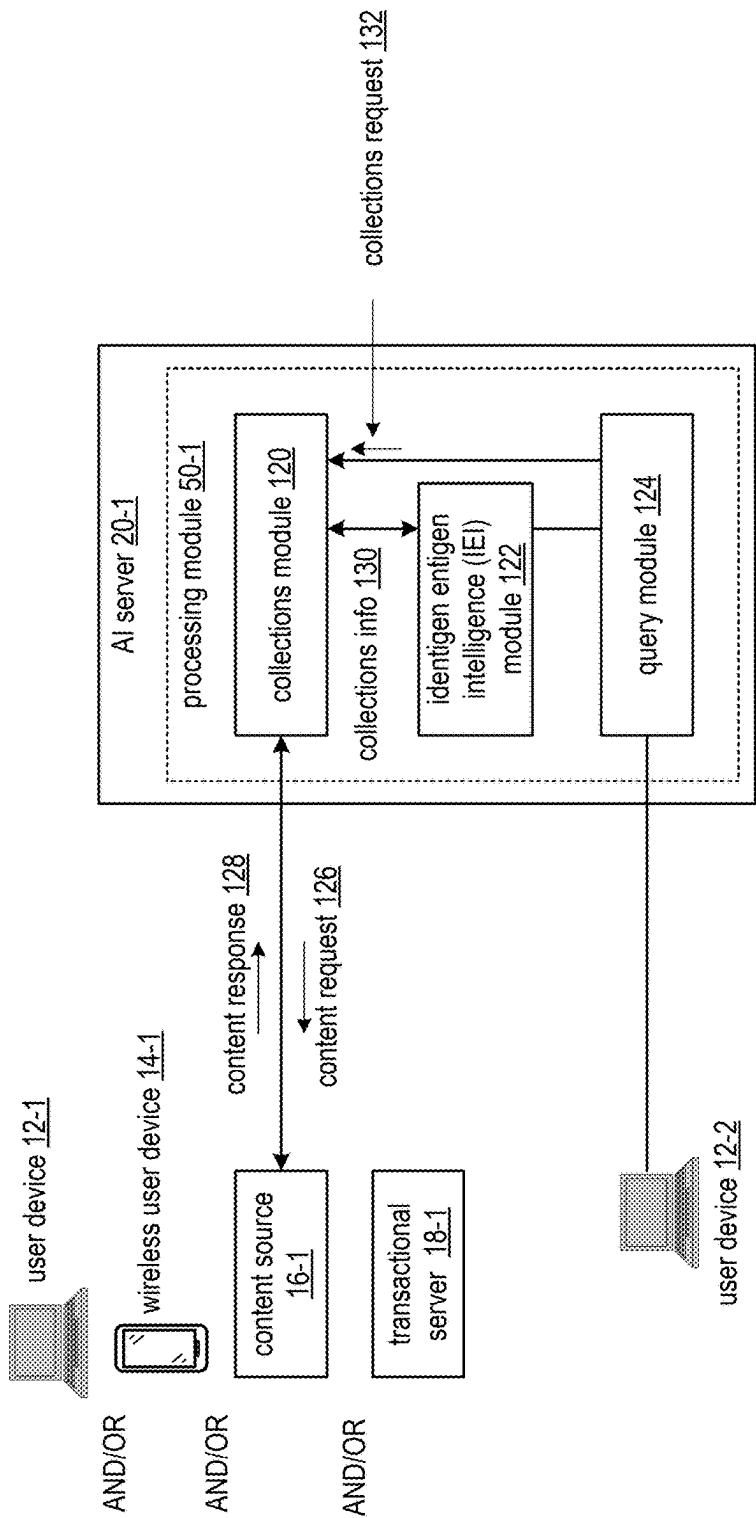


FIG. 4A

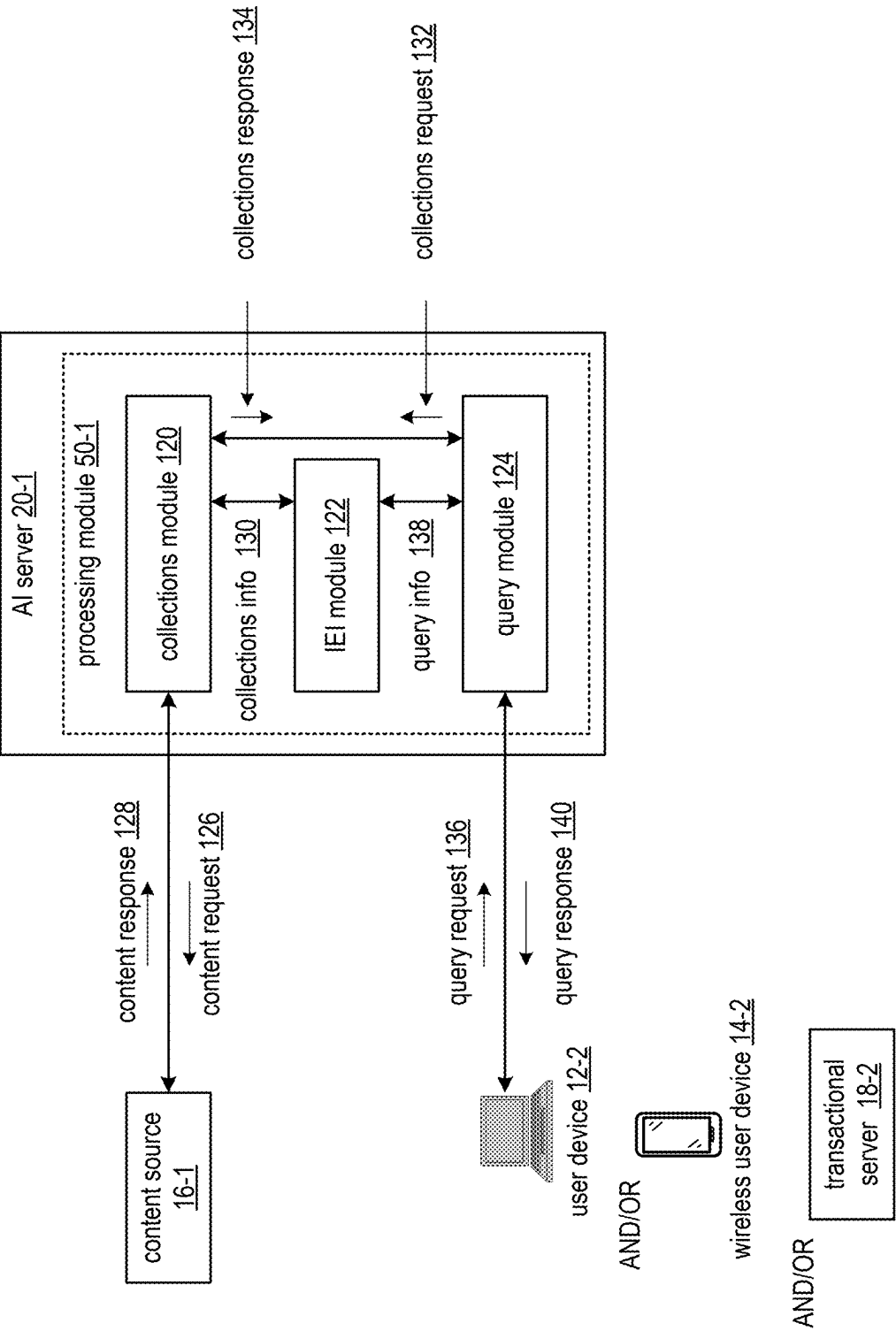


FIG. 4B

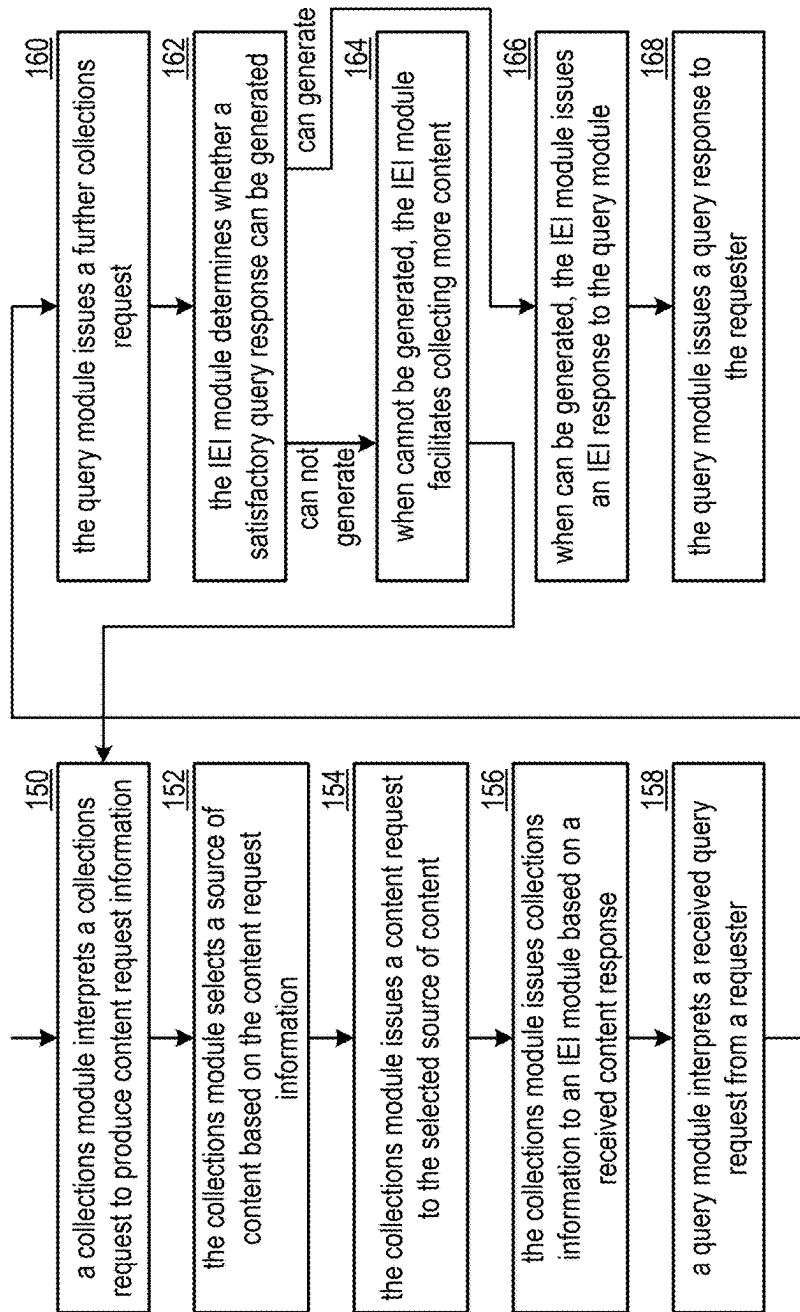


FIG. 4C

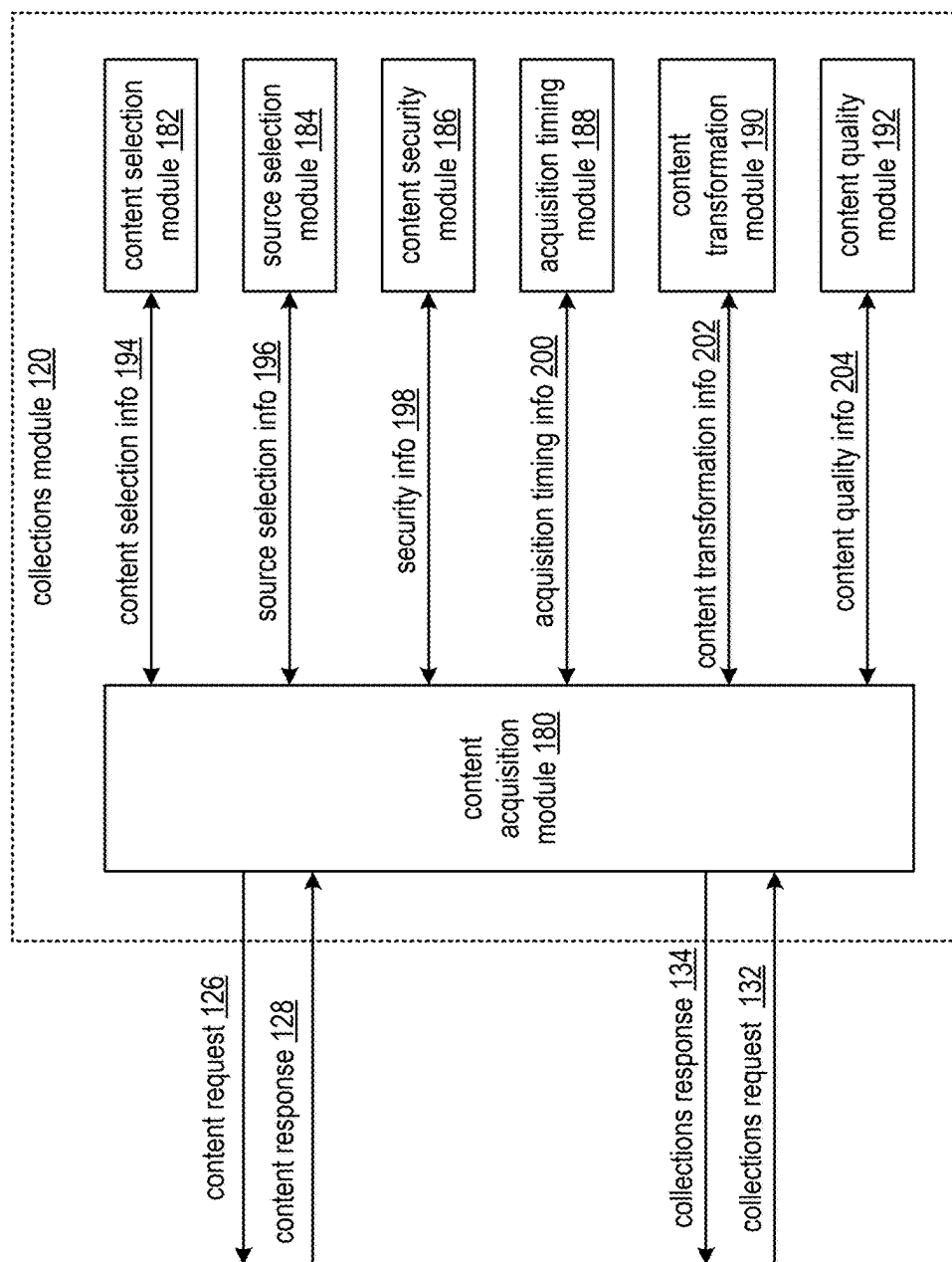


FIG. 5A

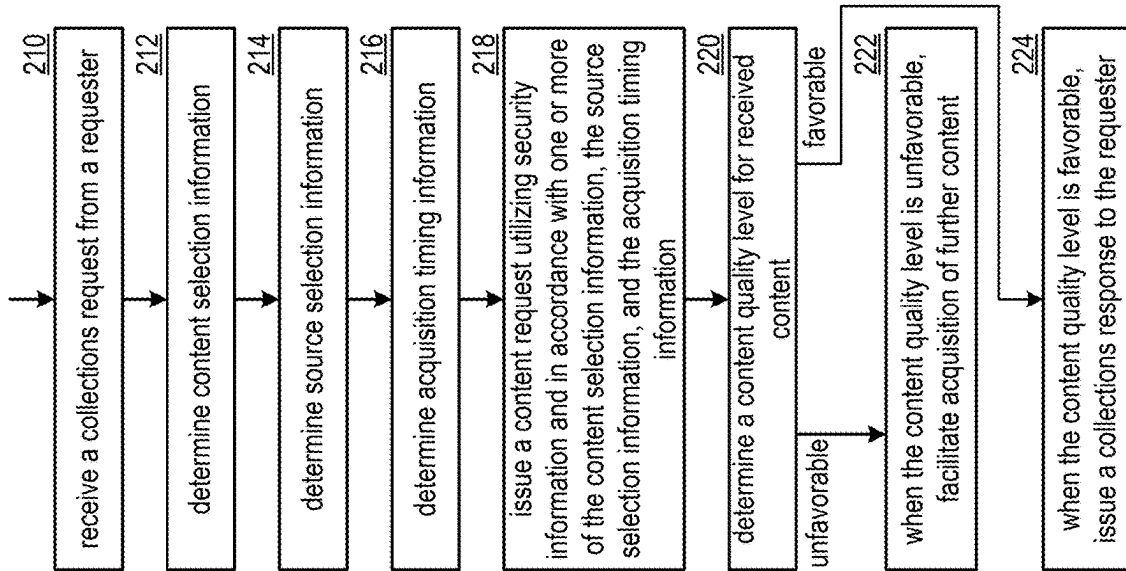


FIG. 5B

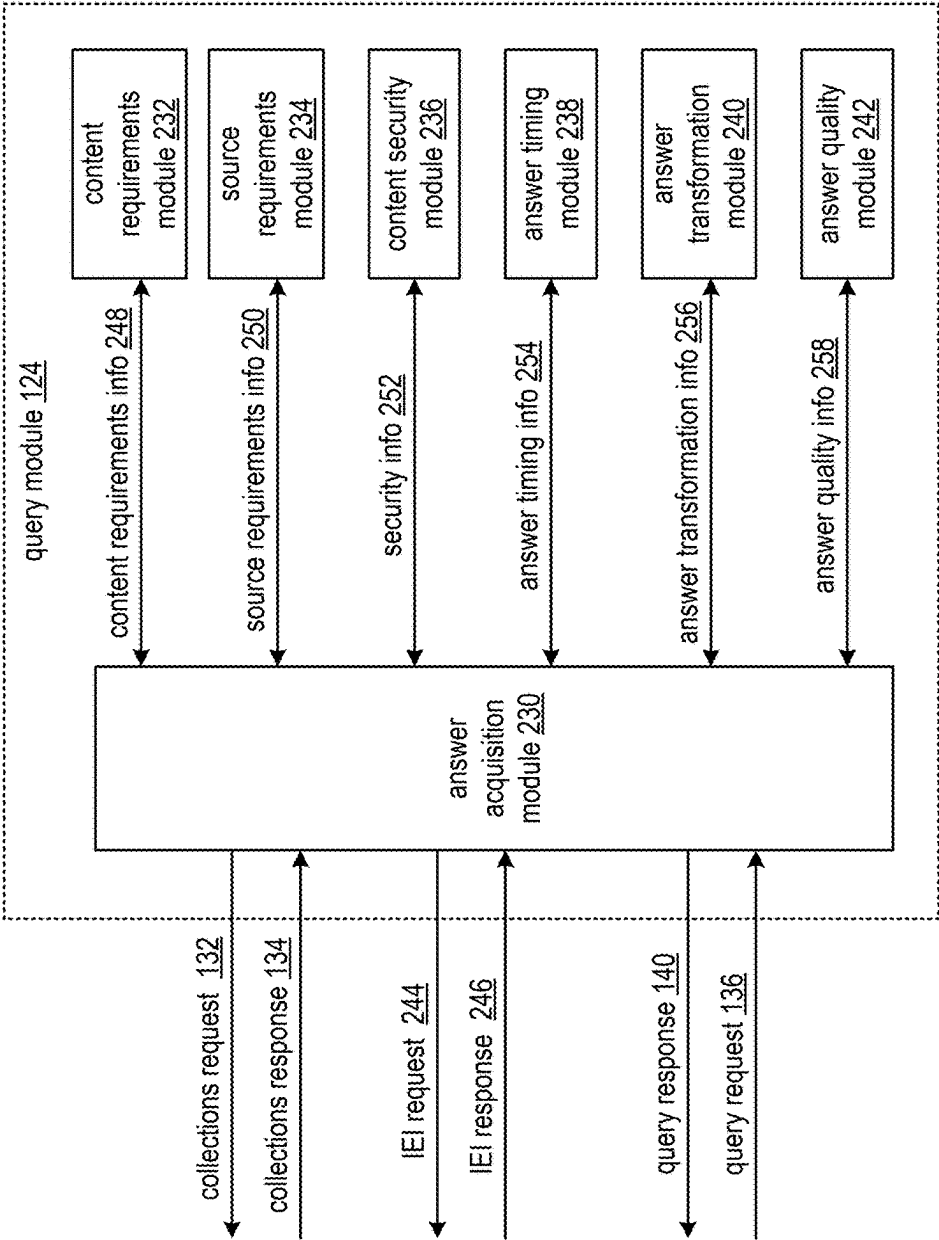


FIG. 5C

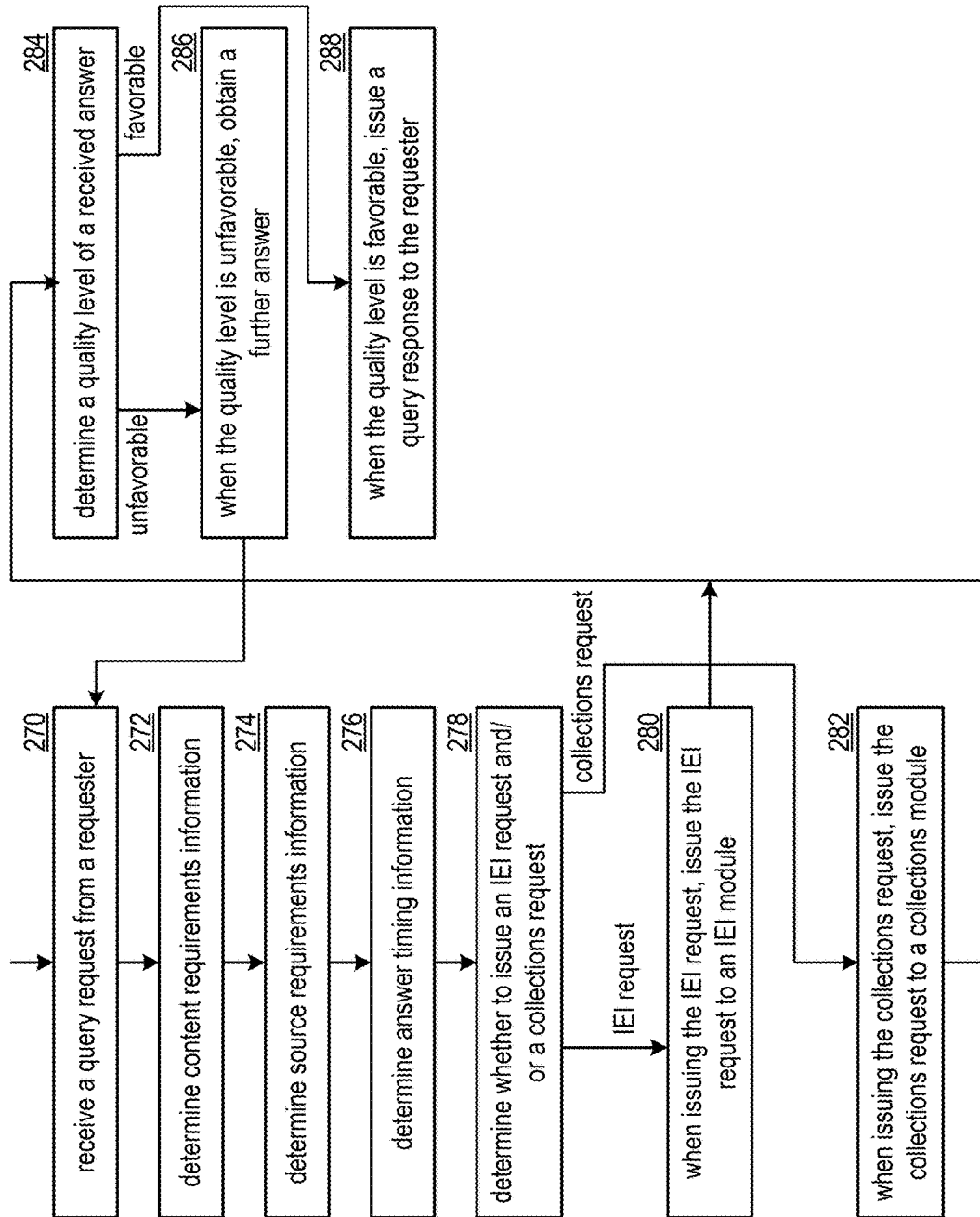


FIG. 5D

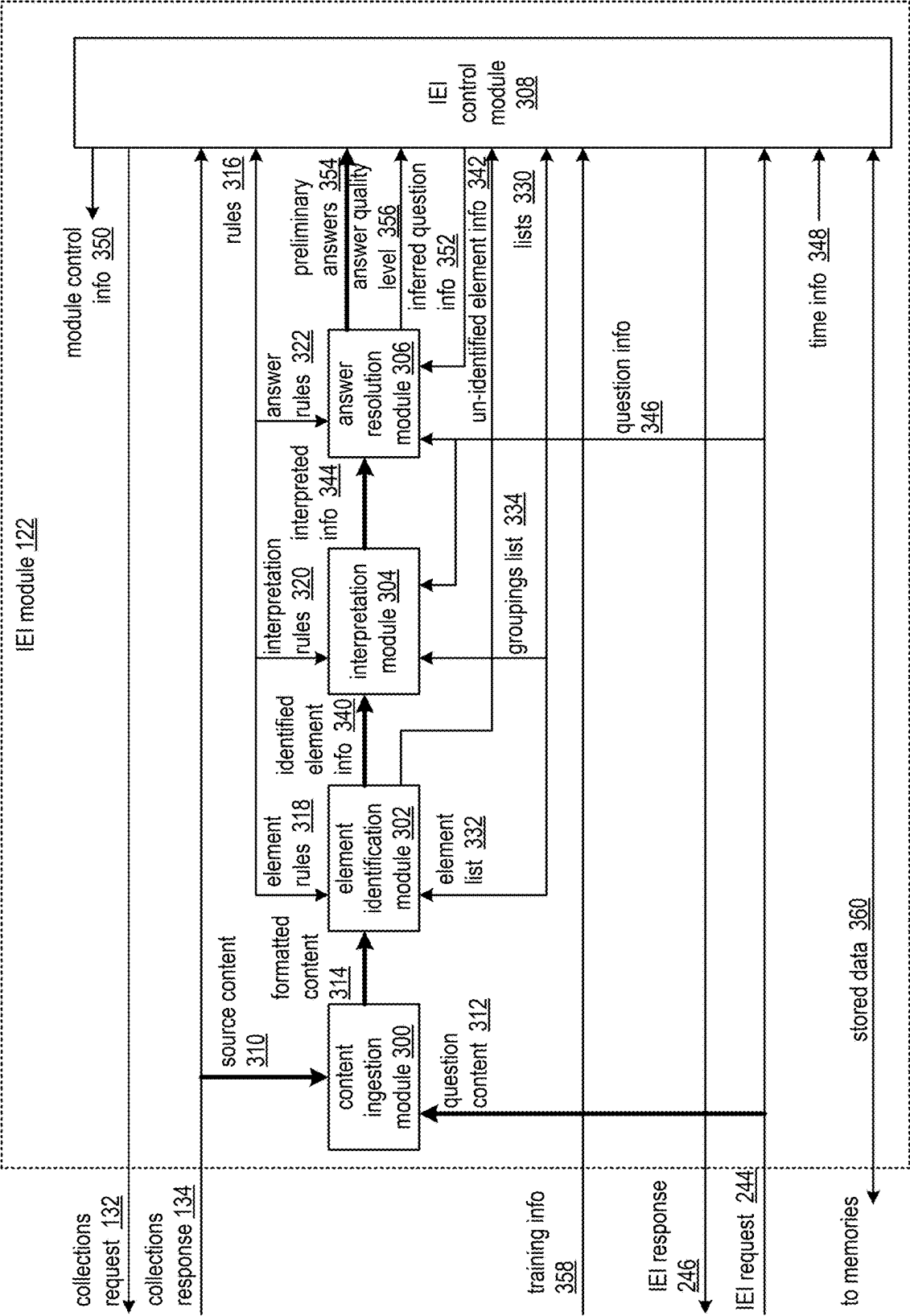


FIG. 5E

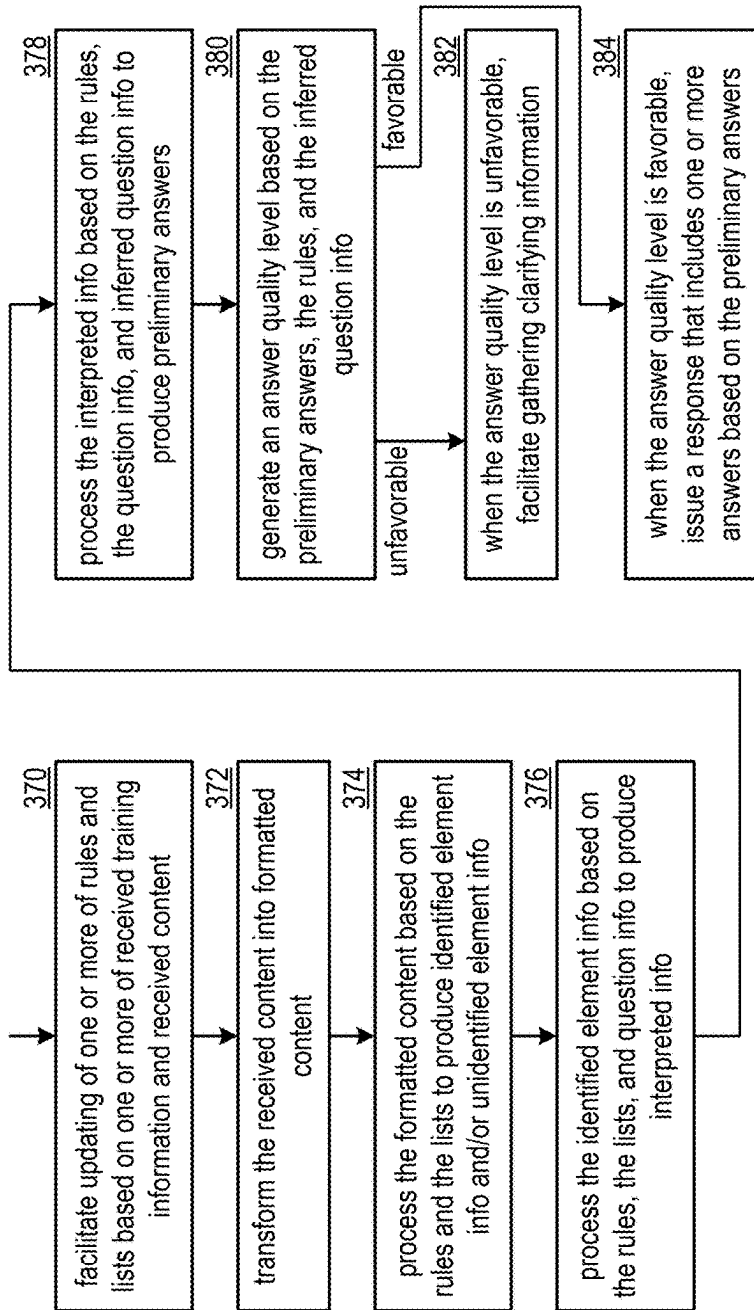


FIG. 5F

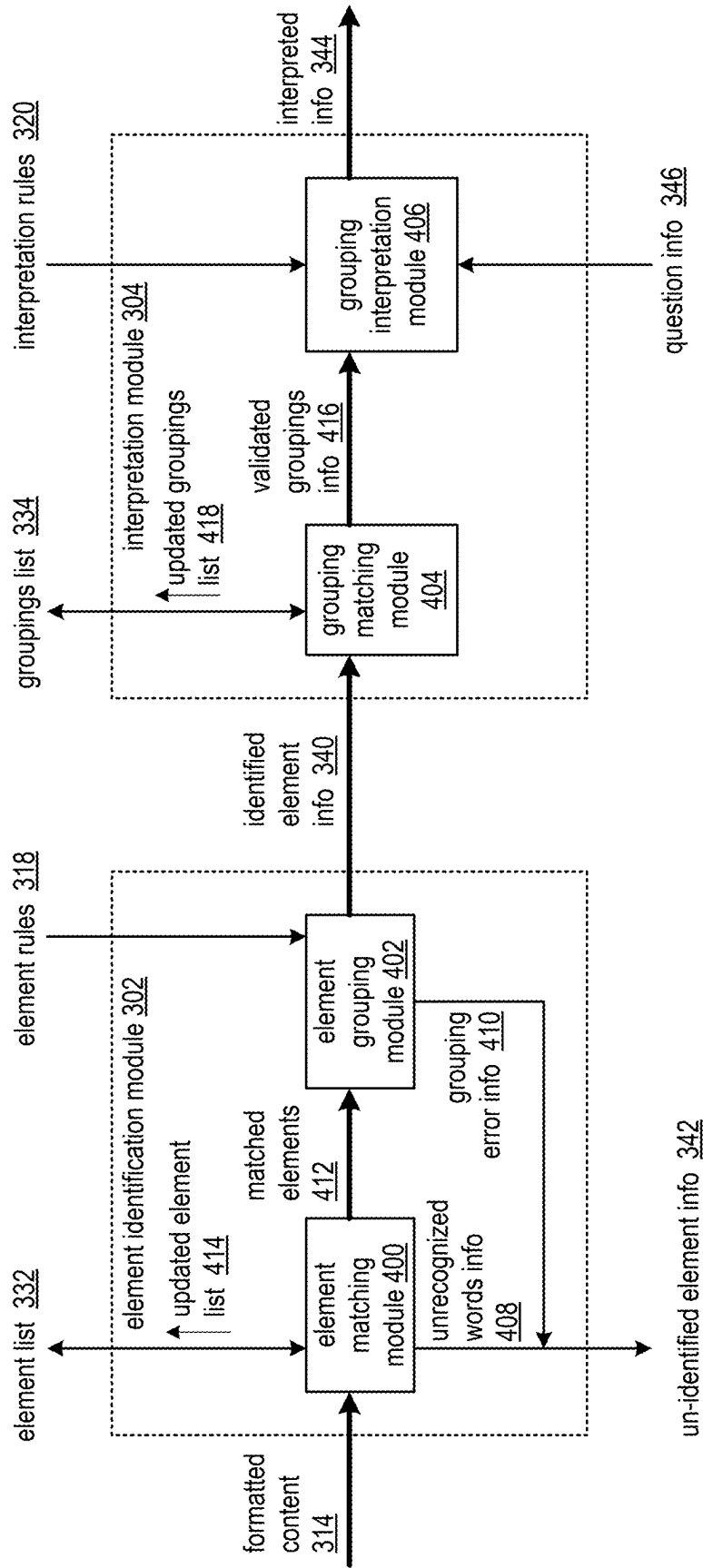


FIG. 6A

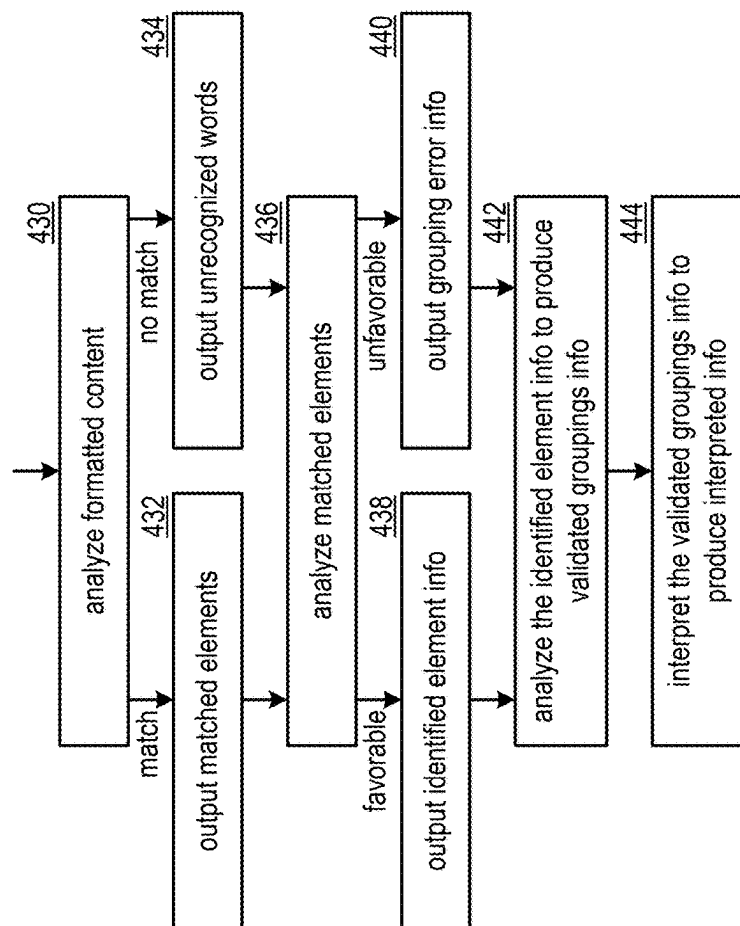


FIG. 6B

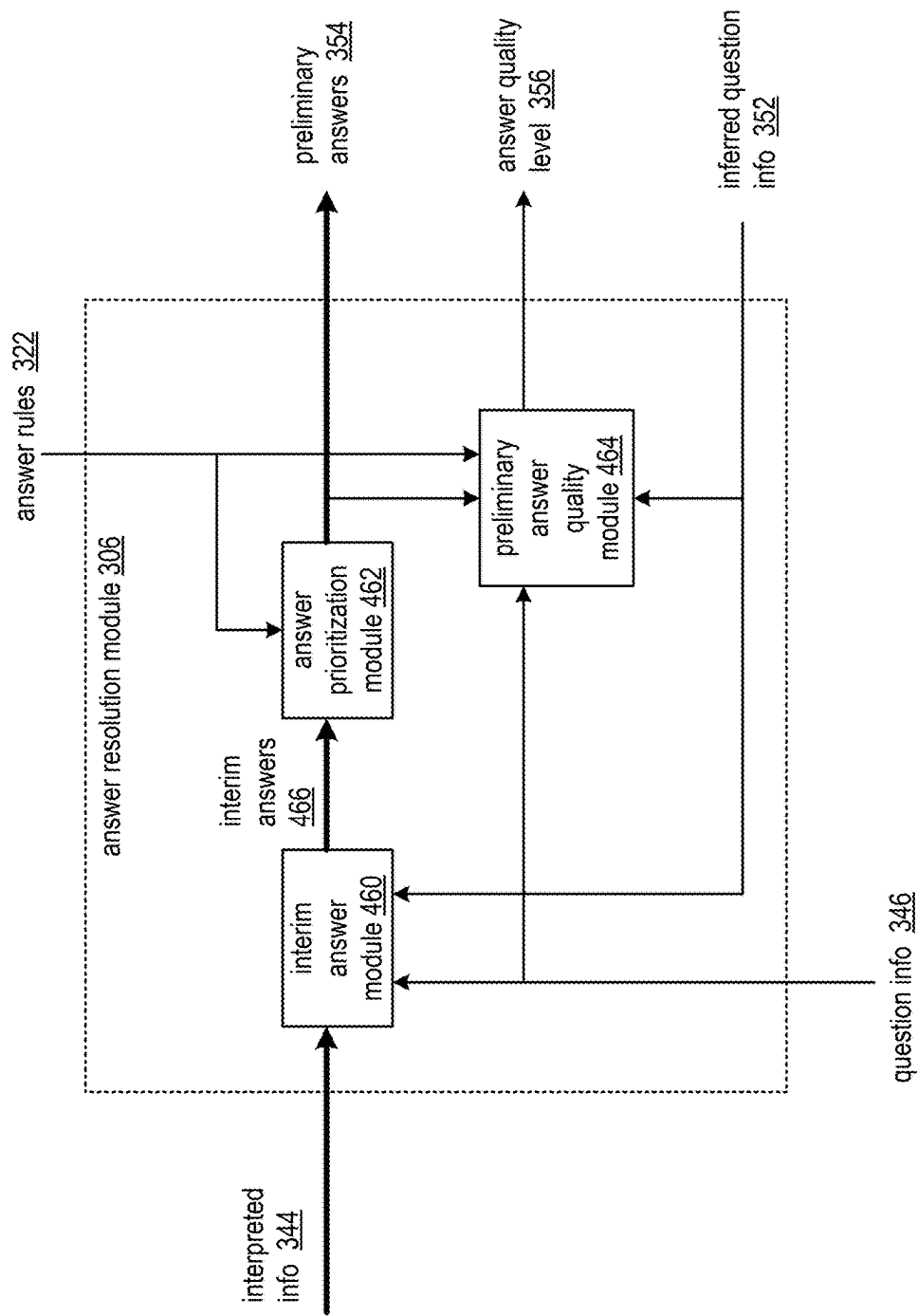


FIG. 6C

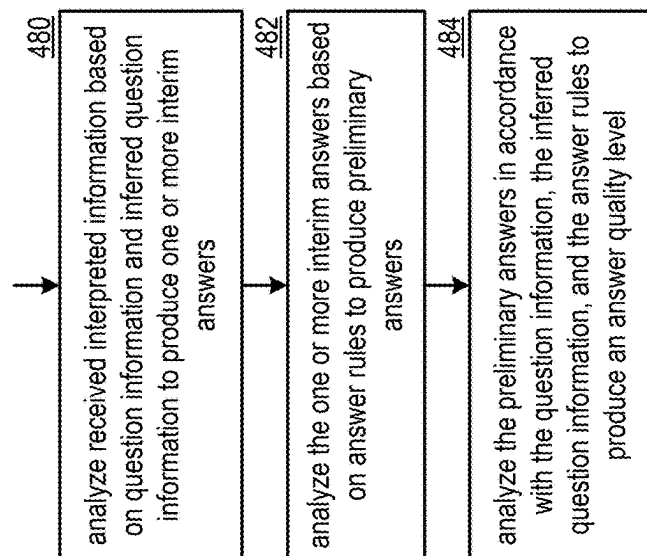


FIG. 6D

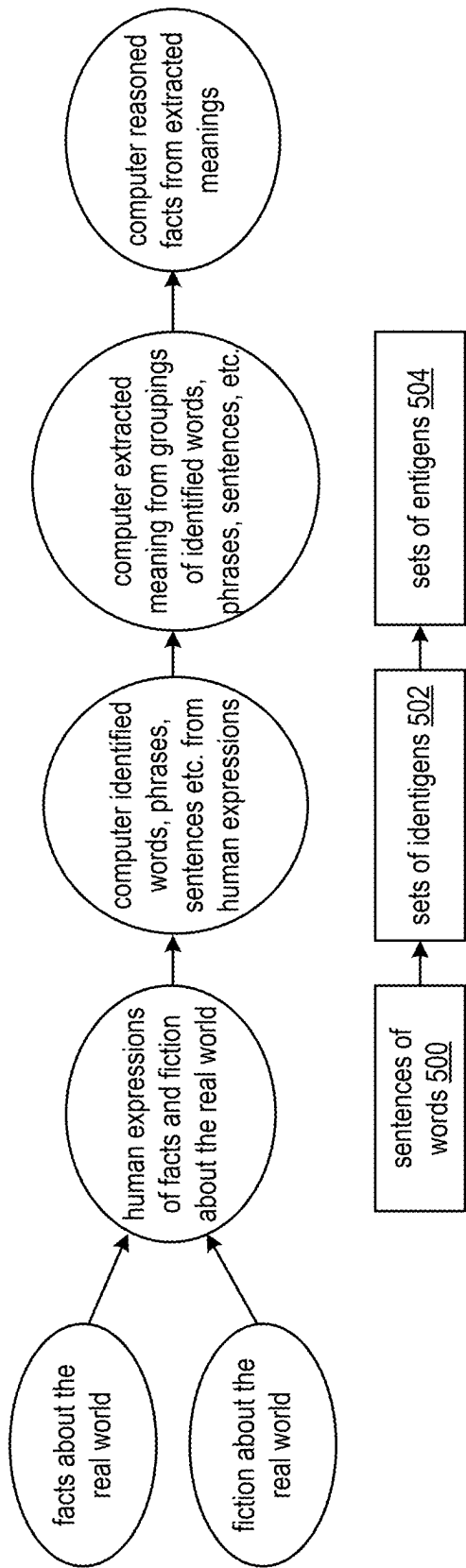


FIG. 7A

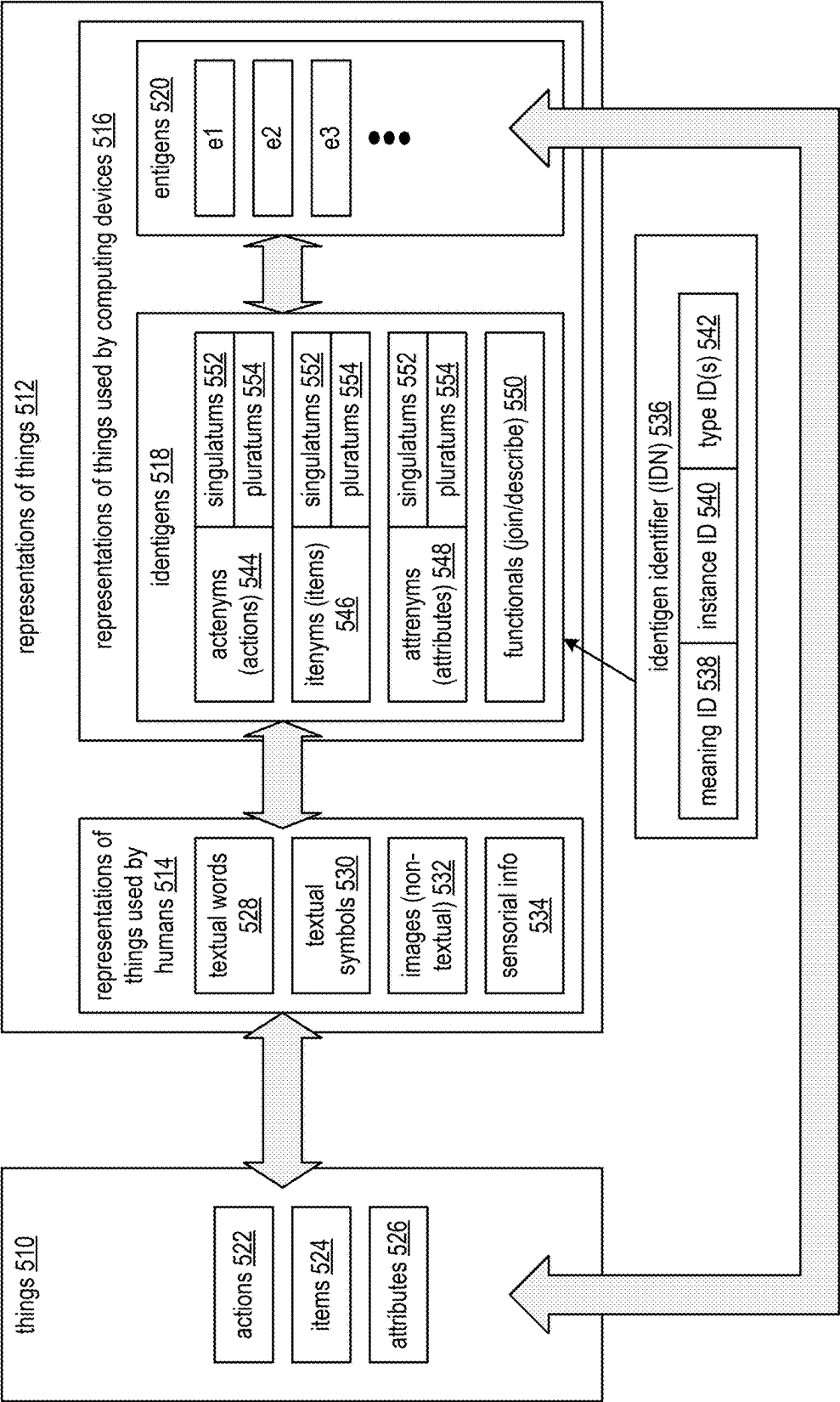


FIG. 7B

FIG. 7C

synonym words table <u>570</u>			
textual words <u>572</u>	identigen <u>518</u>		entigens <u>520</u>
	meaning ID <u>538</u>	instance ID <u>540</u>	
car	50	001	e1000
automobile		002	
auto		003	
bil (Swedish)		004	
carro (Spanish)		005	
bil (Danish)		006	

FIG. 7D

polysemous words table <u>576</u>			
textual words <u>572</u>	identigen <u>518</u>		entigens <u>520</u>
	meaning ID <u>538</u>	instance ID <u>540</u>	
duck (bird)	116937	001	e2000
duck (fabric)	116938	001	e2001
duck (to submerge)	112195	056	e2002
duck (to elude)	010654	051	e2003

words table 580									
textual words 572		identigen (IDN) 518			entigens (ENI)				
		meaning ID 538	instance ID 540	type ID 542	520				
pilot (flyer)		150	001	itenym (item)	e717				
pilot (to fly)		291	001	actenym (action)	e4320				
Tom (person)		457	001	itenym (item)	e61				
Tom (male animal)		648	001	itenym (item)	e930				
tall (attribute - height)		823	001	attrenym (attribute)	e90				
tall (attribute - unlikely)		399	001	attrenym (attribute)	e729				

→ apply rules to identigens of word strings to validate

pilot (to fly)	Tom (male an)	IDN 291.001	IDN 648.001	X	tall (attribute - height)	Tom (male an)	IDN 823.001	IDN 648.001	✓
pilot (flyer)	Tom (male an)	IDN 150.001	IDN 648.001	X	tall (attribute - unlikely)	Tom (male an)	IDN 399.001	IDN 648.001	X
pilot (to fly)	Tom (person)	IDN 291.001	IDN 457.001	X	tall (attribute - height)	Tom (person)	IDN 823.001	IDN 457.001	✓
pilot (flyer)	Tom (person)	IDN 150.001	IDN 457.001	✓	tall (attribute - unlikely)	Tom (person)	IDN 399.001	IDN 457.001	X

→ list valid groupings

groupings table 584									
grouping ID 586		word strings 588		identigens (IDN) 518			entigens (ENI) 520		
3001		pilot Tom		IDN 150.001	IDN 457.001		e717	e61	
3040		tall Tom		IDN 823.001	IDN 648.001		e90	e930	
3041		tall Tom		IDN 823.001	IDN 457.001		e90	e61	
3070		tall pilot Tom		IDN 823.001	IDN 150.001	IDN 457.001	e90	e717	e61

FIG. 7E

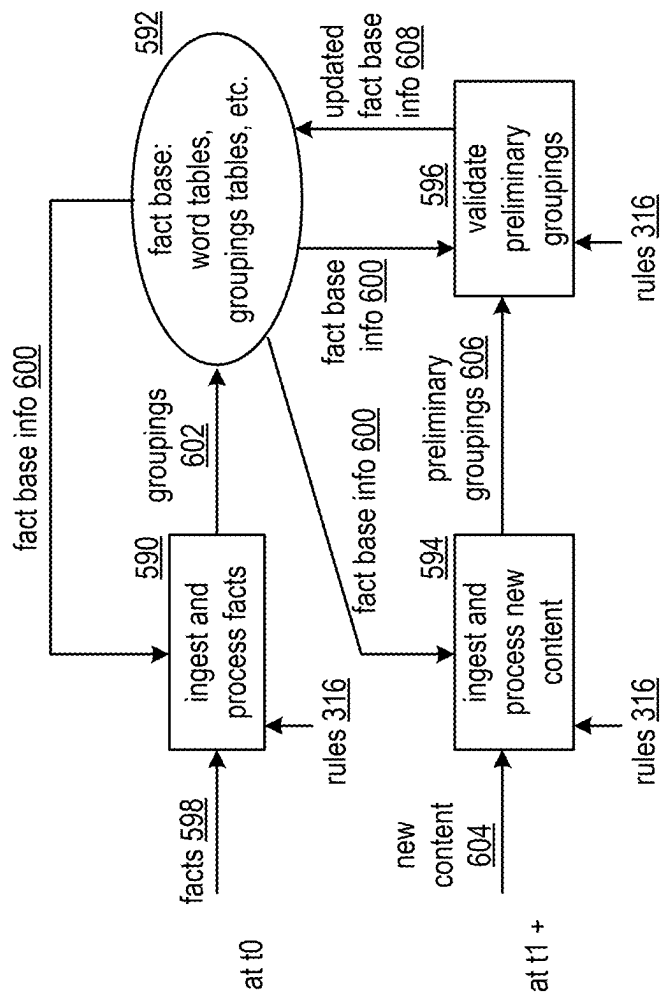


FIG. 8A

groupings table 620						
grouping (GRP) ID 586	word strings 588	IF string 622		THEN string 624		
		IDN string 626	ENI 628	IDN string 626	ENI 628	
5493	If someone has a tumor, then someone is sick.	someone has	12aa	someone sick	12js	
5494	If someone is sick, then someone is possibly sad.	tumor has	06aa			
		someone sick	12js	someone sad	12ja	
new knowledge inference 630						
5495	If someone has a tumor, then someone is possibly sad.	someone has	12aa	someone sad	12ja	
		tumor has	06aa	possibly sad	05b	

FIG. 8B

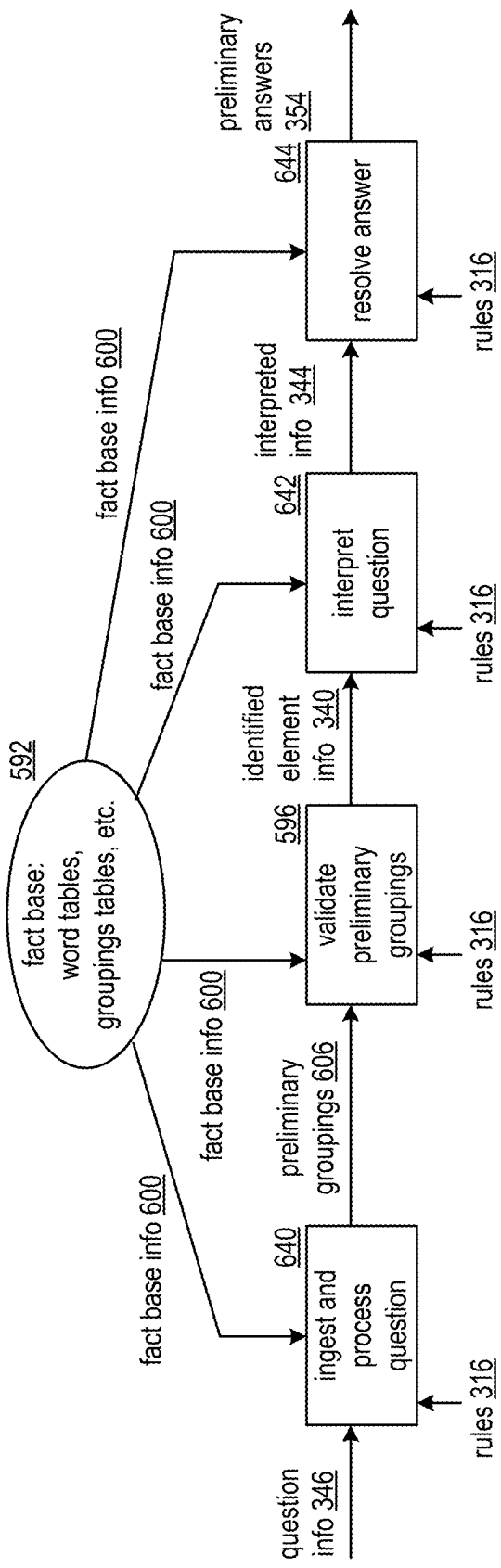


FIG. 8C

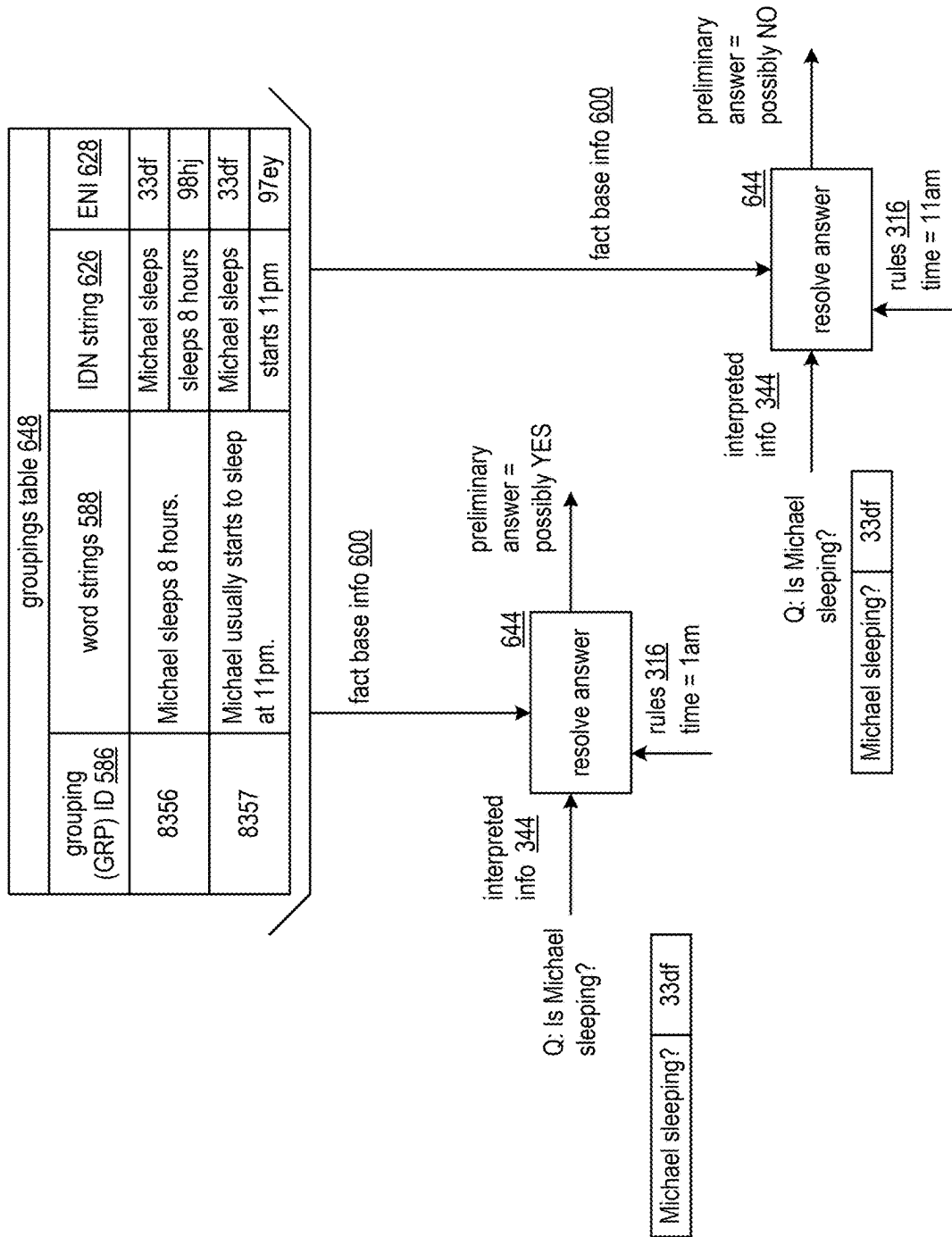


FIG. 8D

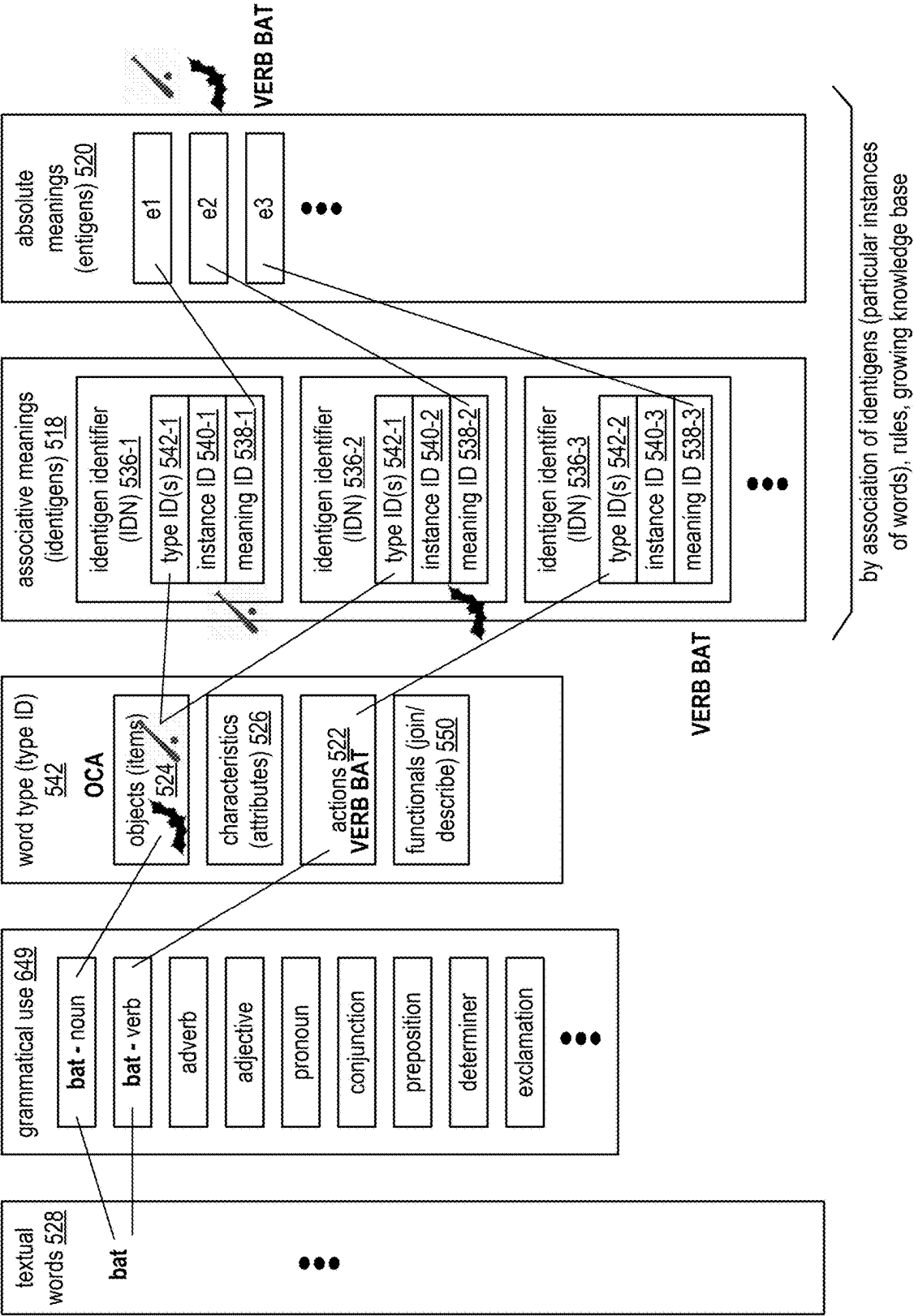


FIG. 8E

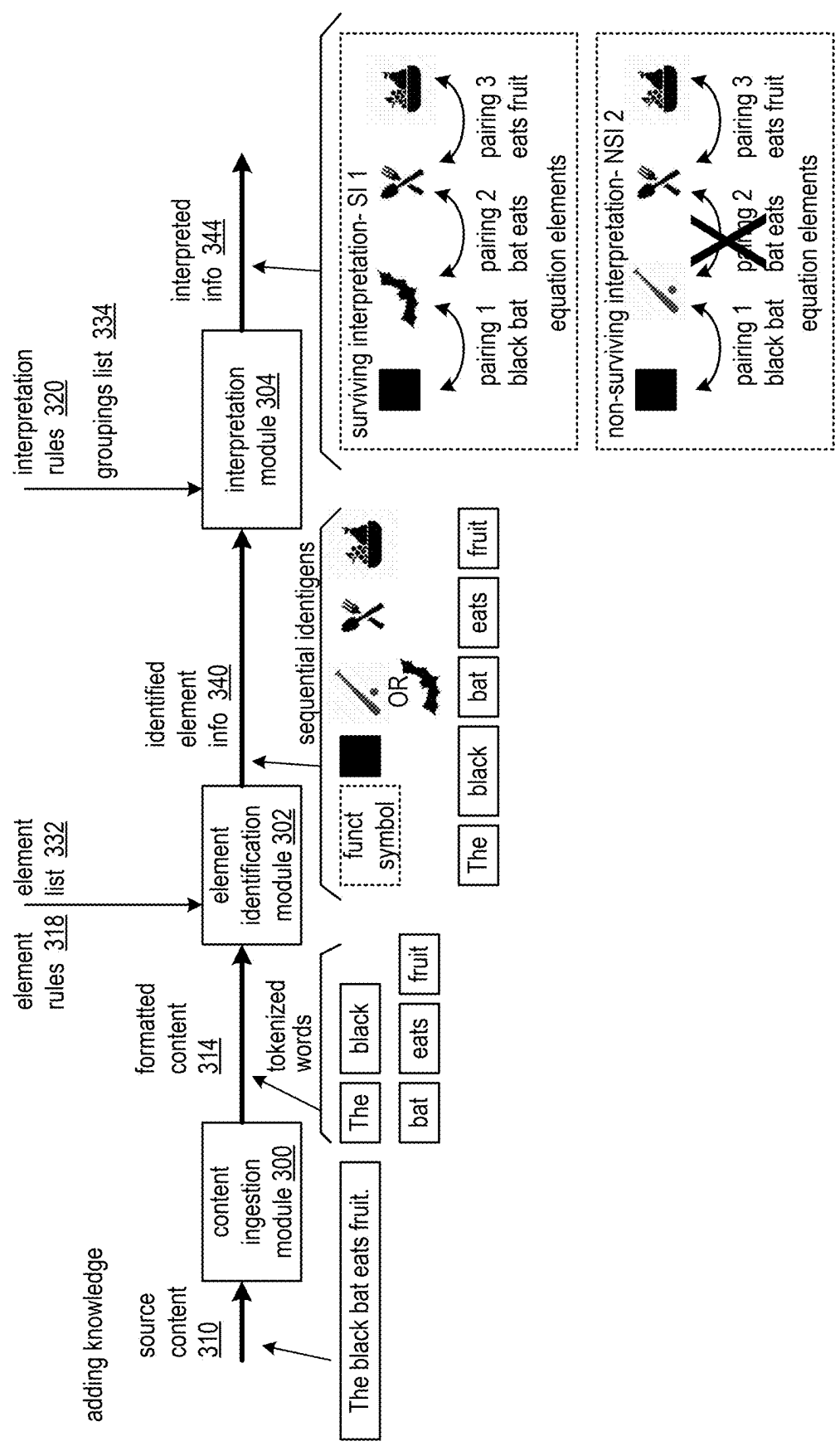
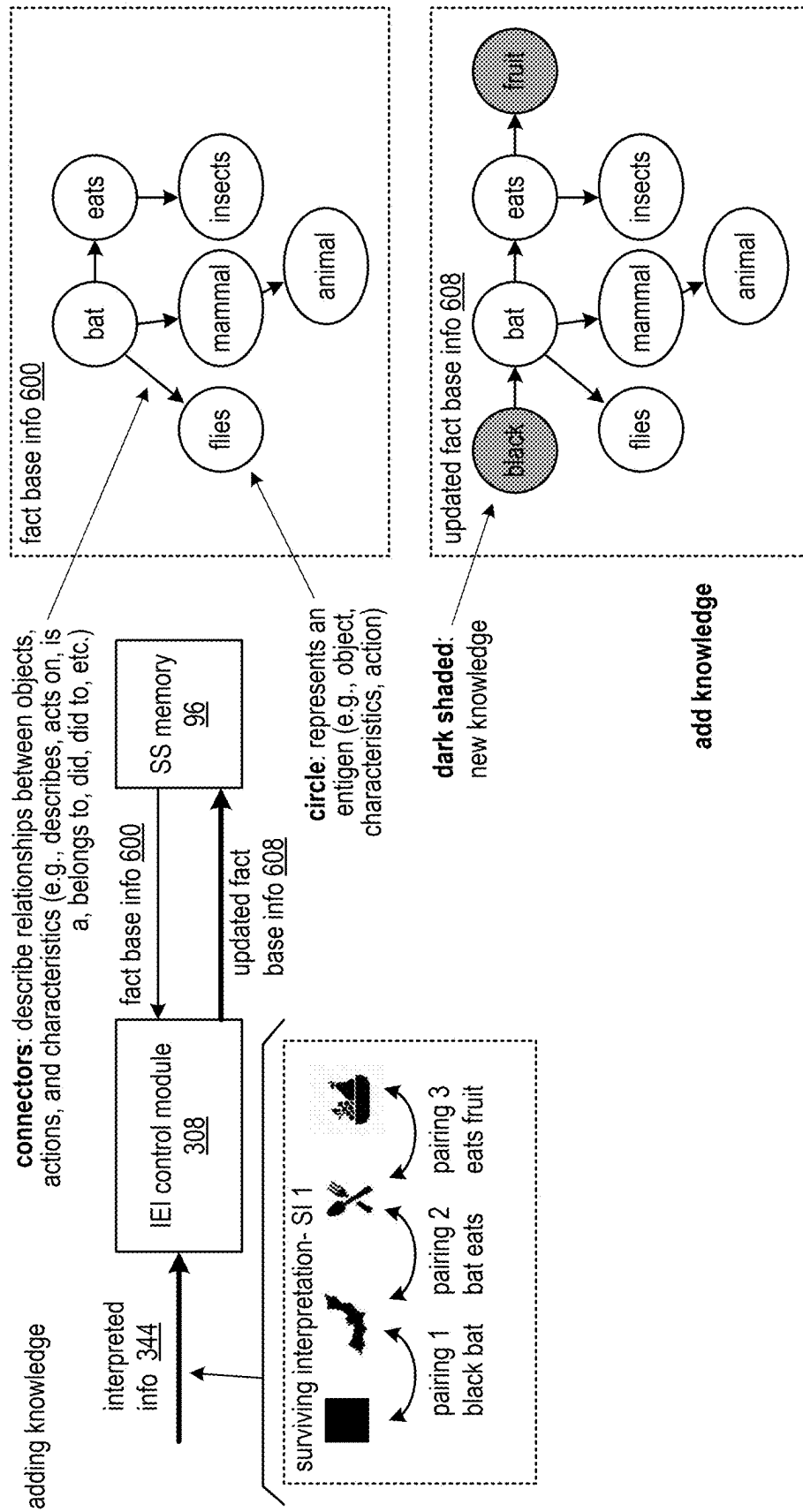


FIG. 8F



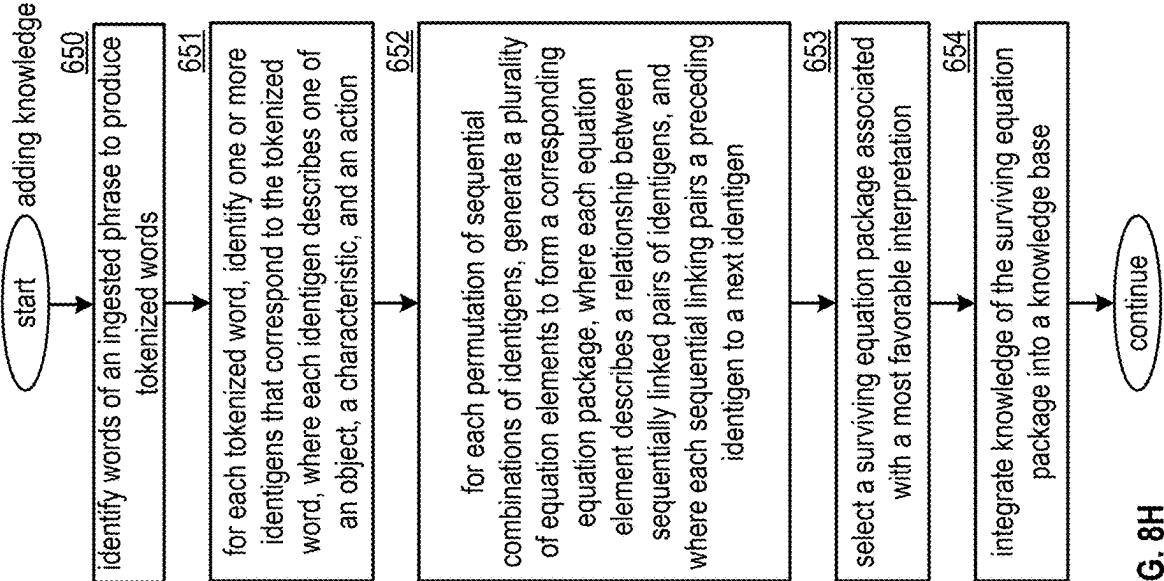


FIG. 8H

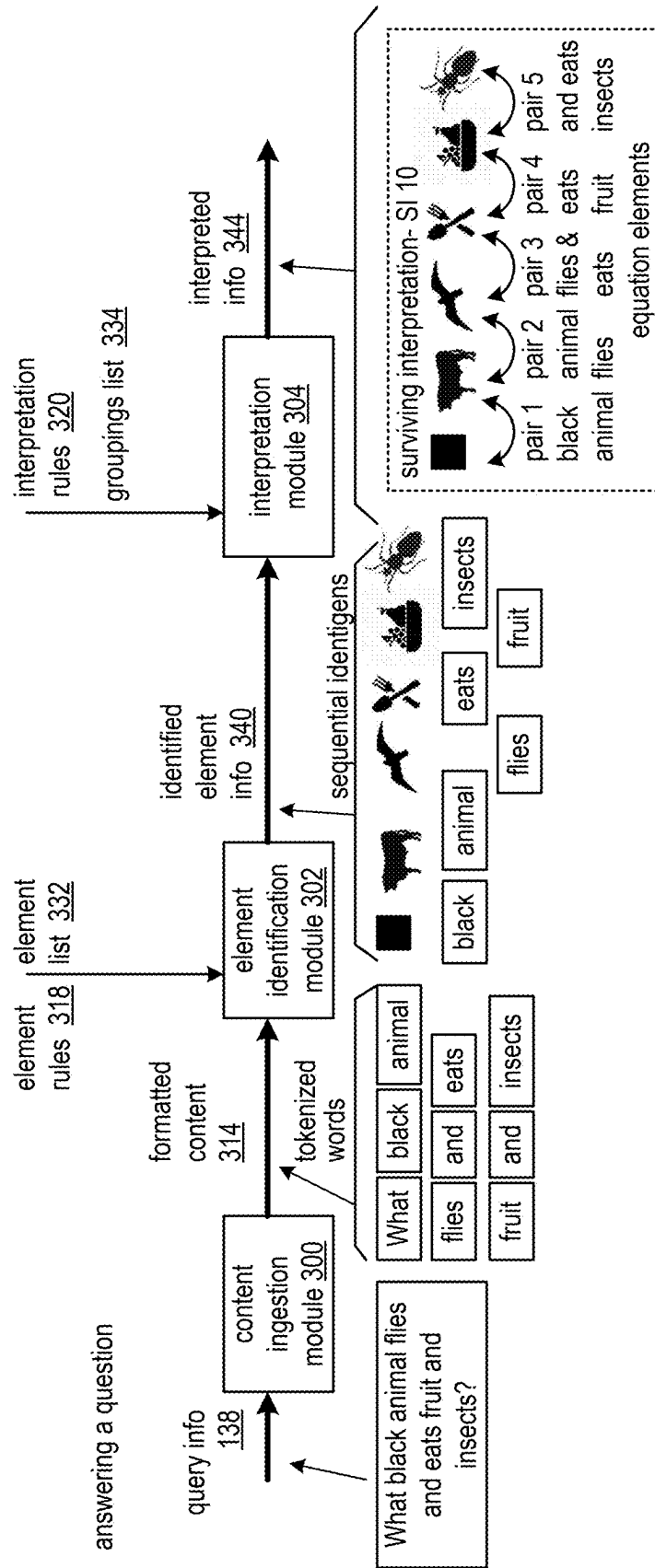


FIG. 8J

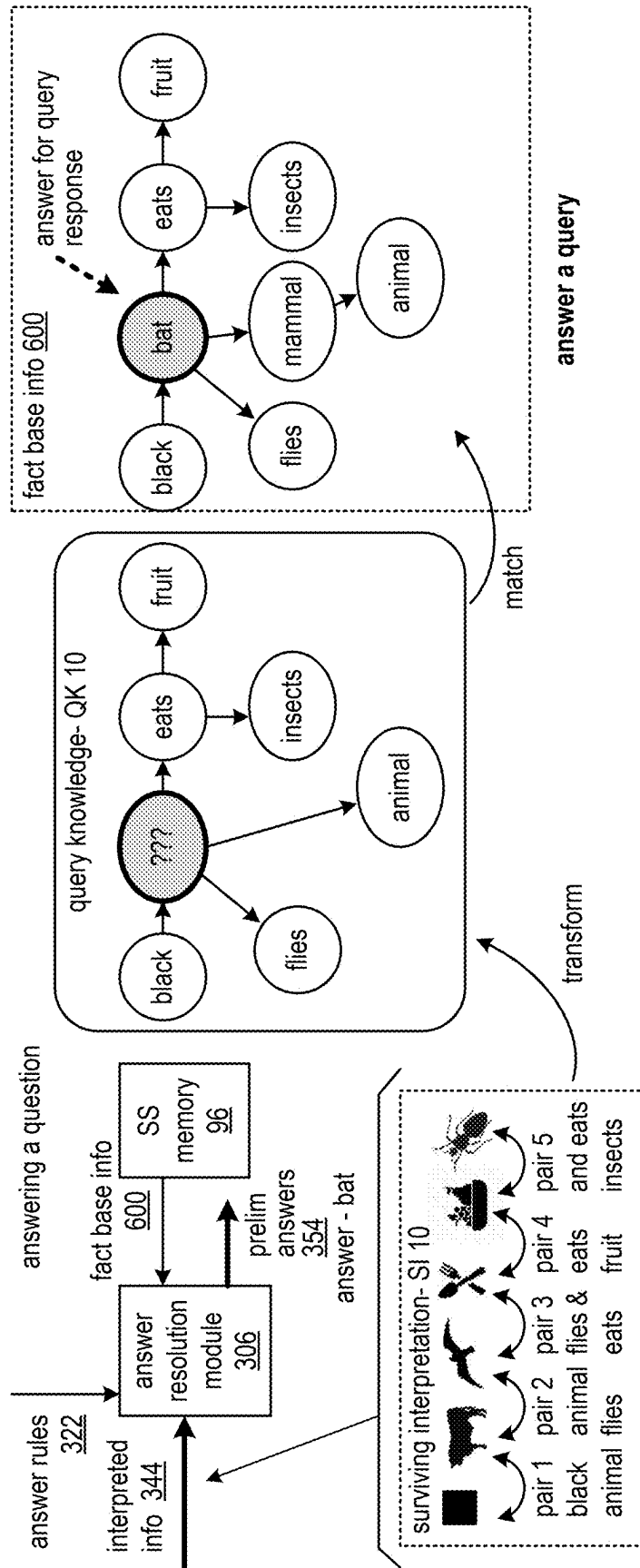


FIG. 8K

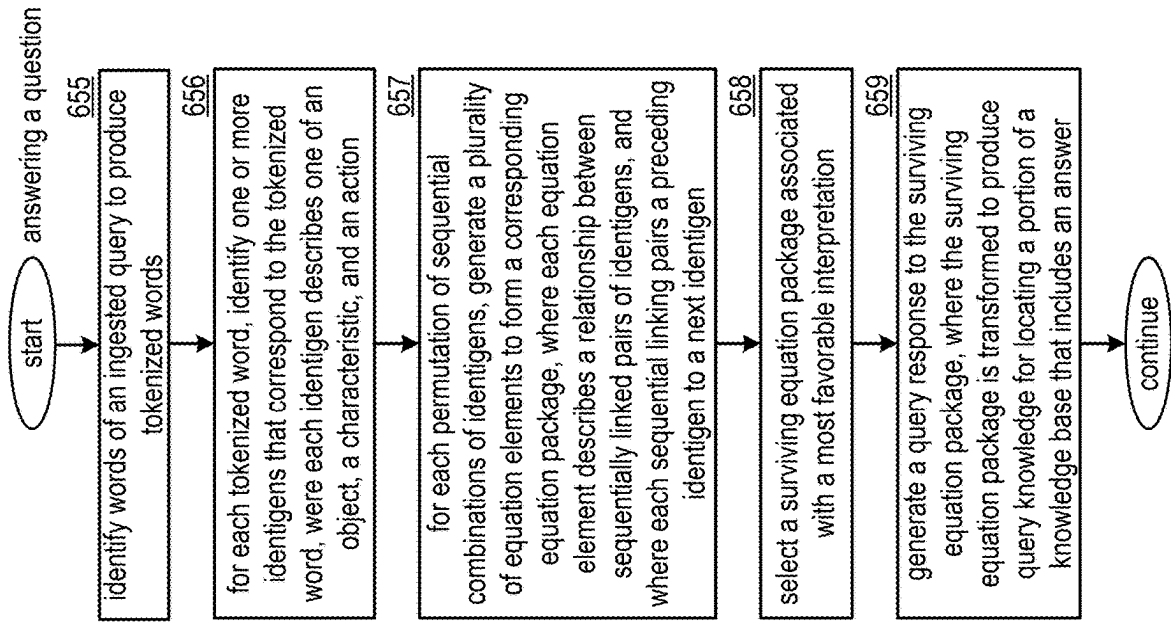


FIG. 8L

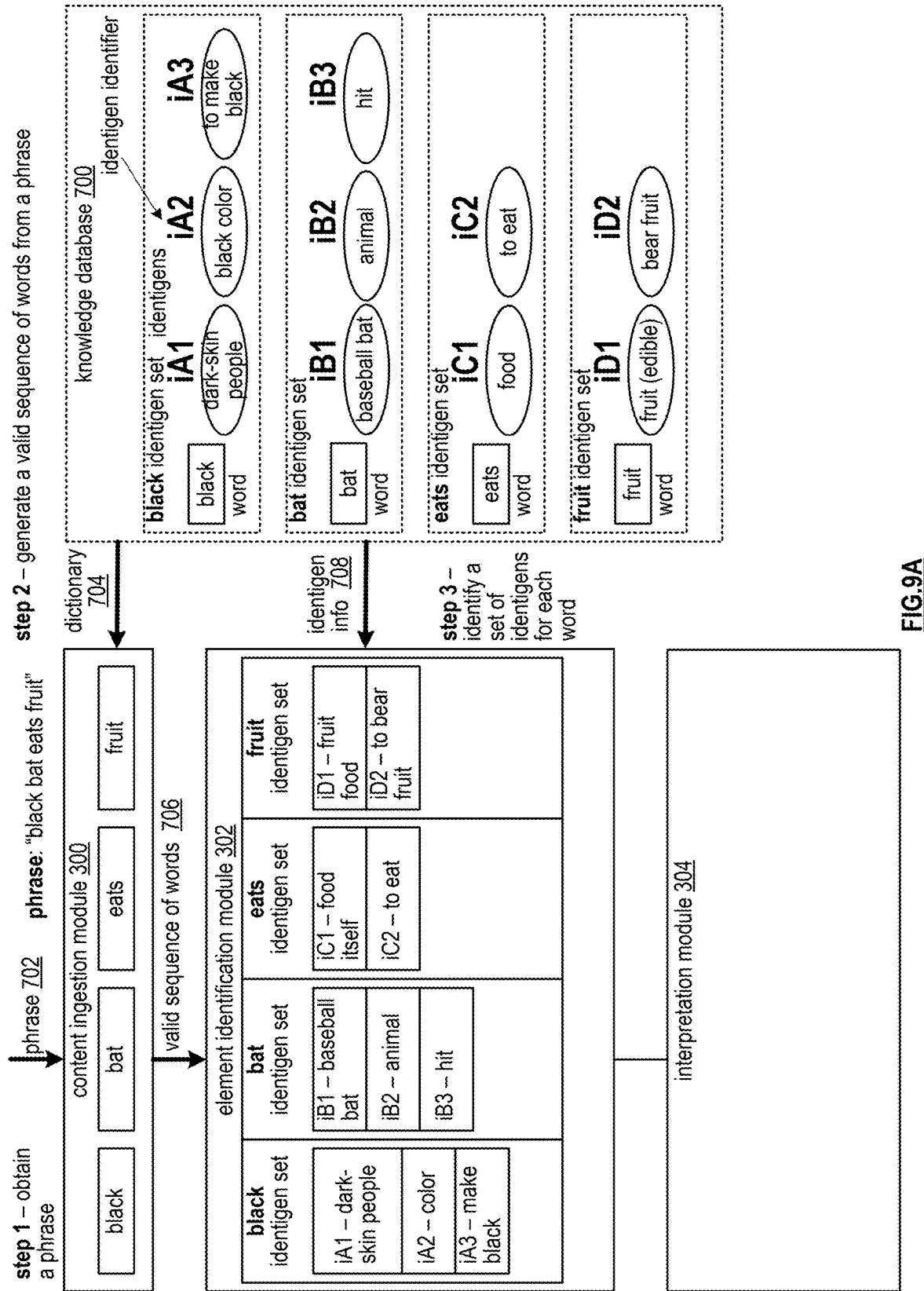


FIG.9A

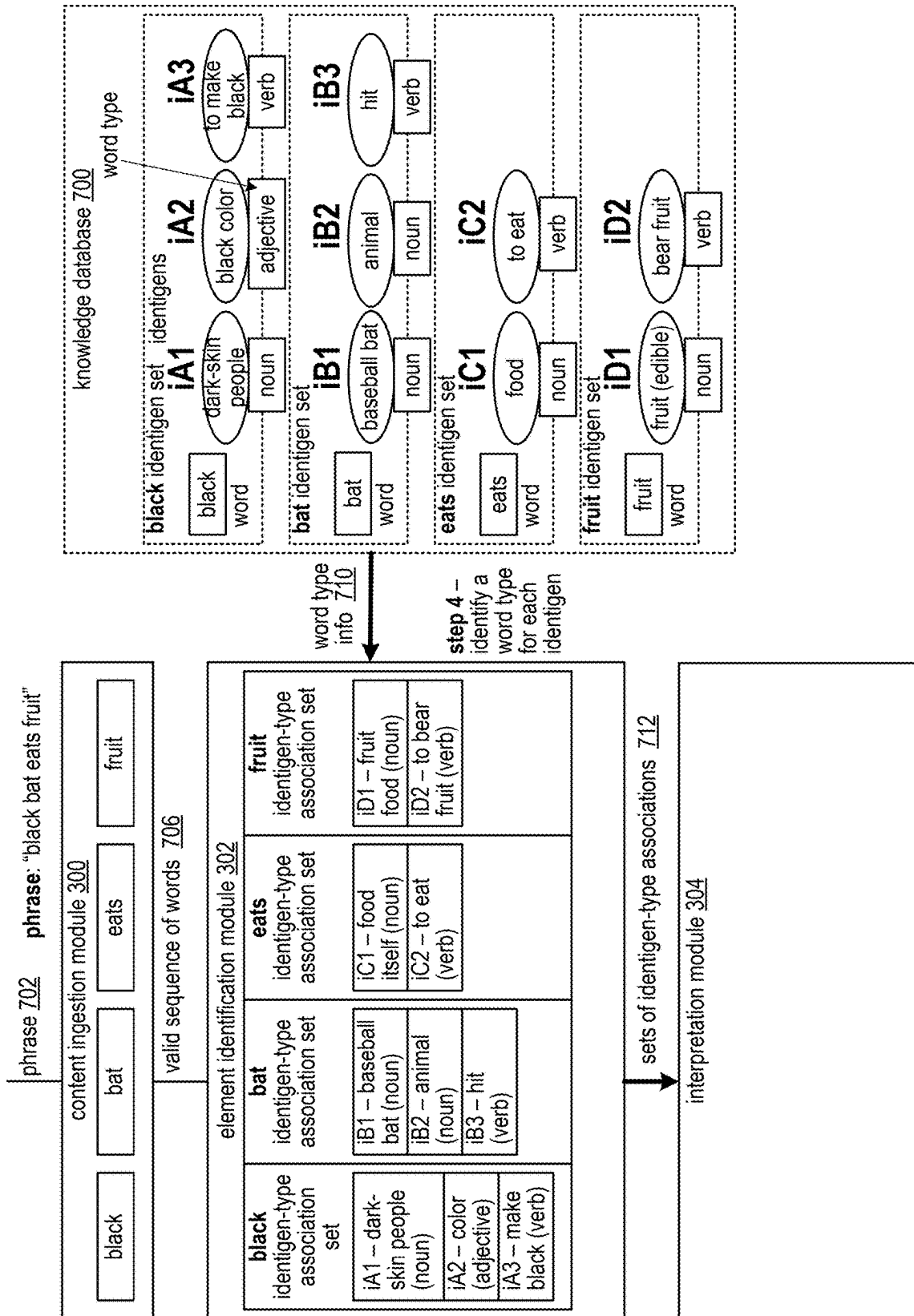
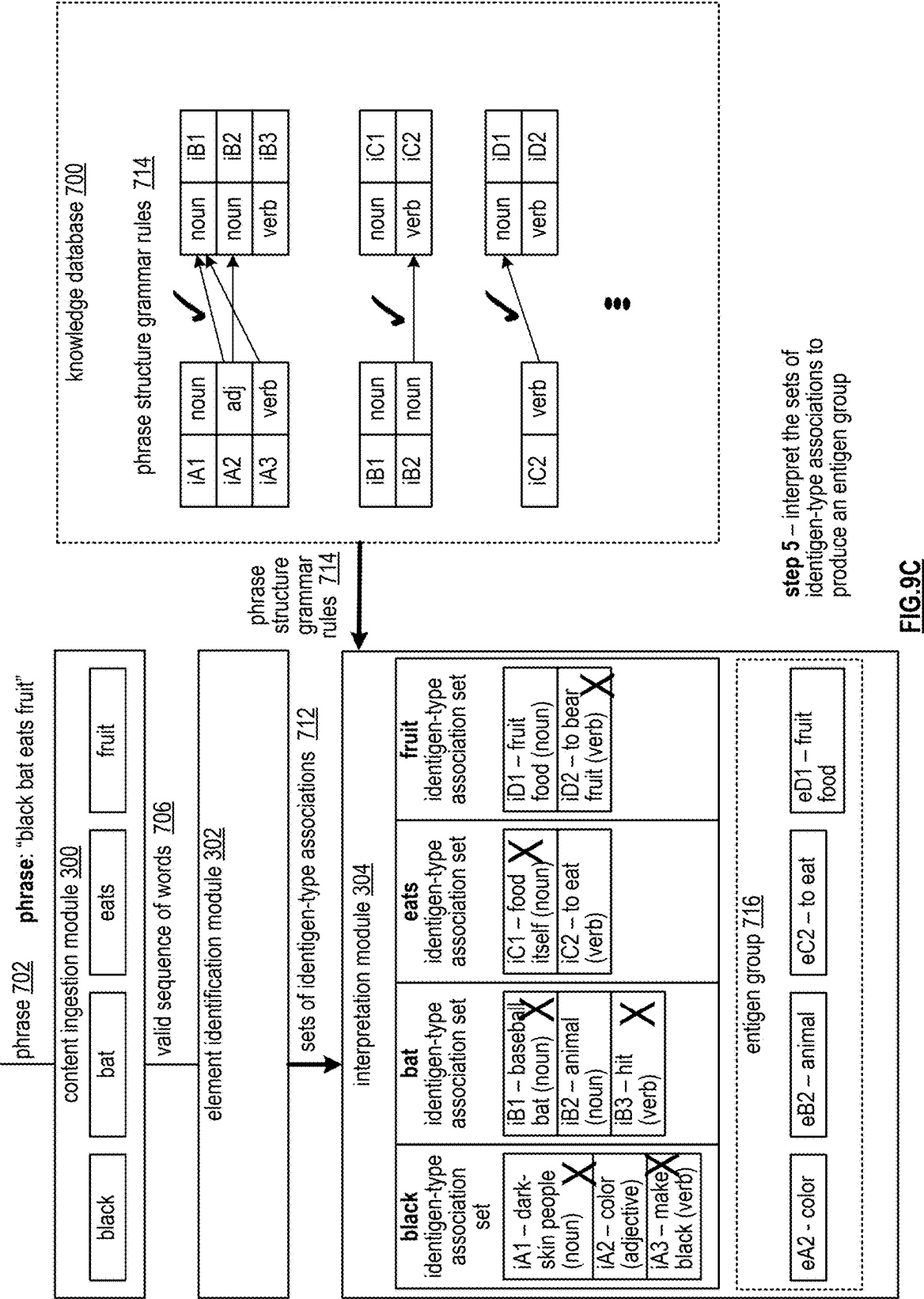
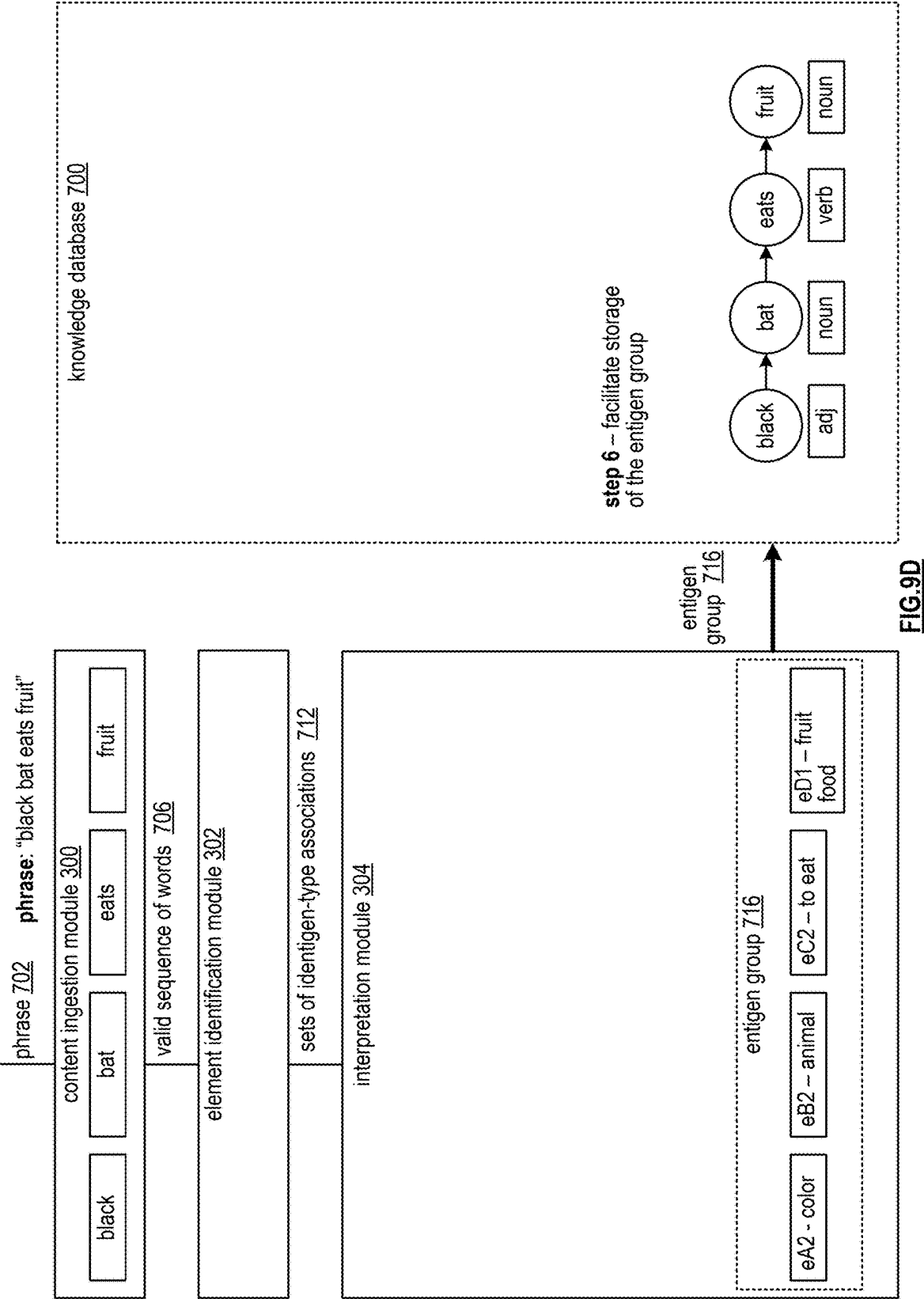


FIG. 9B





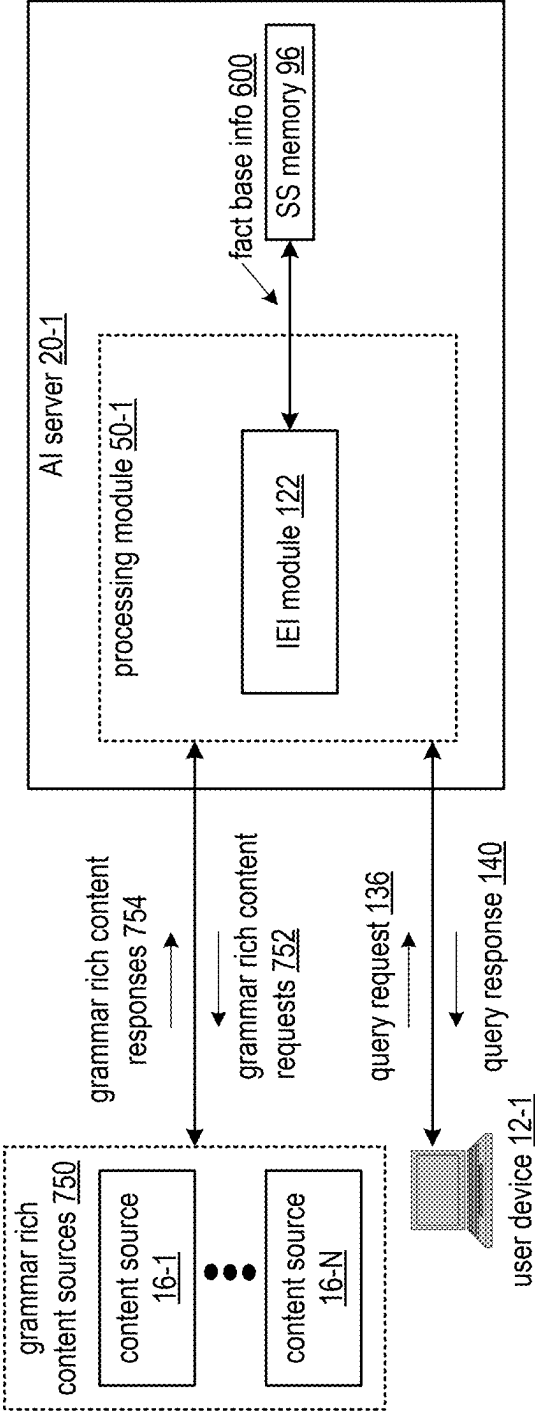


FIG. 10A

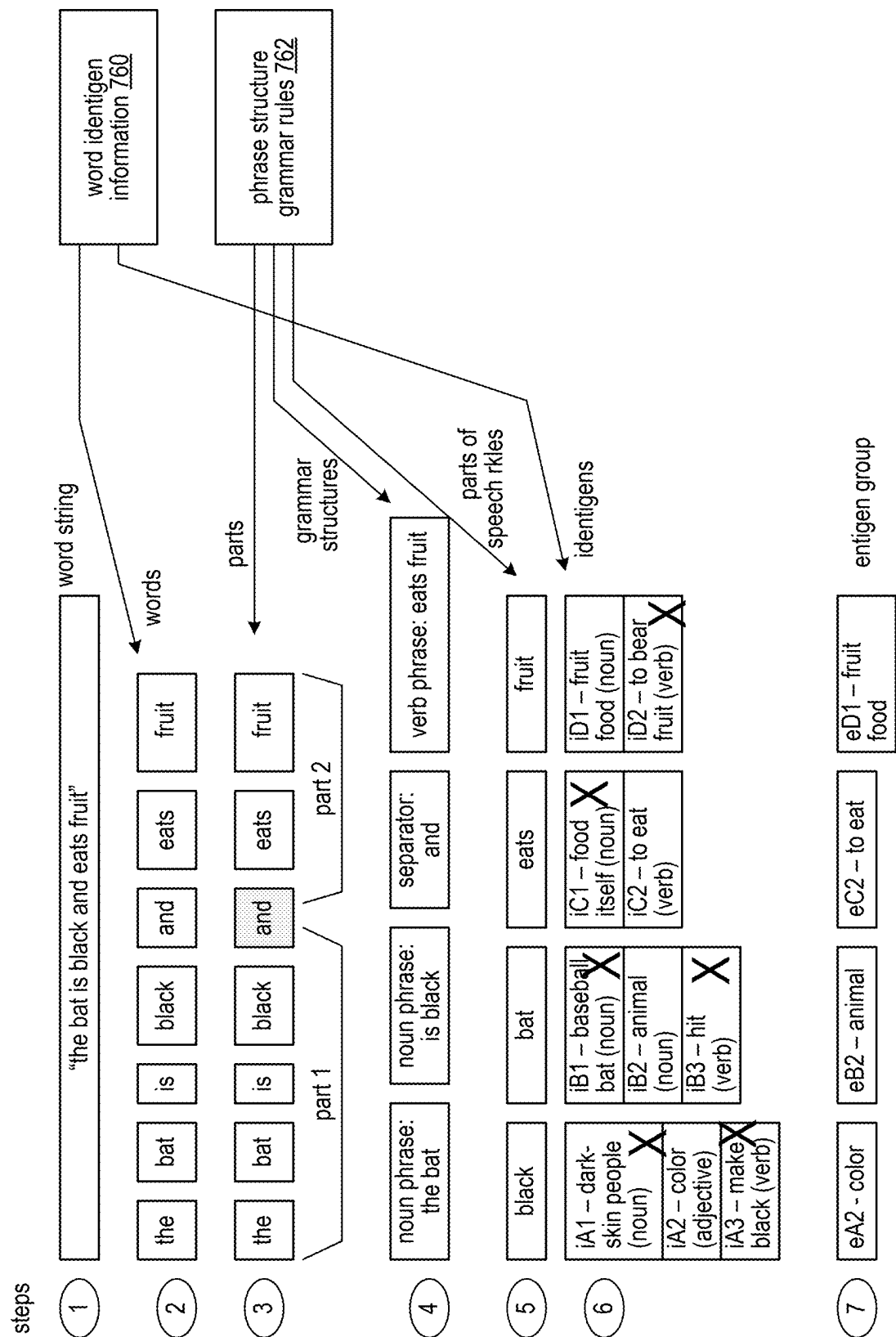
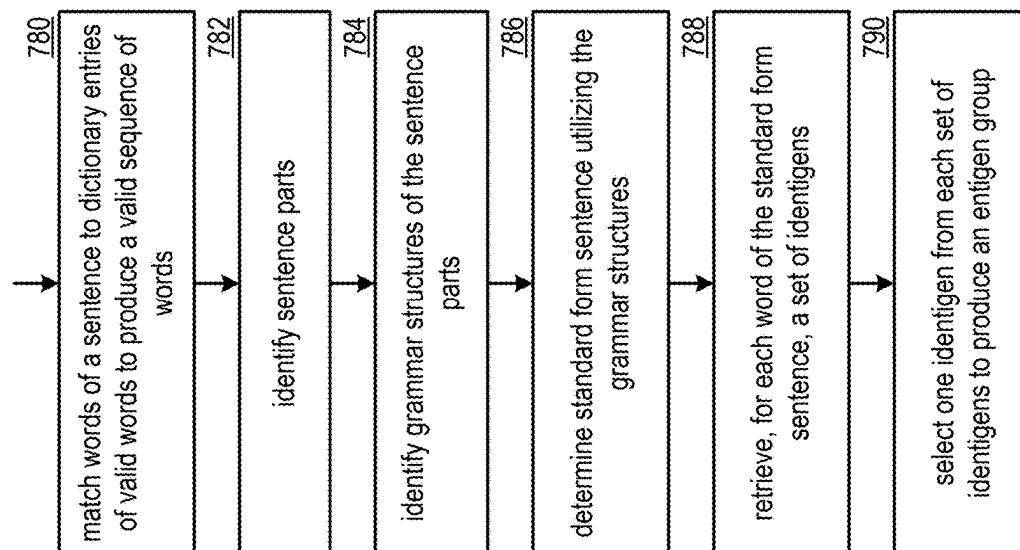


FIG. 10B

**FIG. 10C**

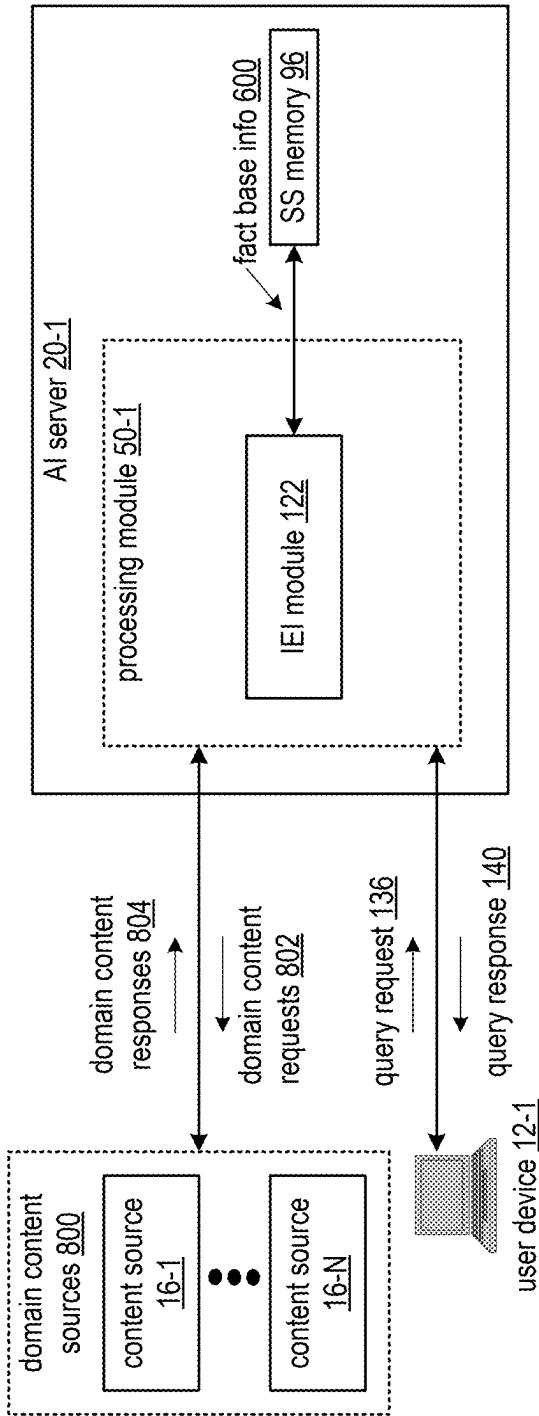


FIG. 11A

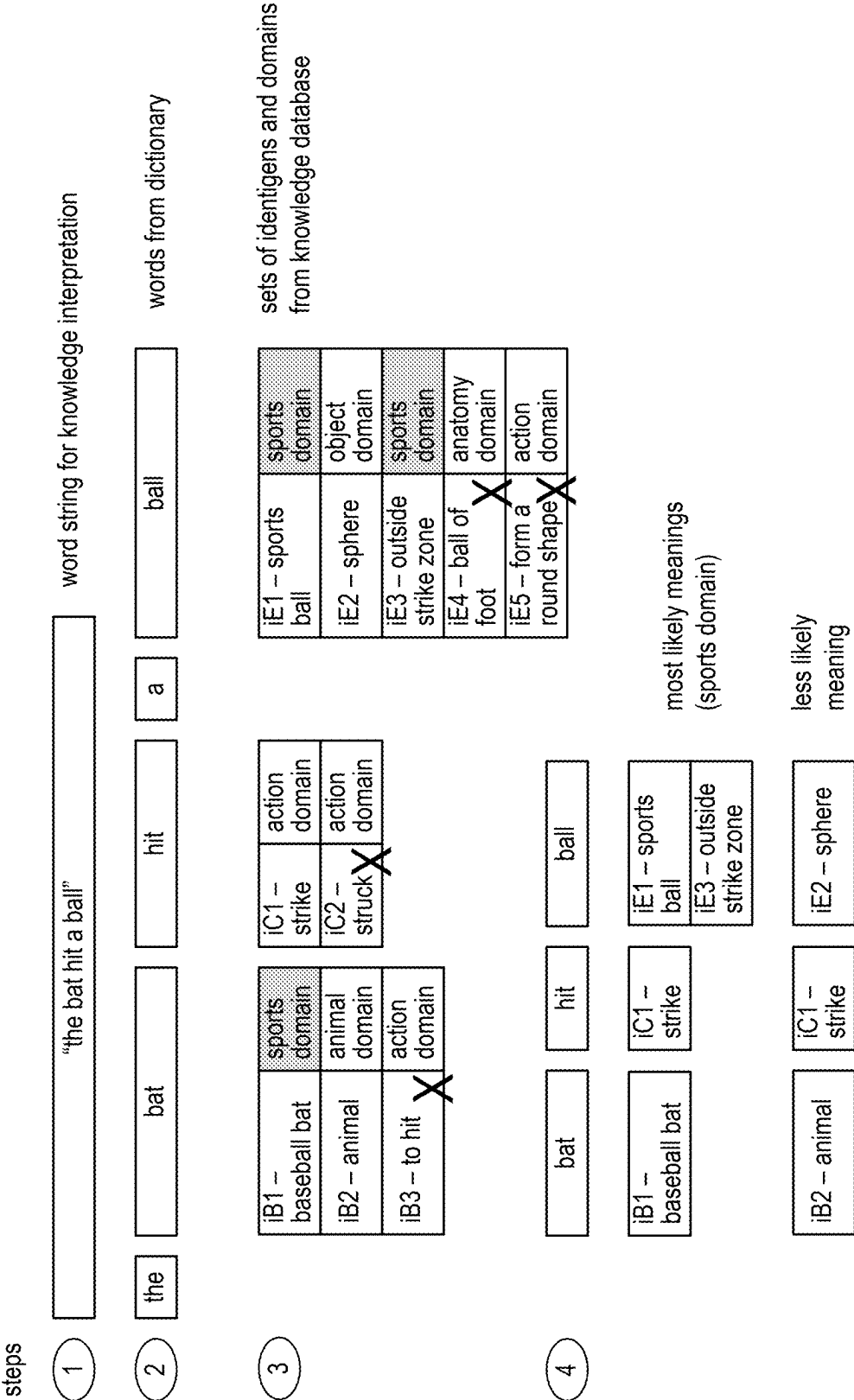


FIG. 11B

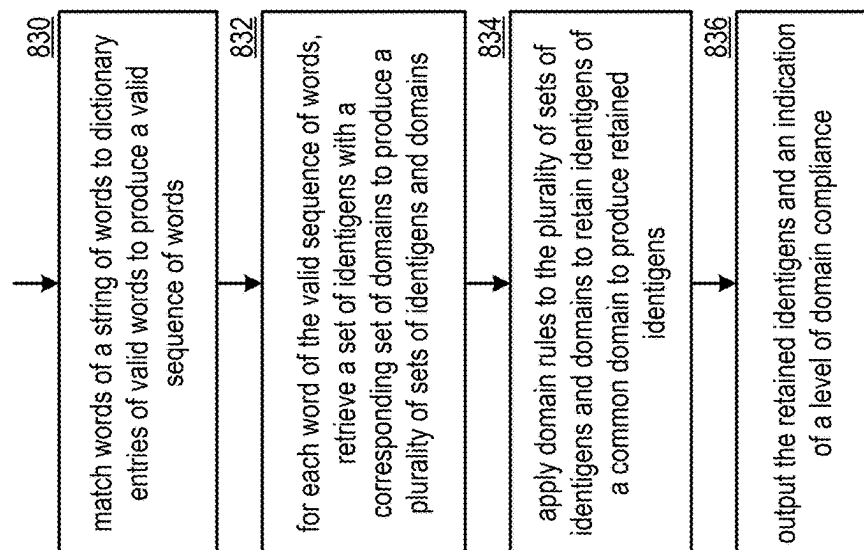


FIG. 11C

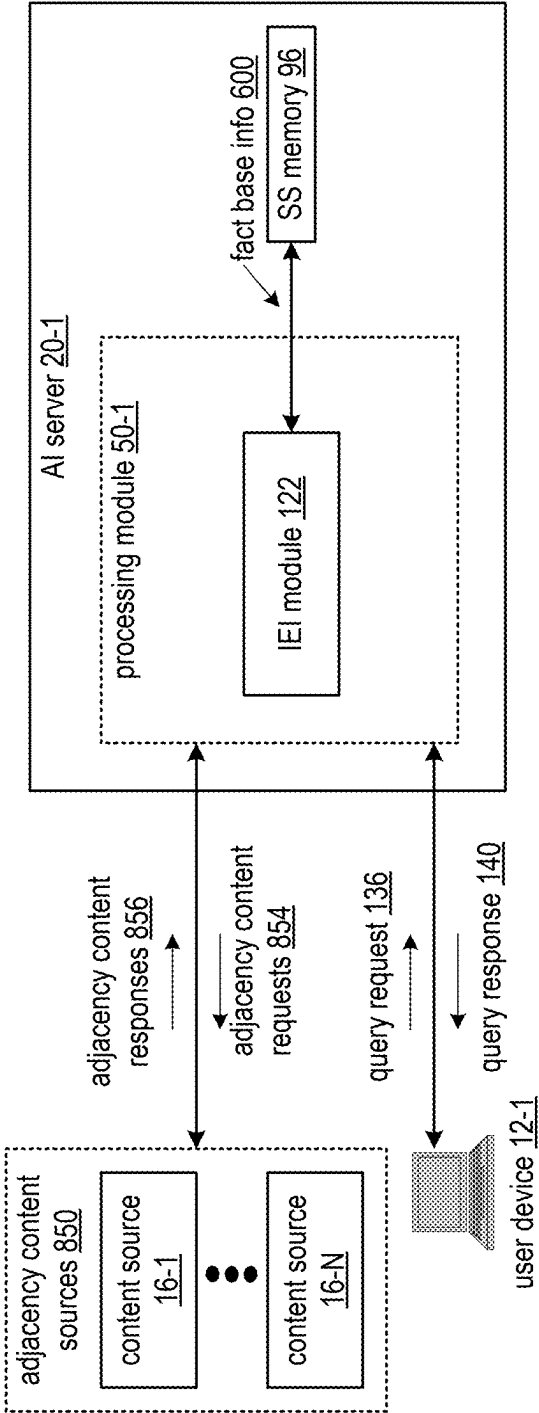


FIG. 12A

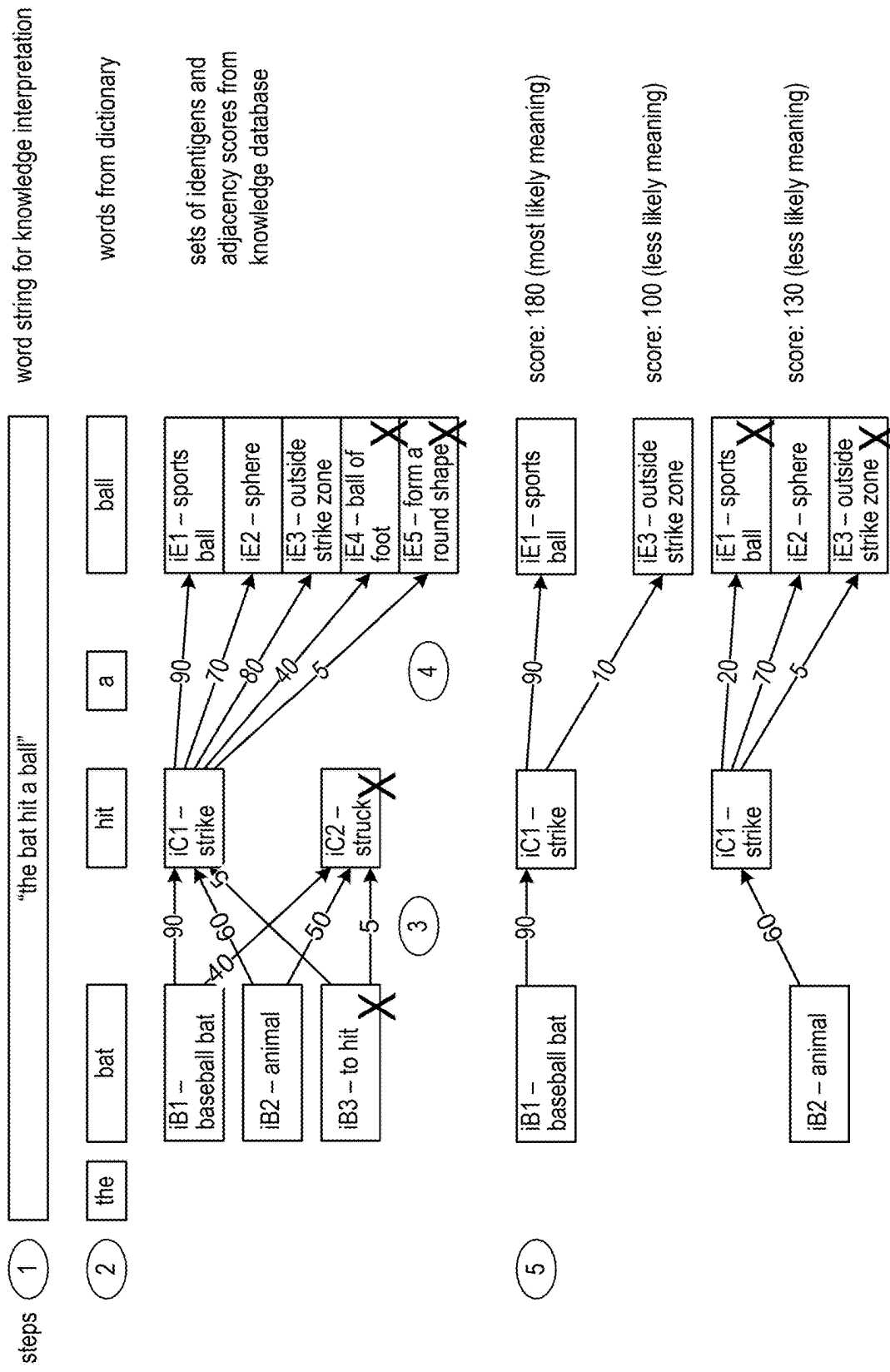


FIG. 12B

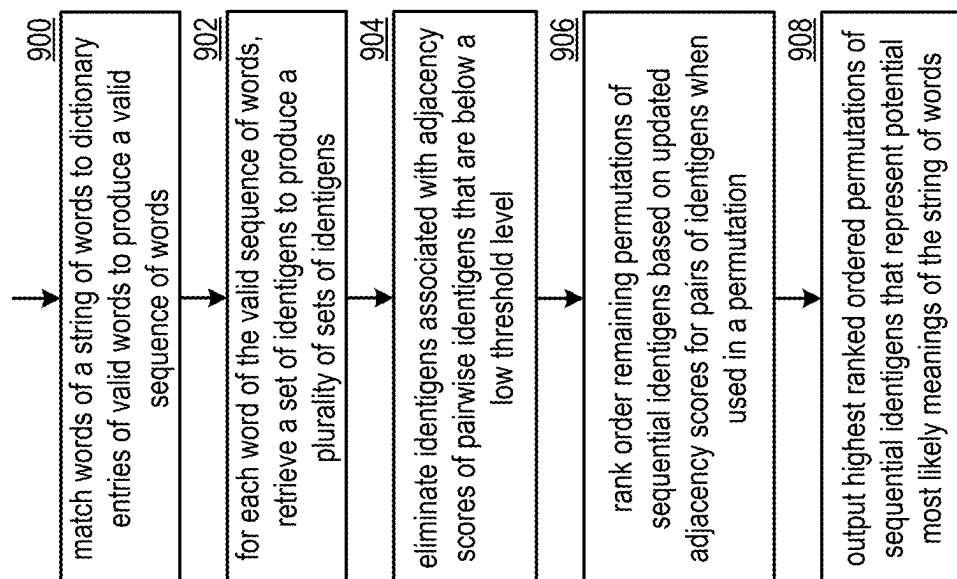


FIG. 12C

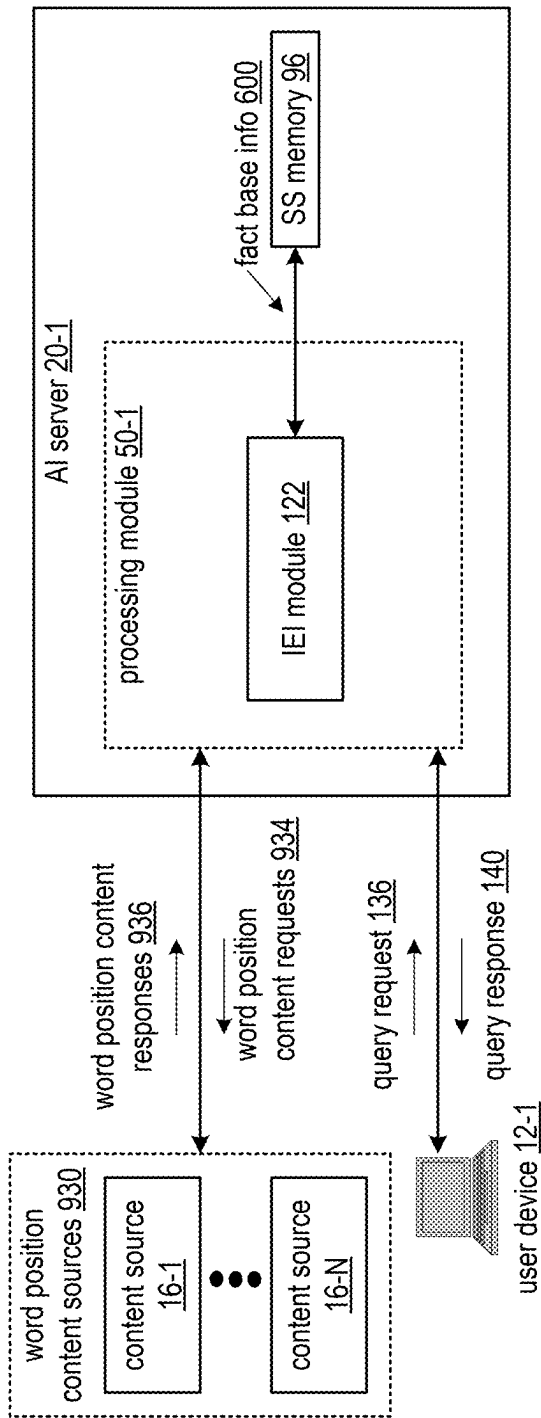


FIG. 13A

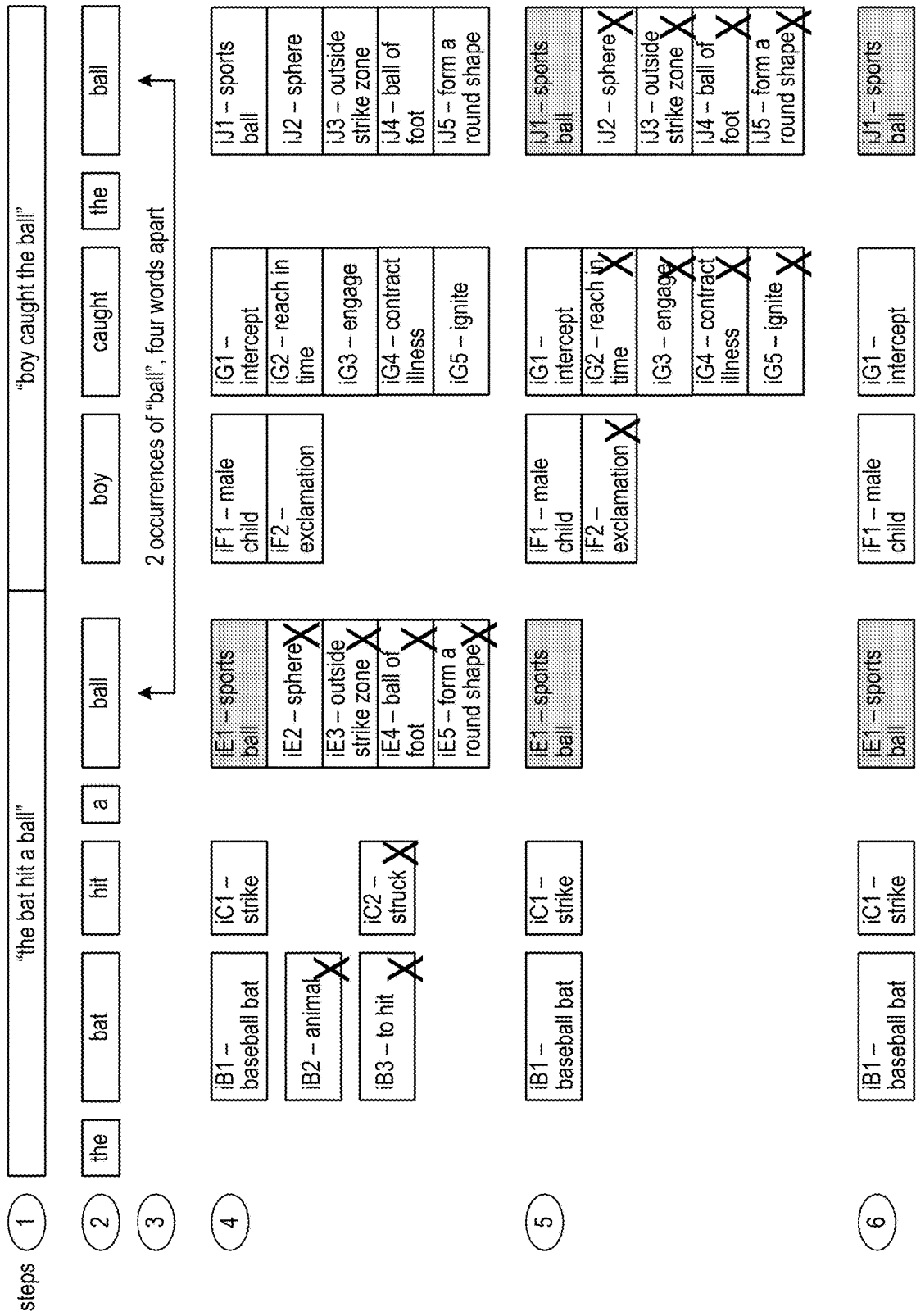


FIG. 13B

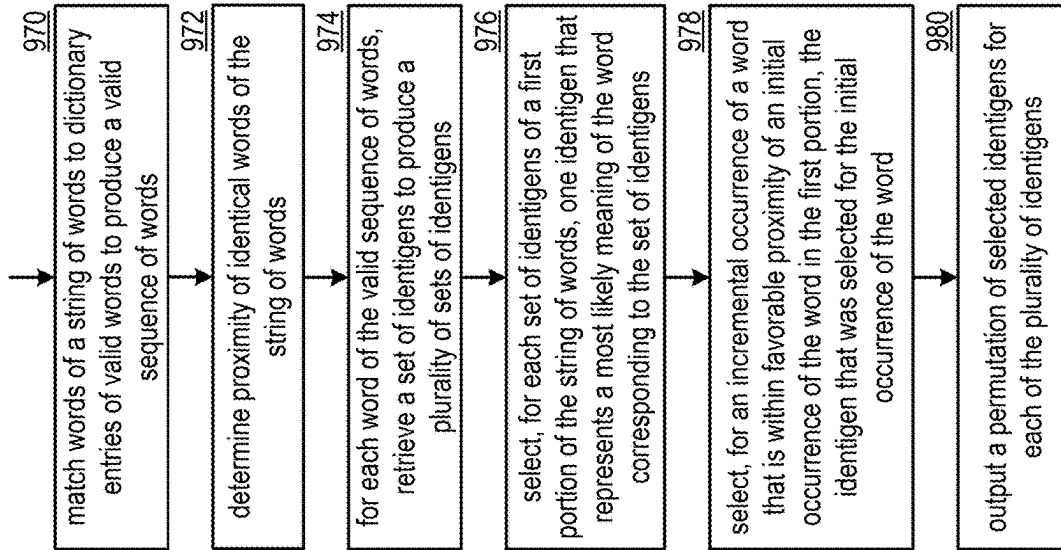


FIG. 13C

INTERPRETING MEANING OF CONTENT**CROSS REFERENCE TO RELATED PATENTS**

[0001] The present U.S. Utility Patent Application claims priority pursuant to 35 U.S.C. § 120 as a continuation of U.S. Utility application Ser. No. 18/403,063, entitled “INTERPRETING MEANING OF CONTENT,” filed Jan. 3, 2024, issuing as U.S. Pat. No. 12,314,662 on May 27, 2025, which claims priority pursuant to 35 U.S.C. § 120 as a continuation of U.S. Utility application Ser. No. 18/103,772, entitled “INTERPRETING MEANING OF CONTENT,” filed Jan. 31, 2023, issued on Jan. 23, 2024 as U.S. Pat. No. 11,880,656, which claims priority pursuant 35 U.S.C. § 120 as a continuation of U.S. Utility application Ser. No. 16/855,445, entitled “INTERPRETING MEANING OF CONTENT,” filed Apr. 22, 2020, issued on Feb. 21, 2023 as U.S. Pat. No. 11,586,819, which claims priority pursuant to 35 U.S.C. § 119(e) to U.S. Provisional Application No. 68/839,892, entitled “INTERPRETING MEANING OF CONTENT,” filed Apr. 29, 2019, all of which are hereby incorporated herein by reference in their entirety and made part of the present U.S. Utility Patent Application for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable.

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC

[0003] Not Applicable.

BACKGROUND OF THE INVENTION**Technical Field of the Invention**

[0004] This invention relates generally to computing systems and more particularly to generating data representations of data and analyzing the data utilizing the data representations.

Description of Related Art

[0005] It is known that data is stored in information systems, such as files containing text. It is often difficult to produce useful information from this stored data due to many factors. The factors include the volume of available data, accuracy of the data, and variances in how text is interpreted to express knowledge. For example, many languages and regional dialects utilize the same or similar words to represent different concepts.

[0006] Computers are known to utilize pattern recognition techniques and apply statistical reasoning to process text to express an interpretation in an attempt to overcome ambiguities inherent in words. One pattern recognition technique includes matching a word pattern of a query to a word pattern of the stored data to find an explicit textual answer. Another pattern recognition technique classifies words into major grammatical types such as functional words, nouns, adjectives, verbs and adverbs. Grammar based techniques then utilize these grammatical types to study how words should be distributed within a string of words to form a properly constructed grammatical sentence where each word

is forced to support a grammatical operation without necessarily identifying what the word is actually trying to describe.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0007] FIG. 1 is a schematic block diagram of an embodiment of a computing system in accordance with the present invention;

[0008] FIG. 2 is a schematic block diagram of an embodiment of various servers of a computing system in accordance with the present invention;

[0009] FIG. 3 is a schematic block diagram of an embodiment of various devices of a computing system in accordance with the present invention;

[0010] FIGS. 4A and 4B are schematic block diagrams of another embodiment of a computing system in accordance with the present invention;

[0011] FIG. 4C is a logic diagram of an embodiment of a method for interpreting content to produce a response to a query within a computing system in accordance with the present invention;

[0012] FIG. 5A is a schematic block diagram of an embodiment of a collections module of a computing system in accordance with the present invention;

[0013] FIG. 5B is a logic diagram of an embodiment of a method for obtaining content within a computing system in accordance with the present invention;

[0014] FIG. 5C is a schematic block diagram of an embodiment of a query module of a computing system in accordance with the present invention;

[0015] FIG. 5D is a logic diagram of an embodiment of a method for providing a response to a query within a computing system in accordance with the present invention;

[0016] FIG. 5E is a schematic block diagram of an embodiment of an identigen entigen intelligence (IEI) module of a computing system in accordance with the present invention;

[0017] FIG. 5F is a logic diagram of an embodiment of a method for analyzing content within a computing system in accordance with the present invention;

[0018] FIG. 6A is a schematic block diagram of an embodiment of an element identification module and an interpretation module of a computing system in accordance with the present invention;

[0019] FIG. 6B is a logic diagram of an embodiment of a method for interpreting information within a computing system in accordance with the present invention;

[0020] FIG. 6C is a schematic block diagram of an embodiment of an answer resolution module of a computing system in accordance with the present invention;

[0021] FIG. 6D is a logic diagram of an embodiment of a method for producing an answer within a computing system in accordance with the present invention;

[0022] FIG. 7A is an information flow diagram for interpreting information within a computing system in accordance with the present invention;

[0023] FIG. 7B is a relationship block diagram illustrating an embodiment of relationships between things and representations of things within a computing system in accordance with the present invention;

[0024] FIG. 7C is a diagram of an embodiment of a synonym words table within a computing system in accordance with the present invention;

[0025] FIG. 7D is a diagram of an embodiment of a polysemous words table within a computing system in accordance with the present invention;

[0026] FIG. 7E is a diagram of an embodiment of transforming words into groupings within a computing system in accordance with the present invention;

[0027] FIG. 8A is a data flow diagram for accumulating knowledge within a computing system in accordance with the present invention;

[0028] FIG. 8B is a diagram of an embodiment of a groupings table within a computing system in accordance with the present invention;

[0029] FIG. 8C is a data flow diagram for answering questions utilizing accumulated knowledge within a computing system in accordance with the present invention;

[0030] FIG. 8D is a data flow diagram for answering questions utilizing interference within a computing system in accordance with the present invention;

[0031] FIG. 8E is a relationship block diagram illustrating another embodiment of relationships between things and representations of things within a computing system in accordance with the present invention;

[0032] FIGS. 8F and 8G are schematic block diagrams of another embodiment of a computing system in accordance with the present invention;

[0033] FIG. 8H is a logic diagram of an embodiment of a method for processing content to produce knowledge within a computing system in accordance with the present invention;

[0034] FIGS. 8J and 8K are schematic block diagrams another embodiment of a computing system in accordance with the present invention;

[0035] FIG. 8L is a logic diagram of an embodiment of a method for generating a query response to a query within a computing system in accordance with the present invention;

[0036] FIGS. 9A, 9B, 9C, and 9D are schematic block diagrams of another embodiment of a computing system illustrating a method for interpreting meaning of content within the computing system in accordance with the present invention;

[0037] FIG. 10A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;

[0038] FIG. 10B is a data flow diagram illustrating an example of a method for interpreting meaning within a computing system in accordance with the present invention;

[0039] FIG. 10C is a logic diagram of an embodiment of a method for interpreting meaning within a computing system in accordance with the present invention;

[0040] FIG. 11A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;

[0041] FIG. 11B is a data flow diagram illustrating an example of a method for interpreting meaning within a computing system in accordance with the present invention;

[0042] FIG. 11C is a logic diagram of an embodiment of a method for interpreting meaning within a computing system in accordance with the present invention;

[0043] FIG. 12A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;

[0044] FIG. 12B is a data flow diagram illustrating an example of a method for interpreting meaning within a computing system in accordance with the present invention;

[0045] FIG. 12C is a logic diagram of an embodiment of a method for interpreting meaning within a computing system in accordance with the present invention;

[0046] FIG. 13A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;

[0047] FIG. 13B is a data flow diagram illustrating an example of a method for interpreting meaning within a computing system in accordance with the present invention; and

[0048] FIG. 13C is a logic diagram of an embodiment of a method for interpreting meaning within a computing system in accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0049] FIG. 1 is a schematic block diagram of an embodiment of a computing system 10 that includes a plurality of user devices 12-1 through 12-N, a plurality of wireless user devices 14-1 through 14-N, a plurality of content sources 16-1 through 16-N, a plurality of transactional servers 18-1 through 18-N, a plurality of artificial intelligence (AI) servers 20-1 through 20-N, and a core network 24. The core network 24 includes at least one of the Internet, a public radio access network (RAN), and any private network. Hereafter, the computing system 10 may be interchangeably referred to as a data network, a data communication network, a system, a communication system, and a data communication system. Hereafter, the user device and the wireless user device may be interchangeably referred to as user devices, and each of the transactional servers and the AI servers may be interchangeably referred to as servers.

[0050] Each user device, wireless user device, transactional server, and AI server includes a computing device that includes a computing core. In general, a computing device is any electronic device that can communicate data, process data, and/or store data. A further generality of a computing device is that it includes one or more of a central processing unit (CPU), a memory system, a sensor (e.g., internal or external), user input/output interfaces, peripheral device interfaces, communication elements, and an interconnecting bus structure.

[0051] As further specific examples, each of the computing devices may be a portable computing device and/or a fixed computing device. A portable computing device may be an embedded controller, a smart sensor, a smart pill, a social networking device, a gaming device, a cell phone, a smart phone, a robot, a personal digital assistant, a digital music player, a digital video player, a laptop computer, a handheld computer, a tablet, a video game controller, an engine controller, a vehicular controller, an aircraft controller, a maritime vessel controller, and/or any other portable device that includes a computing core. A fixed computing device may be security camera, a sensor device, a household appliance, a machine, a robot, an embedded controller, a personal computer (PC), a computer server, a cable set-top box, a satellite receiver, a television set, a printer, a fax machine, home entertainment equipment, a camera controller, a video game console, a critical infrastructure controller, and/or any type of home or office computing equipment that includes a computing core. An embodiment of the various servers is discussed in greater detail with reference to FIG. 2. An embodiment of the various devices is discussed in greater detail with reference to FIG. 3.

[0052] Each of the content sources **16-1** through **16-N** includes any source of content, where the content includes one or more of data files, a data stream, a tech stream, a text file, an audio stream, an audio file, a video stream, a video file, etc. Examples of the content sources include a weather service, a multi-language online dictionary, a fact server, a big data storage system, the Internet, social media systems, an email server, a news server, a schedule server, a traffic monitor, a security camera system, audio monitoring equipment, an information server, a service provider, a data aggregator, and airline traffic server, a shipping and logistics server, a banking server, a financial transaction server, etc. Alternatively, or in addition to, one or more of the various user devices may provide content. For example, a wireless user device may provide content (e.g., issued as a content message) when the wireless user device is able to capture data (e.g., text input, sensor input, etc.).

[0053] Generally, an embodiment of this invention presents solutions where the computing system **10** supports the generation and utilization of knowledge extracted from content. For example, the AI servers **20-1** through **20-N** ingest content from the content sources **16-1** through **16-N** by receiving, via the core network **24** content messages **28-1** through **28-N** as AI messages **32-1** through **32-N**, extract the knowledge from the ingested content, and interact with the various user devices to utilize the extracted knowledge by facilitating the issuing, via the core network **24**, user messages **22-1** through **22-N** to the user devices **12-1** through **12-N** and wireless signals **26-1** through **26-N** to the wireless user devices **14-1** through **14-N**.

[0054] Each content message **28-1** through **28-N** includes a content request (e.g., requesting content related to a topic, content type, content timing, one or more domains, etc.) or a content response, where the content response includes real-time or static content such as one or more of dictionary information, facts, non-facts, weather information, sensor data, news information, blog information, social media content, user daily activity schedules, traffic conditions, community event schedules, school schedules, user schedules airline records, shipping records, logistics records, banking records, census information, global financial history information, etc. Each AI message **32-1** through **32-N** includes one or more of content messages, user messages (e.g., a query request, a query response that includes an answer to a query request), and transaction messages (e.g., transaction information, requests and responses related to transactions). Each user message **22-1** through **22-N** includes one or more of a query request, a query response, a trigger request, a trigger response, a content collection, control information, software information, configuration information, security information, routing information, addressing information, presence information, analytics information, protocol information, all types of media, sensor data, statistical data, user data, error messages, etc.

[0055] When utilizing a wireless signal capability of the core network **24**, each of the wireless user devices **14-1** through **14-N** encodes/decodes data and/or information messages (e.g., user messages such as user messages **22-1** through **22-N**) in accordance with one or more wireless standards for local wireless data signals (e.g., Wi-Fi, Bluetooth, ZigBee) and/or for wide area wireless data signals (e.g., 2G, 3G, 4G, 5G, satellite, point-to-point, etc.) to produce wireless signals **26-1** through **26-N**. Having encoded/decoded the data and/or information messages, the

wireless user devices **14-1** through **14-N** and/receive the wireless signals to/from the wireless capability of the core network **24**.

[0056] As another example of the generation and utilization of knowledge, the transactional servers **18-1** through **18-N** communicate, via the core network **24**, transaction messages **30-1** through **30-N** as further AI messages **32-1** through **32-N** to facilitate ingesting of transactional type content (e.g., real-time crypto currency transaction information) and to facilitate handling of utilization of the knowledge by one or more of the transactional servers (e.g., for a transactional function) in addition to the utilization of the knowledge by the various user devices. Each transaction message **30-1** through **30-N** includes one or more of a query request, a query response, a trigger request, a trigger response, a content message, and transactional information, where the transactional information may include one or more of consumer purchasing history, crypto currency ledgers, stock market trade information, other investment transaction information, etc.

[0057] In another specific example of operation of the generation and utilization of knowledge extracted from the content, the user device **12-1** issues a user message **22-1** to the AI server **20-1**, where the user message **22-1** includes a query request and where the query request includes a question related to a first domain of knowledge. The issuing includes generating the user message **22-1** based on the query request (e.g., the question), selecting the AI server **20-1** based on the first domain of knowledge, and sending, via the core network **24**, the user message **22-1** as a further AI message **32-1** to the AI server **20-1**. Having received the AI message **32-1**, the AI server **20-1** analyzes the question within the first domain, generates further knowledge, generates a preliminary answer, generates a quality level indicator of the preliminary answer, and determines to gather further content when the quality level indicator is below a minimum quality threshold level.

[0058] When gathering the further content, the AI server **20-1** issues, via the core network **24**, a still further AI message **32-1** as a further content message **28-1** to the content source **16-1**, where the content message **28-1** includes a content request for more content associated with the first domain of knowledge and in particular the question. Alternatively, or in addition to, the AI server **20-1** issues the content request to another AI server to facilitate a response within a domain associated with the other AI server. Further alternatively, or in addition to, the AI server **20-1** issues the content request to one or more of the various user devices to facilitate a response from a subject matter expert.

[0059] Having received the content message **28-1**, the contents or **16-1** issues, via the core network **24**, a still further content message **28-1** to the AI server **20-1** as a yet further AI message **32-1**, where the still further content message **28-1** includes requested content. The AI server **20-1** processes the received content to generate further knowledge. Having generated the further knowledge, the AI server **20-1** re-analyzes the question, generates still further knowledge, generates another preliminary answer, generates another quality level indicator of the other preliminary answer, and determines to issue a query response to the user device **12-1** when the quality level indicator is above the minimum quality threshold level. When issuing the query response, the AI server **20-1** generates an AI message **32-1** that includes another user message **22-1**, where the other

user message 22-1 includes the other preliminary answer as a query response including the answer to the question. Having generated the AI message 32-1, the AI server 20-1 sends, via the core network 24, the AI message 32-1 as the user message 22-1 to the user device 12-1 thus providing the answer to the original question of the query request.

[0060] FIG. 2 is a schematic block diagram of an embodiment of the AI servers 20-1 through 20-N and the transactional servers 18-1 through 18-N of the computing system 10 of FIG. 1. The servers include a computing core 52, one or more visual output devices 74 (e.g., video graphics display, touchscreen, LED, etc.), one or more user input devices 76 (e.g., keypad, keyboard, touchscreen, voice to text, a push button, a microphone, a card reader, a door position switch, a biometric input device, etc.), one or more audio output devices 78 (e.g., speaker(s), headphone jack, a motor, etc.), and one or more visual input devices 80 (e.g., a still image camera, a video camera, photocell, etc.).

[0061] The servers further include one or more universal serial bus (USB) devices (USB devices 1-U), one or more peripheral devices (e.g., peripheral devices 1-P), one or more memory devices (e.g., one or more flash memory devices 92, one or more hard drive (HD) memories 94, and one or more solid state (SS) memory devices 96, and/or cloud memory 98). The servers further include one or more wireless location modems 84 (e.g., global positioning satellite (GPS), Wi-Fi, angle of arrival, time difference of arrival, signal strength, dedicated wireless location, etc.), one or more wireless communication modems 86-1 through 86-N (e.g., a cellular network transceiver, a wireless data network transceiver, a Wi-Fi transceiver, a Bluetooth transceiver, a 315 MHz transceiver, a zig bee transceiver, a 60 GHz transceiver, etc.), a telco interface 102 (e.g., to interface to a public switched telephone network), and a wired local area network (LAN) 88 (e.g., optical, electrical), and a wired wide area network (WAN) 90 (e.g., optical, electrical).

[0062] The computing core 52 includes a video graphics module 54, one or more processing modules 50-1 through 50-N (e.g., which may include one or more secure co-processors), a memory controller 56 and one or more main memories 58-1 through 58-N (e.g., RAM serving as local memory). The computing core 52 further includes one or more input/output (I/O) device interfaces 62, an input/output (I/O) controller 60, a peripheral interface 64, one or more USB interfaces 66, one or more network interfaces 72, one or more memory interfaces 70, and/or one or more peripheral device interfaces 68.

[0063] The processing modules may be a single processing device or a plurality of processing devices where the processing device may further be referred to as one or more of a “processing circuit”, a “processor”, and/or a “processing unit”. Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions.

[0064] The processing module, module, processing circuit, and/or processing unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, and/or processing unit. Such a

memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network).

[0065] Further note that if the processing module, module, processing circuit, and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture.

[0066] Each of the interfaces 62, 66, 68, 70, and 72 includes a combination of hardware (e.g., connectors, wiring, etc.) and may further include operational instructions stored on memory (e.g., driver software) that are executed by one or more of the processing modules 50-1 through 50-N and/or a processing circuit within the interface. Each of the interfaces couples to one or more components of the servers. For example, one of the IO device interfaces 62 couples to an audio output device 78. As another example, one of the memory interfaces 70 couples to flash memory 92 and another one of the memory interfaces 70 couples to cloud memory 98 (e.g., an on-line storage system and/or on-line backup system). In other embodiments, the servers may include more or less devices and modules than shown in this example embodiment of the servers.

[0067] FIG. 3 is a schematic block diagram of an embodiment of the various devices of the computing system 10 of FIG. 1, including the user devices 12-1 through 12-N and the wireless user devices 14-1 through 14-N. The various devices include the visual output device 74 of FIG. 2, the user input device 76 of FIG. 2, the audio output device 78 of FIG. 2, the visual input device 80 of FIG. 2, and one or more sensors 82.

[0068] The sensor may be implemented internally and/or externally to the device. Example sensors includes a still camera, a video camera, servo motors associated with a camera, a position detector, a smoke detector, a gas detector, a motion sensor, an accelerometer, velocity detector, a compass, a gyro, a temperature sensor, a pressure sensor, an altitude sensor, a humidity detector, a moisture detector, an imaging sensor, and a biometric sensor. Further examples of the sensor include an infrared sensor, an audio sensor, an ultrasonic sensor, a proximity detector, a magnetic field detector, a biomaterial detector, a radiation detector, a weight detector, a density detector, a chemical analysis detector, a fluid flow volume sensor, a DNA reader, a wind speed sensor, a wind direction sensor, and an object detection sensor.

[0069] Further examples of the sensor include an object identifier sensor, a motion recognition detector, a battery level detector, a room temperature sensor, a sound detector, a smoke detector, an intrusion detector, a motion detector, a door position sensor, a window position sensor, and a sunlight detector. Still further sensor examples include medical category sensors including: a pulse rate monitor, a heart rhythm monitor, a breathing detector, a blood pressure monitor, a blood glucose level detector, blood type, an electrocardiogram sensor, a body mass detector, an imaging sensor, a microphone, body temperature, etc.

[0070] The various devices further include the computing core 52 of FIG. 2, the one or more universal serial bus (USB) devices (USB devices 1-U) of FIG. 2, the one or more peripheral devices (e.g., peripheral devices 1-P) of FIG. 2, and the one or more memories of FIG. 2 (e.g., flash memories 92, HD memories 94, SS memories 96, and/or cloud memories 98). The various devices further include the one or more wireless location modems 84 of FIG. 2, the one or more wireless communication modems 86-1 through 86-N of FIG. 2, the telco interface 102 of FIG. 2, the wired local area network (LAN) 88 of FIG. 2, and the wired wide area network (WAN) 90 of FIG. 2. In other embodiments, the various devices may include more or less internal devices and modules than shown in this example embodiment of the various devices.

[0071] FIGS. 4A and 4B are schematic block diagrams of another embodiment of a computing system that includes one or more of the user device 12-1 of FIG. 1, the wireless user device 14-1 of FIG. 1, the content source 16-1 of FIG. 1, the transactional server 18-1 of FIG. 1, the user device 12-2 of FIG. 1, and the AI server 20-1 of FIG. 1. The AI server 20-1 includes the processing module 50-1 (e.g., associated with the servers) of FIG. 2, where the processing module 50-1 includes a collections module 120, an intelligent entity intelligence (IEI) module 122, and a query module 124. Alternatively, the collections module 120, the IEI module 122, and the query module 124 may be implemented by the processing module 50-1 (e.g., associated with the various user devices) of FIG. 3. The computing system functions to interpret content to produce a response to a query.

[0072] FIG. 4A illustrates an example of the interpreting of the content to produce the response to the query where the collections module 120 interprets (e.g., based on an interpretation approach such as rules) at least one of a collections request 132 from the query module 124 and a collections request within collections information 130 from the IEI module 122 to produce content request information (e.g., potential sources, content descriptors of desired content). Alternatively, or in addition to, the collections module 120 may facilitate gathering further content based on a plurality of collection requests from a plurality of devices of the computing system 10 of FIG. 1.

[0073] The collections request 132 is utilized to facilitate collection of content, where the content may be received in a real-time fashion once or at desired intervals, or in a static fashion from previous discrete time frames. For instance, the query module 124 issues the collections request 132 to facilitate collection of content as a background activity to support a long-term query (e.g., how many domestic airline flights over the next seven days include travelers between the age of 18 and 35 years old). The collections request 132 may include one or more of a requester identifier (ID), a

content type (e.g., language, dialect, media type, topic, etc.), a content source indicator, security credentials (e.g., an authorization level, a password, a user ID, parameters utilized for encryption, etc.), a desired content quality level, trigger information (e.g., parameters under which to collect content based on a pre-event, an event (i.e., content quality level reaches a threshold to cause the trigger, trueness), or a timeframe), a desired format, and a desired timing associated with the content.

[0074] Having interpreted the collections request 132, the collections module 120 selects a source of content based on the content request information. The selecting includes one or more of identifying one or more potential sources based on the content request information, selecting the source of content from the potential sources utilizing a selection approach (e.g., favorable history, a favorable security level, favorable accessibility, favorable cost, favorable performance, etc.). For example, the collections module 120 selects the content source 16-1 when the content source 16-1 is known to provide a favorable content quality level for a domain associated with the collections request 132.

[0075] Having selected the source of content, the collections module 120 issues a content request 126 to the selected source of content. The issuing includes generating the content request 126 based on the content request information for the selected source of content and sending the content request 126 to the selected source of content. The content request 126 may include one or more of a content type indicator, a requester ID, security credentials for content access, and any other information associated with the collections request 132. For example, the collections module 120 sends the content request 126, via the core network 24 of FIG. 1, to the content source 16-1. Alternatively, or in addition to, the collections module 120 may send a similar content request 126 to one or more of the user device 12-1, the wireless user device 14-1, and the transactional server 18-1 to facilitate collecting of further content.

[0076] In response to the content request 126, the collections module 120 receives one or more content responses 128. The content response 128 includes one or more of content associated with the content source, a content source identifier, security credential processing information, and any other information pertaining to the desired content. Having received the content response 128, the collections module 120 interprets the received content response 128 to produce collections information 130, where the collections information 130 further includes a collections response from the collections module 120 to the IEI module 122.

[0077] The collections response includes one or more of transformed content (e.g., completed sentences and paragraphs), timing information associated with the content, a content source ID, and a content quality level. Having generated the collections response of the collections information 130, the collections module 120 sends the collections information 130 to the IEI module 122. Having received the collections information 130 from the collections module 120, the IEI module 122 interprets the further content of the content response to generate further knowledge, where the further knowledge is stored in a memory associated with the IEI module 122 to facilitate subsequent answering of questions posed in received queries.

[0078] FIG. 4B further illustrates the example of the interpreting of the content to produce the response to the query where, the query module 124 interprets a received

query request **136** from a requester to produce an interpretation of the query request. For example, the query module **124** receives the query request **136** from the user device **12-2**, and/or from one or more of the wireless user device **14-2** and the transactional server **18-2**. The query request **136** includes one or more of an identifier (ID) associated with the request (e.g., requester ID, ID of an entity to send a response to), a question, question constraints (e.g., within a timeframe, within a geographic area, within a domain of knowledge, etc.), and content associated with the question (e.g., which may be analyzed for new knowledge itself).

[0079] The interpreting of the query request **136** includes determining whether to issue a request to the IEI module **122** (e.g., a question, perhaps with content) and/or to issue a request to the collections module **120** (e.g., for further background content). For example, the query module **124** produces the interpretation of the query request to indicate to send the request directly to the IEI module **122** when the question is associated with a simple non-time varying function answer (e.g., question: “how many hydrogen atoms does a molecule of water have?”).

[0080] Having interpreted the query request **136**, the query module **124** issues at least one of an IEI request as query information **138** to the IEI module **122** (e.g., when receiving a simple new query request) and a collections request **132** to the collections module **120** (e.g., based on two or more query requests **136** requiring more substantive content gathering). The IEI request of the query information **138** includes one or more of an identifier (ID) of the query module **124**, an ID of the requester (e.g., the user device **12-2**), a question (e.g., with regards to content for analysis, with regards to knowledge minded by the AI server from general content), one or more constraints (e.g., assumptions, restrictions, etc.) associated with the question, content for analysis of the question, and timing information (e.g., a date range for relevance of the question).

[0081] Having received the query information **138** that includes the IEI request from the query module **124**, the IEI module **122** determines whether a satisfactory response can be generated based on currently available knowledge, including that of the query request **136**. The determining includes indicating that the satisfactory response cannot be generated when an estimated quality level of an answer falls below a minimum quality threshold level. When the satisfactory response cannot be generated, the IEI module **122** facilitates collecting more content. The facilitating includes issuing a collections request to the collections module **120** of the AI server **20-1** and/or to another server or user device, and interpreting a subsequent collections response **134** of collections information **130** that includes further content to produce further knowledge to enable a more favorable answer.

[0082] When the IEI module **122** indicates that the satisfactory response can be generated, the IEI module **122** issues an IEI response as query information **138** to the query module **124**. The IEI response includes one or more of one or more answers, timing relevance of the one or more answers, an estimated quality level of each answer, and one or more assumptions associated with the answer. The issuing includes generating the IEI response based on the collections response **134** of the collections information **130** and the IEI request, and sending the IEI response as the query information **138** to the query module **124**. Alternatively, or in addition to, at least some of the further content collected by

the collections module **120** is utilized to generate a collections response **134** issued by the collections module **120** to the query module **124**. The collections response **134** includes one or more of further content, a content availability indicator (e.g., when, where, required credentials, etc.), a content freshness indicator (e.g., timestamps, predicted time availability), content source identifiers, and a content quality level.

[0083] Having received the query information **138** from the IEI module **122**, the query module **124** issues a query response **140** to the requester based on the IEI response and/or the collections response **134** directly from the collections module **120**, where the collection module **120** generates the collections response **134** based on collected content and the collections request **132**. The query response **140** includes one or more of an answer, answer timing, an answer quality level, and answer assumptions.

[0084] FIG. 4C is a logic diagram of an embodiment of a method for interpreting content to produce a response to a query within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4B, and also FIG. 4C. The method includes step **150** where a collections module of a processing module of one or more computing devices (e.g., of one or more servers) interprets a collections request to produce content request information. The interpreting may include one or more of identifying a desired content source, identifying a content type, identifying a content domain, and identifying content timing requirements.

[0085] The method continues at step **152** where the collections module selects a source of content based on the content request information. For example, the collections module identifies one or more potential sources based on the content request information and selects the source of content from the potential sources utilizing a selection approach (e.g., based on one or more of favorable history, a favorable security level, favorable accessibility, favorable cost, favorable performance, etc.). The method continues at step **154** where the collections module issues a content request to the selected source of content. The issuing includes generating a content request based on the content request information for the selected source of content and sending the content request to the selected source of content.

[0086] The method continues at step **156** where the collections module issues collections information to an intelligent entity intelligence (IEI) module based on a received content response, where the IEI module extracts further knowledge from newly obtained content from the one or more received content responses. For example, the collections module generates the collections information based on newly obtained content from the one or more received content responses of the selected source of content.

[0087] The method continues at step **158** where a query module interprets a received query request from a requester to produce an interpretation of the query request. The interpreting may include determining whether to issue a request to the IEI module (e.g., a question) or to issue a request to the collections module to gather further background content. The method continues at step **160** where the query module issues a further collections request. For example, when receiving a new query request, the query module generates a request for the IEI module. As another example, when receiving a plurality of query requests for

similar questions, the query module generates a request for the collections module to gather further background content.

[0088] The method continues at step 162 where the IEI module determines whether a satisfactory query response can be generated when receiving the request from the query module. For example, the IEI module indicates that the satisfactory query response cannot be generated when an estimated quality level of an answer is below a minimum answer quality threshold level. The method branches to step 166 when the IEI module determines that the satisfactory query response can be generated. The method continues to step 164 when the IEI module determines that the satisfactory query response cannot be generated. When the satisfactory query response cannot be generated, the method continues at step 164 where the IEI module facilitates collecting more content. The method loops back to step 150.

[0089] When the satisfactory query response can be generated, the method continues at step 166 where the IEI module issues an IEI response to the query module. The issuing includes generating the IEI response based on the collections response and the IEI request, and sending the IEI response to the query module. The method continues at step 168 where the query module issues a query response to the requester. For example, the query module generates the query response based on the IEI response and/or a collections response from the collections module and sends the query response to the requester, where the collections module generates the collections response based on collected content and the collections request.

[0090] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

[0091] FIG. 5A is a schematic block diagram of an embodiment of the collections module 120 of FIG. 4A that includes a content acquisition module 180, a content selection module 182, a source selection module 184, a content security module 186, an acquisition timing module 188, a content transformation module 190, and a content quality module 192. Generally, an embodiment of this invention presents solutions where the collections module 120 supports collecting content.

[0092] In an example of operation of the collecting of the content, the content acquisition module 180 receives a collections request 132 from a requester. The content acquisition module 180 obtains content selection information 194 based on the collections request 132. The content selection information 194 includes one or more of content requirements, a desired content type indicator, a desired content source identifier, a content type indicator, a candidate source identifier (ID), and a content profile (e.g., a template of typical parameters of the content). For example, the content acquisition module 180 receives the content selection infor-

mation 194 from the content selection module 182, where the content selection module 182 generates the content selection information 194 based on a content selection information request from the content acquisition module 180 and where the content acquisition module 180 generates the content selection information request based on the collections request 132.

[0093] The content acquisition module 180 obtains source selection information 196 based on the collections request 132. The source selection information 196 includes one or more of candidate source identifiers, a content profile, selected sources, source priority levels, and recommended source access timing. For example, the content acquisition module 180 receives the source selection information 196 from the source selection module 184, where the source selection module 184 generates the source selection information 196 based on a source selection information request from the content acquisition module 180 and where the content acquisition module 180 generates the source selection information request based on the collections request 132.

[0094] The content acquisition module 180 obtains acquisition timing information 200 based on the collections request 132. The acquisition timing information 200 includes one or more of recommended source access timing, confirmed source access timing, source access testing results, estimated velocity of content update's, content precious, timestamps, predicted time availability, required content acquisition triggers, content acquisition trigger detection indicators, and a duplicative indicator with a pending content request. For example, the content acquisition module 180 receives the acquisition timing information 200 from the acquisition timing module 188, where the acquisition timing module 188 generates the acquisition timing information 200 based on an acquisition timing information request from the content acquisition module 180 and where the content acquisition module 180 generates the acquisition timing information request based on the collections request 132.

[0095] Having obtained the content selection information 194, the source selection information 196, and the acquisition timing information 200, the content acquisition module 180 issues a content request 126 to a content source utilizing security information 198 from the content security module 186, where the content acquisition module 180 generates the content request 126 in accordance with the content selection information 194, the source selection information 196, and the acquisition timing information 200. The security information 198 includes one or more of source priority requirements, requester security information, available security procedures, and security credentials for trust and/or encryption. For example, the content acquisition module 180 generates the content request 126 to request a particular content type in accordance with the content selection information 194 and to include security parameters of the security information 198, initiates sending of the content request 126 in accordance with the acquisition timing information 200, and sends the content request 126 to a particular targeted content source in accordance with the source selection information 196.

[0096] In response to receiving a content response 128, the content acquisition module 180 determines the quality level of received content extracted from the content response 128. For example, the content acquisition module 180 receives

content quality information **204** from the content quality module **192**, where the content quality module **192** generates the quality level of the received content based on receiving a content quality request from the content acquisition module **180** and where the content acquisition module **180** generates the content quality request based on content extracted from the content response **128**. The content quality information includes one or more of a content reliability threshold range, a content accuracy threshold range, a desired content quality level, a predicted content quality level, and a predicted level of trust.

[0097] When the quality level is below a minimum desired quality threshold level, the content acquisition module **180** facilitates acquisition of further content. The facilitating includes issuing another content request **126** to a same content source and/or to another content source to receive and interpret further received content. When the quality level is above the minimum desired quality threshold level, the content acquisition module **180** issues a collections response **134** to the requester. The issuing includes processing the content in accordance with a transformation approach to produce transformed content, generating the collections response **134** to include the transformed content, and sending the collections response **134** to the requester. The processing of the content to produce the transformed content includes receiving content transformation information **202** from the content transformation module **190**, where the content transformation module **190** transforms the content in accordance with the transformation approach to produce the transformed content. The content transformation information includes a desired format, available formats, recommended formatting, the received content, transformation instructions, and the transformed content.

[0098] FIG. 5B is a logic diagram of an embodiment of a method for obtaining content within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4C, 5A, and also FIG. 5B. The method includes step **210** where a processing module of one or more processing modules of one or more computing devices of the computing system receives a collections request from the requester. The method continues at step **212** where the processing module determines content selection information. The determining includes interpreting the collections request to identify requirements of the content.

[0099] The method continues at step **214** where the processing module determines source selection information. The determining includes interpreting the collections request to identify and select one or more sources for the content to be collected. The method continues at step **216** where the processing module determines acquisition timing information. The determining includes interpreting the collections request to identify timing requirements for the acquisition of the content from the one or more sources. The method continues at step **218** where the processing module issues a content request utilizing security information and in accordance with one or more of the content selection information, the source selection information, and the acquisition timing information. For example, the processing module issues the content request to the one or more sources for the content in accordance with the content requirements, where the sending of the request is in accordance with the acquisition timing information.

[0100] The method continues at step **220** where the processing module determines a content quality level for received content area. The determining includes receiving the content from the one or more sources, obtaining content quality information for the received content based on a quality analysis of the received content. The method branches to step **224** when the content quality level is favorable and the method continues to step **222** when the quality level is unfavorable. For example, the processing module determines that the content quality level is favorable when the content quality level is equal to or above a minimum quality threshold level and determines that the content quality level is unfavorable when the content quality level is less than the minimum quality threshold level.

[0101] When the content quality level is unfavorable, the method continues at step **222** where the processing module facilitates acquisition and further content. For example, the processing module issues further content requests and receives further content for analysis. When the content quality level is favorable, the method continues at step **224** where the processing module issues a collections response to the requester. The issuing includes generating the collections response and sending the collections response to the requester. The generating of the collections response may include transforming the received content into transformed content in accordance with a transformation approach (e.g., reformatting, interpreting absolute meaning and translating into another language in accordance with the absolute meaning, etc.).

[0102] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0103] FIG. 5C is a schematic block diagram of an embodiment of the query module **124** of FIG. 4A that includes an answer acquisition module **230**, a content requirements module **232**, a source requirements module **234**, a content security module **236**, an answer timing module **238**, an answer transformation module **240**, and an answer quality module **242**. Generally, an embodiment of this invention presents solutions where the query module **124** supports responding to a query.

[0104] In an example of operation of the responding to the query, the answer acquisition module **230** receives a query request **136** from a requester. The answer acquisition module **230** obtains content requirements information **248** based on the query request **136**. The content requirements information **248** includes one or more of content parameters, a desired content type, a desired content source if any, a content type if any, candidate source identifiers, a content profile, and a question of the query request **136**. For example, the answer acquisition module **230** receives the content requirements information **248** from the content requirements module **232**,

where the content requirements module 232 generates the content requirements information 248 based on a content requirements information request from the answer acquisition module 230 and where the answer acquisition module 230 generates the content requirements information request based on the query request 136.

[0105] The answer acquisition module 230 obtains source requirements information 250 based on the query request 136. The source requirements information 250 includes one or more of candidate source identifiers, a content profile, a desired source parameter, recommended source parameters, source priority levels, and recommended source access timing. For example, the answer acquisition module 230 receives the source requirements information 250 from the source requirements module 234, where the source requirements module 234 generates the source requirements information 250 based on a source requirements information request from the answer acquisition module 230 and where the answer acquisition module 230 generates the source requirements information request based on the query request 136.

[0106] The answer acquisition module 230 obtains answer timing information 254 based on the query request 136. The answer timing information 254 includes one or more of requested answer timing, confirmed answer timing, source access testing results, estimated velocity of content updates, content freshness, timestamps, predicted time available, requested content acquisition trigger, and a content acquisition trigger detected indicator. For example, the answer acquisition module 230 receives the answer timing information 254 from the answer timing module 238, where the answer timing module 238 generates the answer timing information 254 based on an answer timing information request from the answer acquisition module 230 and where the answer acquisition module 230 generates the answer timing information request based on the query request 136.

[0107] Having obtained the content requirements information 248, the source requirements information 250, and the answer timing information 254, the answer acquisition module 230 determines whether to issue an IEI request 244 and/or a collections request 132 based on one or more of the content requirements information 248, the source requirements information 250, and the answer timing information 254. For example, the answer acquisition module 230 selects the IEI request 244 when an immediate answer to a simple query request 136 is required and is expected to have a favorable quality level. As another example, the answer acquisition module 230 selects the collections request 132 when a longer-term answer is required as indicated by the answer timing information to before and/or when the query request 136 has an unfavorable quality level.

[0108] When issuing the IEI request 244, the answer acquisition module 230 generates the IEI request 244 in accordance with security information 252 received from the content security module 236 and based on one or more of the content requirements information 248, the source requirements information 250, and the answer timing information 254. Having generated the IEI request 244, the answer acquisition module 230 sends the IEI request 244 to at least one IEI module.

[0109] When issuing the collections request 132, the answer acquisition module 230 generates the collections request 132 in accordance with the security information 252 received from the content security module 236 and based on

one or more of the content requirements information 248, the source requirements information 250, and the answer timing information 254. Having generated the collections request 132, the answer acquisition module 230 sends the collections request 132 to at least one collections module. Alternatively, the answer acquisition module 230 facilitate sending of the collections request 132 to one or more various user devices (e.g., to access a subject matter expert).

[0110] The answer acquisition module 230 determines a quality level of a received answer extracted from a collections response 134 and/or an IEI response 246. For example, the answer acquisition module 230 extracts the quality level of the received answer from answer quality information 258 received from the answer quality module 242 in response to an answer quality request from the answer acquisition module 230. When the quality level is unfavorable, the answer acquisition module 230 facilitates obtaining a further answer. The facilitation includes issuing at least one of a further IEI request 244 and a further collections request 132 to generate a further answer for further quality testing. When the quality level is favorable, the answer acquisition module 230 issues a query response 140 to the requester. The issuing includes generating the query response 140 based on answer transformation information 256 received from the answer transformation module 240, where the answer transformation module 240 generates the answer transformation information 256 to include a transformed answer based on receiving the answer from the answer acquisition module 230. The answer transformation information 256 6A further include the question, a desired format of the answer, available formats, recommended formatting, received IEI responses, transformation instructions, and transformed IEI responses into an answer.

[0111] FIG. 5D is a logic diagram of an embodiment of a method for providing a response to a query within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4C, 5C, and also FIG. 5D. The method includes step 270 where a processing module of one or more processing modules of one or more computing devices of the computing system receives a query request (e.g., a question) from a requester. The method continues at step 272 where the processing module determines content requirements information. The determining includes interpreting the query request to produce the content requirements. The method continues at step 274 where the processing module determines source requirements information. The determining includes interpreting the query request to produce the source requirements. The method continues at step 276 where the processing module determines answer timing information. The determining includes interpreting the query request to produce the answer timing information.

[0112] The method continues at step 278 the processing module determines whether to issue an IEI request and/or a collections request. For example, the determining includes selecting the IEI request when the answer timing information indicates that a simple one-time answer is appropriate. As another example, the processing module selects the collections request when the answer timing information indicates that the answer is associated with a series of events over an event time frame.

[0113] When issuing the IEI request, the method continues at step 280 where the processing module issues the IEI

request to an IEI module. The issuing includes generating the IEI request in accordance with security information and based on one or more of the content requirements information, the source requirements information, and the answer timing information.

[0114] When issuing the collections request, the method continues at step 282 where the processing module issues the collections request to a collections module. The issuing includes generating the collections request in accordance with the security information and based on one or more of the content requirements information, the source requirements information, and the answer timing information. Alternatively, the processing module issues both the IEI request and the collections request when a satisfactory partial answer may be provided based on a corresponding IEI response and a further more generalized and specific answer may be provided based on a corresponding collections response and associated further IEI response.

[0115] The method continues at step 284 where the processing module determines a quality level of a received answer. The determining includes extracting the answer from the collections response and/or the IEI response and interpreting the answer in accordance with one or more of the content requirements information, the source requirements information, the answer timing information, and the query request to produce the quality level. The method branches to step 288 when the quality level is favorable and the method continues to step 286 when the quality level is unfavorable. For example, the processing module indicates that the quality level is favorable when the quality level is equal to or greater than a minimum answer quality threshold level. As another example, the processing module indicates that the quality level is unfavorable when the quality level is less than the minimum answer quality threshold level.

[0116] When the quality level is unfavorable, the method continues at step 286 where the processing module obtains a further answer. The obtaining includes at least one of issuing a further IEI request and a further collections request to facilitate obtaining of a further answer for further answer quality level testing as the method loops back to step 270. When the quality level is favorable, the method continues at step 288 where the processing module issues a query response to the requester. The issuing includes transforming the answer into a transformed answer in accordance with an answer transformation approach (e.g., formatting, further interpretations of the virtual question in light of the answer and further knowledge) and sending the transformed answer to the requester as the query response.

[0117] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

[0118] FIG. 5E is a schematic block diagram of an embodiment of the identigen entigen intelligence (IEI) module 122 of FIG. 4A that includes a content ingestion module 300, an element identification module 302, and interpretation module 304, and answer resolution module 306, and an IEI control module 308. Generally, an embodiment of this invention presents solutions where the IEI module 122 supports interpreting content to produce knowledge that may be utilized to answer questions.

[0119] In an example of operation of the producing and utilizing of the knowledge, the content ingestion module 300 generates formatted content 314 based on question content 312 and/or source content 310, where the IEI module 122 receives an IEI request 244 that includes the question content 312 and the IEI module 122 receives a collections response 134 that includes the source content 310. The source content 310 includes content from a source extracted from the collections response 134. The question content 312 includes content extracted from the IEI request 244 (e.g., content paired with a question). The content ingestion module 300 generates the formatted content 314 in accordance with a formatting approach (e.g., creating proper sentences from words of the content). The formatted content 314 includes modified content that is compatible with subsequent element identification (e.g., complete sentences, combinations of words and interpreted sounds and/or inflection cues with temporal associations of words).

[0120] The element identification module 302 processes the formatted content 314 based on element rules 318 and an element list 332 to produce identified element information 340. Rules 316 includes the element rules 318 (e.g., match, partial match, language translation, etc.). Lists 330 includes the element list 332 (e.g., element ID, element context ID, element usage ID, words, characters, symbols etc.). The IEI control module 308 may provide the rules 316 and the lists 330 by accessing stored data 360 from a memory associated with the IEI module 122. Generally, an embodiment of this invention presents solutions where the stored data 360 may further include one or more of a descriptive dictionary, categories, representations of element sets, element list, sequence data, pending questions, pending request, recognized elements, unrecognized elements, errors, etc.

[0121] The identified element information 340 includes one or more of identifiers of elements identified in the formatted content 314, may include ordering and/or sequencing and grouping information. For example, the element identification module 302 compares elements of the formatted content 314 to known elements of the element list 332 to produce identifiers of the known elements as the identified element information 340 in accordance with the element rules 318. Alternatively, the element identification module 302 outputs un-identified element information 342 to the IEI control module 308, where the un-identified element information 342 includes temporary identifiers for elements not identifiable from the formatted content 314 when compared to the element list 332.

[0122] The interpretation module 304 processes the identified element information 340 in accordance with interpretation rules 320 (e.g., potentially valid permutations of various combinations of identified elements), question information 346 (e.g., a question extracted from the IEI request 244 which may be paired with content associated with the question), and a groupings list 334 (e.g., representations of associated groups of representations of things, a set of

element identifiers, valid element usage IDs in accordance with similar, an element context, permutations of sets of identifiers for possible interpretations of a sentence or other) to produce interpreted information 344. The interpreted information 344 includes potentially valid interpretations of combinations of identified elements. Generally, an embodiment of this invention presents solutions where the interpretation module 304 supports producing the interpreted information 344 by considering permutations of the identified element information 340 in accordance with the interpretation rules 320 and the groupings list 334.

[0123] The answer resolution module 306 processes the interpreted information 344 based on answer rules 322 (e.g., guidance to extract a desired answer), the question information 346, and inferred question information 352 (e.g., posed by the IEI control module or analysis of general collections of content or refinement of a stated question from a request) to produce preliminary answers 354 and an answer quality level 356. The answer generally lies in the interpreted information 344 as both new content received and knowledge based on groupings list 334 generated based on previously received content. The preliminary answers 354 includes an answer to a stated or inferred question that subject further refinement. The answer quality level 356 includes a determination of a quality level of the preliminary answers 354 based on the answer rules 322. The inferred question information 352 may further be associated with time information 348, where the time information includes one or more of current real-time, a time reference associated with entity submitting a request, and a time reference of a collections response.

[0124] When the IEI control module 308 determines that the answer quality level 356 is below an answer quality threshold level, the IEI control module 308 facilitates collecting of further content (e.g., by issuing a collections request 132 and receiving corresponding collections responses 134 for analysis). When the answer quality level 356 compares favorably to the answer quality threshold level, the IEI control module 308 issues an IEI response 246 based on the preliminary answers 354. When receiving training information 358, the IEI control module 308 facilitates updating of one or more of the lists 330 and the rules 316 and stores the updated list 330 and the updated rules 316 in the memories as updated stored data 360.

[0125] FIG. 5F is a logic diagram of an embodiment of a method for analyzing content within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4C, 5E, and also FIG. 5F. The method includes step 370 where a processing module of one or more processing modules of one or more computing devices of the computing system facilitates updating of one or more rules and lists based on one or more of received training information and received content. For example, the processing module updates rules with received rules to produce updated rules and updates element lists with received elements to produce updated element lists. As another example, the processing module interprets the received content to identify a new word for at least temporary inclusion in the updated element list.

[0126] The method continues at step 372 where the processing module transforms at least some of the received content into formatted content. For example, the processing module processes the received content in accordance with a

transformation approach to produce the formatted content, where the formatted content supports compatibility with subsequent element identification (e.g., typical sentence structures of groups of words).

[0127] The method continues at step 374 where the processing module processes the formatted content based on the rules and the lists to produce identified element information and/or an identified element information. For example, the processing module compares the formatted content to element lists to identify a match producing identifiers for identified elements or new identifiers for unidentified elements when there is no match.

[0128] The method continues at step 376 with a processing module processes the identified element information based on rules, the lists, and question information to produce interpreted information. For example, the processing module compares the identified element information to associated groups of representations of things to generate potentially valid interpretations of combinations of identified elements.

[0129] The method continues at step 378 where the processing module processes the interpreted information based on the rules, the question information, and inferred question information to produce preliminary answers. For example, the processing module matches the interpreted information to one or more answers (e.g., embedded knowledge based on a fact base built from previously received content) with highest correctness likelihood levels that is subject to further refinement.

[0130] The method continues at step 380 where the processing module generates an answer quality level based on the preliminary answers, the rules, and the inferred question information. For example, the processing module predicts the answer correctness likelihood level based on the rules, the inferred question information, and the question information. The method branches to step 384 when the answer quality level is favorable and the method continues to step 382 when the answer quality level is unfavorable. For example, the generating of the answer quality level further includes the processing module indicating that the answer quality level is favorable when the answer quality level is greater than or equal to a minimum answer quality threshold level. As another example, the generating of the answer quality level further includes the processing module indicating that the answer quality level is unfavorable when the answer quality level is less than the minimum answer quality threshold level.

[0131] When the answer quality level is unfavorable, the method continues at step 382 where the processing module facilitates gathering clarifying information. For example, the processing module issues a collections request to facilitate receiving further content and or request question clarification from a question requester. When the answer quality level is favorable, the method continues at step 384 where the processing module issues a response that includes one or more answers based on the preliminary answers and/or further updated preliminary answers based on gathering further content. For example, the processing module generates a response that includes one or more answers and the answer quality level and issues the response to the requester.

[0132] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g.,

a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

[0133] FIG. 6A is a schematic block diagram of an embodiment of the element identification module 302 of FIG. 5A and the interpretation module 304 of FIG. 5A. The element identification module 302 includes an element matching module 400 and an element grouping module 402. The interpretation module 304 includes a grouping matching module 404 and a grouping interpretation module 406. Generally, an embodiment of this invention presents solutions where the element identification module 302 supports identifying potentially valid permutations of groupings of elements while the interpretation module 304 interprets the potentially valid permutations of groupings of elements to produce interpreted information that includes the most likely of groupings based on a question.

[0134] In an example of operation of the identifying of the potentially valid permutations of groupings of elements, when matching elements of the formatted content 314, the element matching module 400 generates matched elements 412 (e.g., identifiers of elements contained in the formatted content 314) based on the element list 332. For example, the element matching module 400 matches a received element to an element of the element list 332 and outputs the matched elements 412 to include an identifier of the matched element. When finding elements that are unidentified, the element matching module 400 outputs un-recognized words information 408 (e.g., words not in the element list 332, may temporarily add) as part of un-identified element information 342. For example, the element matching module 400 indicates that a match cannot be made between a received element of the formatted content 314, generates the unrecognized words info 408 to include the received element and/or a temporary identifier, and issues an updated element list 414 that includes the temporary identifier and the corresponding unidentified received element.

[0135] The element grouping module 402 analyzes the matched elements 412 in accordance with element rules 318 to produce grouping error information 410 (e.g., incorrect sentence structure indicators) when a structural error is detected. The element grouping module 402 produces identified element information 340 when favorable structure is associated with the matched elements in accordance with the element rules 318. The identified element information 340 may further include grouping information of the plurality of permutations of groups of elements (e.g., several possible interpretations), where the grouping information includes one or more groups of words forming an associated set and/or super-group set of two or more subsets when subsets share a common core element.

[0136] In an example of operation of the interpreting of the potentially valid permutations of groupings of elements to produce the interpreted information, the grouping matching module 404 analyzes the identified element information 340 in accordance with a groupings list 334 to produce

validated groupings information 416. For example, the grouping matching module 404 compares a grouping aspect of the identified element information 340 (e.g., for each permutation of groups of elements of possible interpretations), generates the validated groupings information 416 to include identification of valid permutations aligned with the groupings list 334. Alternatively, or in addition to, the grouping matching module 404 generates an updated groupings list 418 when determining a new valid grouping (e.g., has favorable structure and interpreted meaning) that is to be added to the groupings list 334.

[0137] The grouping interpretation module 406 interprets the validated groupings information 416 based on the question information 346 and in accordance with the interpretation rules 320 to produce interpreted information 344 (e.g., most likely interpretations, next most likely interpretations, etc.). For example, the grouping interpretation module 406 obtains context, obtains favorable historical interpretations, processes the validated groupings based on interpretation rules 320, where each interpretation is associated with a correctness likelihood level.

[0138] FIG. 6B is a logic diagram of an embodiment of a method for interpreting information within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4C, 5E-5F, 6A, and also FIG. 6B. The method includes step 430 where a processing module of one or more processing modules of one or more computing devices of the computing system analyzes formatted content. For example, the processing module attempt to match a received element of the formatted content to one or more elements of an elements list. When there is no match, the method branches to step 434 and when there is a match, the method continues to step 432. When there is a match, the method continues at step 432 where the processing module outputs matched elements (e.g., to include the matched element and/or an identifier of the matched element). When there is no match, the method continues at step 434 where the processing module outputs unrecognized words (e.g., elements and/or a temporary identifier for the unmatched element).

[0139] The method continues at step 436 where the processing module analyzes matched elements. For example, the processing module attempt to match a detected structure of the matched elements (e.g., chained elements as in a received sequence) to favorable structures in accordance with element rules. The method branches to step 440 when the analysis is unfavorable and the method continues to step 438 when the analysis is favorable. When the analysis is favorable matching a detected structure to the favorable structure of the element rules, the method continues at step 438 where the processing module outputs identified element information (e.g., an identifier of the favorable structure, identifiers of each of the detected elements). When the analysis is unfavorable matching a detected structure to the favorable structure of the element rules, the method continues at step 440 where the processing module outputs grouping error information (e.g., a representation of the incorrect structure, identifiers of the elements of the incorrect structure, a temporary new identifier of the incorrect structure).

[0140] The method continues at step 442 where the processing module analyzes the identified element information to produce validated groupings information. For example, the processing module compares a grouping aspect of the

identified element information and generates the validated groupings information to include identification of valid permutations that align with the groupings list. Alternatively, or in addition to, the processing module generates an updated groupings list when determining a new valid grouping.

[0141] The method continues at step 444 where the processing module interprets the validated groupings information to produce interpreted information. For example, the processing module obtains one or more of context and historical interpretations and processes the validated groupings based on interpretation rules to generate the interpreted information, where each interpretation is associated with a correctness likelihood level (e.g., a quality level).

[0142] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

[0143] FIG. 6C is a schematic block diagram of an embodiment of the answer resolution module 306 of FIG. 5A that includes an interim answer module 460, and answer prioritization module 462, and a preliminary answer quality module 464. Generally, an embodiment of this invention presents solutions where the answer resolution module 306 supports producing an answer for interpreted information 344.

[0144] In an example of operation of the providing of the answer, the interim answer module 460 analyzes the interpreted information 344 based on question information 346 and inferred question information 352 to produce interim answers 466 (e.g., answers to stated and/or inferred questions without regard to rules that is subject to further refinement). The answer prioritization module 462 analyzes the interim answers 466 based on answer rules 322 to produce preliminary answer 354. For example, the answer prioritization module 462 identifies all possible answers from the interim answers 466 that conform to the answer rules 322.

[0145] The preliminary answer quality module 464 analyzes the preliminary answers 354 in accordance with the question information 346, the inferred question information 352, and the answer rules 322 to produce an answer quality level 356. For example, for each of the preliminary answers 354, the preliminary answer quality module 464 may compare a fit of the preliminary answer 354 to a corresponding previous answer and question quality level, calculate the answer quality level 356 based on a level of conformance to the answer rules 322, calculate the answer quality level 356 based on alignment with the inferred question information 352, and determine the answer quality level 356 based on an interpreted correlation with the question information 346.

[0146] FIG. 6D is a logic diagram of an embodiment of a method for producing an answer within a computing system.

In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-3, 4A-4C, 5E-5F, 6C, and also FIG. 6D. The method includes step 480 where a processing module of one or more processing modules of one or more computing devices of the computing system analyzes received interpreted information based on question information and inferred question information to produce one or more interim answers. For example, the processing module generates potential answers based on patterns consistent with previously produced knowledge and likelihood of correctness.

[0147] The method continues at step 482 where the processing module analyzes the one or more interim answers based on answer rules to produce preliminary answers. For example, the processing module identifies all possible answers from the interim answers that conform to the answer rules. The method continues at step 484 where the processing module analyzes the preliminary answers in accordance with the question information, the inferred question information, and the answer rules to produce an answer quality level. For example, for each of the elementary answers, the processing module may compare a fit of the preliminary answer to a corresponding previous answer-and-answer quality level, calculate the answer quality level based on performance to the answer rules, calculate answer quality level based on alignment with the inferred question information, and determine the answer quality level based on interpreted correlation with the question information.

[0148] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

[0149] FIG. 7A is an information flow diagram for interpreting information within a computing system, where sets of entigens 504 are interpreted from sets of identigens 502 which are interpreted from sentences of words 500. Such identigen entigen intelligence (IEI) processing of the words (e.g., to IEI process) includes producing one or more of interim knowledge, a preliminary answer, and an answer quality level. For example, the IEI processing includes identifying permutations of identigens of a phrase of a sentence (e.g., interpreting human expressions to produce identigen groupings for each word of ingested content), reducing the permutations of identigens (e.g., utilizing rules to eliminate unfavorable permutations), mapping the reduced permutations of identigens to at least one set of entigens (e.g., most likely identigens become the entigens) to produce the interim knowledge, processing the knowledge in accordance with a knowledge database (e.g., comparing the set of entigens to the knowledge database) to

produce a preliminary answer, and generating the answer quality level based on the preliminary answer for a corresponding domain.

[0150] Human expressions are utilized to portray facts and fiction about the real world. The real-world includes items, actions, and attributes. The human expressions include textual words, textual symbols, images, and other sensorial information (e.g., sounds). It is known that many words, within a given language, can mean different things based on groupings and orderings of the words. For example, the sentences of words **500** can include many different forms of sentences that mean vastly different things even when the words are very similar.

[0151] The present invention presents solutions where the computing system **10** supports producing a computer-based representation of a truest meaning possible of the human expressions given the way that multitudes of human expressions relate to these meanings. As a first step of the flow diagram to transition from human representations of things to a most precise computer representation of the things, the computer identifies the words, phrases, sentences, etc. from the human expressions to produce the sets of identigens **502**. Each identigen includes an identifier of their meaning and an identifier of an instance for each possible language, culture, etc. For example, the words car and automobile share a common meaning identifier but have different instance identifiers since they are different words and are spelled differently. As another example, the word duck is associated both with a bird and an action to elude even though they are spelled the same. In this example the bird duck has a different meaning than the elude duck and as such each has a different meaning identifier of the corresponding identigens.

[0152] As a second step of the flow diagram to transition from human representations of things to the most precise computer representation of the things, the computer extracts meaning from groupings of the identified words, phrases, sentences, etc. to produce the sets of entigens **504**. Each entigen includes an identifier of a single conceivable and perceivable thing in space and time (e.g., independent of language and other aspects of the human expressions). For example, the words car and automobile are different instances of the same meaning and point to a common shared entigen. As another example, the word duck for the bird meaning has an associated unique entigen that is different than the entigen for the word duck for the elude meaning.

[0153] As a third step of the flow diagram to transition from human expressions of things to the most precise computer representation of the things, the computer reasons facts from the extracted meanings. For example, the computer maintains a fact-based of the valid meanings from the valid groupings or sets of entigens so as to support subsequent inferences, deductions, rationalizations of posed questions to produce answers that are aligned with a most factual view. As time goes on, and as an entigen has been identified, it can encounter an experience transformations in time, space, attributes, actions, and words which are used to identify it without creating contradictions or ever losing its identity.

[0154] FIG. 7B is a relationship block diagram illustrating an embodiment of relationships between things **510** and representations of things **512** within a computing system. The things **510** includes conceivable and perceivable things

including actions **522**, items **524**, and attributes **526**. The representation of things **512** includes representations of things used by humans **514** and representation of things used by of computing devices **516** of embodiments of the present invention. The things **510** relates to the representations of things used by humans **514** where the invention presents solutions where the computing system **10** supports mapping the representations of things used by humans **514** to the representations of things used by computing devices **516**, where the representations of things used by computing devices **516** map back to the things **510**.

[0155] The representations of things used by humans **514** includes textual words **528**, textual symbols **530**, images (e.g., non-textual) **532**, and other sensorial information **534** (e.g., sounds, sensor data, electrical fields, voice inflections, emotion representations, facial expressions, whistles, etc.). The representations of things used by computing devices **516** includes identigens **518** and entigens **520**. The representations of things used by humans **514** maps to the identigens **518** and the identigens **518** map to the entigens **520**. The entigens **520** uniquely maps back to the things **510** in space and time, a truest meaning the computer is looking for to create knowledge and answer questions based on the knowledge.

[0156] To accommodate the mapping of the representations of things used by humans **514** to the identigens **518**, the identigens **518** is partitioned into actenymys **544** (e.g., actions), itenymys **546** (e.g., items), attrenymys **548** (e.g., attributes), and functionals **550** (e.g., that join and/or describe). Each of the actenymys **544**, itenymys **546**, and attrenymys **548** may be further classified into singulatums **552** (e.g., identify one unique entigen) and pluratums **554** (e.g., identify a plurality of entigens that have similarities).

[0157] Each identigen **518** is associated with an identigens identifier (IDN) **536**. The IDN **536** includes a meaning identifier (ID) **538** portion, an instance ID **540** portion, and a type ID **542** portion. The meaning ID **538** includes an identifier of common meaning. The instance ID **540** includes an identifier of a particular word and language. The type ID **542** includes one or more identifiers for actenymys, itenymys, attrenymys, singulatums, pluratums, a time reference, and any other reference to describe the IDN **536**. The mapping of the representations of things used by humans **514** to the identigens **518** by the computing system of the present invention includes determining the identigens **518** in accordance with logic and instructions for forming groupings of words.

[0158] Generally, an embodiment of this invention presents solutions where the identigens **518** map to the entigens **520**. Multiple identigens may map to a common unique entigen. The mapping of the identigens **518** to the entigens **520** by the computing system of the present invention includes determining entigens in accordance with logic and instructions for forming groupings of identigens.

[0159] FIG. 7C is a diagram of an embodiment of a synonym words table **570** within a computing system, where the synonym words table **570** includes multiple fields including textual words **572**, identigens **518**, and entigens **520**. The identigens **518** includes fields for the meaning identifier (ID) **538** and the instance ID **540**. The computing system of the present invention may utilize the synonym words table **570** to map textual words **572** to identigens **518** and map the identigens **518** to entigens **520**. For example, the words car, automobile, auto, bil (Swedish), carro (Spanish), and bil (Danish) all share a common meaning but are

different instances (e.g., different words and languages). The words map to a common meaning ID but to individual unique instant identifiers. Each of the different identigens map to a common entigen since they describe the same thing.

[0160] FIG. 7D is a diagram of an embodiment of a polysemous words table **576** within a computing system, where the polysemous words table **576** includes multiple fields including textual words **572**, identigens **518**, and entigens **520**. The identigens **518** includes fields for the meaning identifier (ID) **538** and the instance ID **540**. The computing system of the present invention may utilize the polysemous words table **576** to map textual words **572** to identigens **518** and map the identigens **518** to entigens **520**. For example, the word duck maps to four different identigens since the word duck has four associated different meanings (e.g., bird, fabric, to submerge, to elude) and instances. Each of the identigens represent different things and hence map to four different entigens.

[0161] FIG. 7E is a diagram of an embodiment of transforming words into groupings within a computing system that includes a words table **580**, a groupings of words section to validate permutations of groupings, and a groupings table **584** to capture the valid groupings. The words table **580** includes multiple fields including textual words **572**, identigens **518**, and entigens **520**. The identigens **518** includes fields for the meaning identifier (ID) **538**, the instance ID **540**, and the type ID **542**. The computing system of the present invention may utilize the words table **580** to map textual words **572** to identigens **518** and map the identigens **518** to entigens **520**. For example, the word pilot may refer to a flyer and the action to fly. Each meaning has a different identigen and different entigen.

[0162] The computing system of the present invention may apply rules to the fields of the words table **580** to validate various groupings of words. Those that are invalid are denoted with a "X" while those that are valid are associated with a check mark. For example, the grouping "pilot Tom" is invalid when the word pilot refers to flying and Tom refers to a person. The identigen combinations for the flying pilot and the person Tom are denoted as invalid by the rules. As another example, the grouping "pilot Tom" is valid when the word pilot refers to a flyer and Tom refers to the person. The identigen combinations for the flyer pilot and the person Tom are denoted as valid by the rules.

[0163] The groupings table **584** includes multiple fields including grouping ID **586**, word strings **588**, identigens **518**, and entigens **520**. The computing system of the present invention may produce the groupings table **584** as a stored fact base for valid and/or invalid groupings of words identified by their corresponding identigens. For example, the valid grouping "pilot Tom" referring to flyer Tom the person is represented with a grouping identifier of **3001** and identity and identifiers 150.001 and 457.001. The entigen field **520** may indicate associated entigens that correspond to the identigens. For example, entigen e717 corresponds to the flyer pilot meaning and entigen e61 corresponds to the time the person meaning. Alternatively, or in addition to, the entigen field **520** may be populated with a single entigen identifier (ENI).

[0164] The word strings field **588** may include any number of words in a string. Different ordering of the same words can produce multiple different strings and even different meanings and hence entigens. More broadly, each entry

(e.g., role) of the groupings table **584** may refer to groupings of words, two or more word strings, an idiom, just identigens, just entigens, and/or any combination of the preceding elements. Each entry has a unique grouping identifier. An idiom may have a unique grouping ID and include identifiers of original word identigens and replacing identigens associated with the meaning of the idiom not just the meaning of the original words. Valid groupings may still have ambiguity on their own and may need more strings and/or context to select a best fit when interpreting a truest meaning of the grouping.

[0165] FIG. 8A is a data flow diagram for accumulating knowledge within a computing system, where a computing device, at a time= t_0 , ingests and processes facts **598** at a step **590** based on rules **316** and fact base information **600** to produce groupings **602** for storage in a fact base **592** (e.g., words, phrases, word groupings, identigens, entigens, quality levels). The facts **598** may include information from books, archive data, Central intelligence agency (CIA) world fact book, trusted content, etc. The ingesting may include filtering to organize and promote better valid groupings detection (e.g., considering similar domains together). The groupings **602** includes one or more of groupings identifiers, identigen identifiers, entigen identifiers, and estimated fit quality levels. The processing step **590** may include identifying identigens from words of the facts **598** in accordance with the rules **316** and the fact base info **600** and identifying groupings utilizing identigens in accordance with rules **316** and fact base info **600**.

[0166] Subsequent to ingestion and processing of the facts **598** to establish the fact base **592**, at a time= t_1+ , the computing device ingests and processes new content **604** at a step **594** in accordance with the rules **316** and the fact base information **600** to produce preliminary grouping **606**. The new content may include updated content (e.g., timewise) from periodicals, newsfeeds, social media, etc. The preliminary grouping **606** includes one or more of preliminary groupings identifiers, preliminary identigen identifiers, preliminary entigen identifiers, estimated fit quality levels, and representations of unidentified words.

[0167] The computing device validates the preliminary groupings **606** at a step **596** based on the rules **316** and the fact base info **600** to produce updated fact base info **608** for storage in the fact base **592**. The validating includes one or more of reasoning a fit of existing fact base info **600** with the new preliminary grouping **606**, discarding preliminary groupings, updating just time frame information associated with an entry of the existing fact base info **600** (e.g., to validate knowledge for the present), creating new entigens, and creating a median entigen to summarize portions of knowledge within a median indicator as a quality level indicator (e.g., suggestive not certain).

[0168] Storage of the updated fact base information **608** captures patterns that develop by themselves instead of searching for patterns as in prior art artificial intelligence systems. Growth of the fact base **592** enables subsequent reasoning to create new knowledge including deduction, induction, inference, and inferential sentiment (e.g., a chain of sentiment sentences). Examples of sentiments includes emotion, beliefs, convictions, feelings, judgments, notions, opinions, and views.

[0169] FIG. 8B is a diagram of an embodiment of a groupings table **620** within a computing system. The groupings table **620** includes multiple fields including grouping ID

586, word strings 588, an IF string 622 and a THEN string 624. Each of the fields for the IF string 622 and the THEN string 624 includes fields for an identigen (IDN) string 626, and an entigen (ENI) string 628. The computing system of the present invention may produce the groupings table 620 as a stored fact base to enable IF THEN based inference to generate a new knowledge inference 630.

[0170] As a specific example, grouping 5493 points out the logic of IF someone has a tumor, THEN someone is sick and the grouping 5494 points out the logic that IF someone is sick, THEN someone is sad. As a result of utilizing inference, the new knowledge inference 630 may produce grouping 5495 where IF someone has a tumor, THEN someone is possibly sad (e.g., or is sad).

[0171] FIG. 8C is a data flow diagram for answering questions utilizing accumulated knowledge within a computing system, where a computing device ingests and processes question information 346 at a step 640 based on rules 316 and fact base info 600 from a fact base 592 to produce preliminary grouping 606. The ingesting and processing questions step 640 includes identifying identigens from words of a question in accordance with the rules 316 and the fact base information 600 and may also include identifying groupings from the identified identigens in accordance with the rules 316 and the fact base information 600.

[0172] The computing device validates the preliminary grouping 606 at a step 596 based on the rules 316 and the fact base information 600 to produce identified element information 340. For example, the computing device reasons fit of existing fact base information with new preliminary groupings 606 to produce the identified element information 340 associated with highest quality levels. The computing device interprets a question of the identified element information 340 at a step 642 based on the rules 316 and the fact base information 600. The interpreting of the question may include separating new content from the question and reducing the question based on the fact base information 600 and the new content.

[0173] The computing device produces preliminary answers 354 from the interpreted information 344 at a resolve answer step 644 based on the rules 316 and the fact base information 600. For example, the computing device compares the interpreted information 344 two the fact base information 600 to produce the preliminary answers 354 with highest quality levels utilizing one or more of deduction, induction, inferencing, and applying inferential sentiments logic. Alternatively, or in addition to, the computing device may save new knowledge identified from the question information 346 to update the fact base 592.

[0174] FIG. 8D is a data flow diagram for answering questions utilizing interference within a computing system that includes a groupings table 648 and the resolve answer step 644 of FIG. 8C. The groupings table 648 includes multiple fields including fields for a grouping (GRP) identifier (ID) 586, word strings 588, an identigen (IDN) string 626, and an entigen (ENI) 628. The groupings table 648 may be utilized to build a fact base to enable resolving a future question into an answer. For example, the grouping 8356 notes knowledge that Michael sleeps eight hours and grouping 8357 notes that Michael usually starts to sleep at 11:00 PM.

[0175] In a first question example that includes a question “Michael sleeping?”, the resolve answer step 644 analyzes the question from the interpreted information 344 in accordance

with the fact base information 600, the rules 316, and a real-time indicator that the current time is 1:00 AM to produce a preliminary answer of “possibly YES” when inferring that Michael is probably sleeping at 1:00 AM when Michael usually starts sleeping at 11:00 PM and Michael usually sleeps for a duration of eight hours.

[0176] In a second question example that includes the question “Michael sleeping?”, the resolve answer step 644 analyzes the question from the interpreted information 344 in accordance with the fact base information 600, the rules 316, and a real-time indicator that the current time is now 11:00 AM to produce a preliminary answer of “possibly NO” when inferring that Michael is probably not sleeping at 11:00 AM when Michael usually starts sleeping at 11:00 PM and Michael usually sleeps for a duration of eight hours.

[0177] FIG. 8E is a relationship block diagram illustrating another embodiment of relationships between things and representations of things within a computing system. While things in the real world are described with words, it is often the case that a particular word has multiple meanings in isolation. Interpreting the meaning of the particular word may hinge on analyzing how the word is utilized in a phrase, a sentence, multiple sentences, paragraphs, and even whole documents or more. Describing and stratifying the use of words, word types, and possible meanings help in interpreting a true meaning.

[0178] Humans utilize textual words 528 to represent things in the real world. Quite often a particular word has multiple instances of different grammatical use when part of a phrase of one or more sentences. The grammatical use 649 of words includes the nouns and the verbs, and also includes adverbs, adjectives, pronouns, conjunctions, prepositions, determiners, exclamations, etc.

[0179] As an example of multiple grammatical use, the word “bat” in the English language can be utilized as a noun or a verb. For instance, when utilized as a noun, the word “bat” may apply to a baseball bat or may apply to a flying “bat.” As another instance, when utilized as a verb, the word “bat” may apply to the action of hitting or batting an object, i.e., “bat the ball.”

[0180] To stratify word types by use, the words are associated with a word type (e.g., type identifier 542). The word types include objects (e.g., items 524), characteristics (e.g., attributes 526), actions 522, and the functionals 550 for joining other words and describing words. For example, when the word “bat” is utilized as a noun, the word is describing the object of either the baseball bat or the flying bat. As another example, when the word “bat” is utilized as a verb, the word is describing the action of hitting.

[0181] To determine possible meanings, the words, by word type, are mapped to associative meanings (e.g., identigens 518). For each possible associative meaning, the word type is documented with the meaning and further with an identifier (ID) of the instance (e.g., an identigen identifier).

[0182] For the example of the word “bat” when utilized as a noun for the baseball bat, a first identigen identifier 536-1 includes a type ID 542-1 associated with the object 524, an instance ID 540-1 associated with the first identigen identifier (e.g., unique for the baseball bat), and a meaning ID 538-1 associated with the baseball bat. For the example of the word “bat” when utilized as a noun for the flying bat, a second identigen identifier 536-2 includes a type ID 542-1 associated with the object 524, an instance ID 540-2 associated with the second identigen identifier (e.g., unique for

the flying bat), and a meaning ID **538-2** associated with the flying bat. For the example of the word “bat” when utilized as a verb for the bat that hits, a third identigen identifier **536-2** includes a type ID **542-2** associated with the actions **522**, an instance ID **540-3** associated with the third identigen identifier (e.g., unique for the bat that hits), and a meaning ID **538-3** associated with the bat that hits.

[0183] With the word described by a type and possible associative meanings, a combination of full grammatical use of the word within the phrase etc., application of rules, and utilization of an ever-growing knowledge database that represents knowledge by linked entigens, the absolute meaning (e.g., entigen **520**) of the word is represented as a unique entigen. For example, a first entigen e1 represents the absolute meaning of a baseball bat (e.g., a generic baseball bat not a particular baseball bat that belongs to anyone), a second entigen e2 represents the absolute meaning of the flying bat (e.g., a generic flying bat not a particular flying bat), and a third entigen e3 represents the absolute meaning of the verb bat (e.g., to hit).

[0184] An embodiment of methods to ingest text to produce absolute meanings for storage in a knowledge database are discussed in greater detail with reference to FIGS. **8F-H**. Those embodiments further discuss the discerning of the grammatical use, the use of the rules, and the utilization of the knowledge database to definitively interpret the absolute meaning of a string of words.

[0185] Another embodiment of methods to respond to a query to produce an answer based on knowledge stored in the knowledge database are discussed in greater detail with reference to FIGS. **8J-L**. Those embodiments further discuss the discerning of the grammatical use, the use of the rules, and the utilization of the knowledge database to interpret the query. The query interpretation is utilized to extract the answer from the knowledge database to facilitate forming the query response.

[0186] FIGS. **8F** and **8G** are schematic block diagrams of another embodiment of a computing system that includes the content ingestion module **300** of FIG. **5E**, the element identification module **302** of FIG. **5E**, the interpretation module **304** of FIG. **5E**, the IEI control module **308** of FIG. **5E**, and the SS memory **96** of FIG. **2**. Generally, an embodiment of this invention provides presents solutions where the computing system **10** supports processing content to produce knowledge for storage in a knowledge database.

[0187] The processing of the content to produce the knowledge includes a series of steps. For example, a first step includes identifying words of an ingested phrase to produce tokenized words. As depicted in FIG. **8F**, a specific example of the first step includes the content ingestion module **300** comparing words of source content **310** to dictionary entries to produce formatted content **314** that includes identifiers of known words. Alternatively, when a comparison is unfavorable, the temporary identifier may be assigned to an unknown word. For instance, the content ingestion module **300** produces identifiers associated with the words “the”, “black”, “bat”, “eats”, and “fruit” when the ingested phrase includes “The black bat eats fruit”, and generates the formatted content **314** to include the identifiers of the words.

[0188] A second step of the processing of the content to produce the knowledge includes, for each tokenized word, identifying one or more identigens that correspond the tokenized word, where each identigen describes one of an

object, a characteristic, and an action. As depicted in FIG. **8F**, a specific example of the second step includes the element identification module **302** performing a look up of identigen identifiers, utilizing an element list **332** and in accordance with element rules **318**, of the one or more identigens associated with each tokenized word of the formatted content **314** to produce identified element information **340**.

[0189] A unique identifier is associated with each of the potential object, the characteristic, and the action (OCA) associated with the tokenized word (e.g., sequential identigens). For instance, the element identification module **302** identifies a functional symbol for “the”, identifies a single identigen for “black”, identifies two identigens for “bat” (e.g., baseball bat and flying bat), identifies a single identigen for “eats”, and identifies a single identigen for “fruit.” When at least one tokenized word is associated with multiple identigens, two or more permutations of sequential combinations of identigens for each tokenized word result. For example, when “bat” is associated with two identigens, two permutations of sequential combinations of identigens result for the ingested phrase.

[0190] A third step of the processing of the content to produce the knowledge includes, for each permutation of sequential combinations of identigens, generating a corresponding equation package (i.e., candidate interpretation), where the equation package includes a sequential linking of pairs of identigens (e.g., relationships), where each sequential linking pairs a preceding identigen to a next identigen, and where an equation element describes a relationship between paired identigens (OCAs) such as describes, acts on, is a, belongs to, did, did to, etc. Multiple OCAs occur for a common word when the word has multiple potential meanings (e.g., a baseball bat, a flying bat).

[0191] As depicted in FIG. **8F**, a specific example of the third step includes the interpretation module **304**, for each permutation of identigens of each tokenized word of the identified element information **340**, the interpretation module **304** generates, in accordance with interpretation rules **320** and a groupings list **334**, an equation package to include one or more of the identifiers of the tokenized words, a list of identifiers of the identigens of the equation package, a list of pairing identifiers for sequential pairs of identigens, and a quality metric associated with each sequential pair of identigens (e.g., likelihood of a proper interpretation). For instance, the interpretation module **304** produces a first equation package that includes a first identigen pairing of a black bat (e.g., flying bat with a higher quality metric level), the second pairing of bat eats (e.g., the flying bat eats, with a higher quality metric level), and a third pairing of eats fruit, and the interpretation module **304** produces a second equation package that includes a first pairing of a black bat (e.g., baseball bat, with a neutral quality metric level), the second pairing of bat eats (e.g., the baseball bat eats, with a lower quality metric level), and a third pairing of eats fruit.

[0192] A fourth step of the processing of the content to produce the knowledge includes selecting a surviving equation package associated with a most favorable confidence level. As depicted in FIG. **8F**, a specific example of the fourth step includes the interpretation module **304** applying interpretation rules **320** (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens) to reduce a number of permutations of the sequential combi-

nations of identigens to produce interpreted information **344** that includes identification of at least one equation package as a surviving interpretation SI (e.g., higher quality metric level).

[0193] Non-surviving equation packages are eliminated that compare unfavorably to pairing rules and/or are associated with an unfavorable quality metric levels to produce a non-surviving interpretation NSI 2 (e.g., lower quality metric level), where an overall quality metric level may be assigned to each equation package based on quality metric levels of each pairing, such that a higher quality metric level of an equation package indicates a higher probability of a most favorable interpretation. For instance, the interpretation module **304** eliminates the equation package that includes the second pairing indicating that the “baseball bat eats” which is inconsistent with a desired quality metric level of one or more of the groupings list **334** and the interpretation rules **320** and selects the equation package associated with the “flying bat eats” which is favorably consistent with the one or more of the quality metric levels of the groupings list **334** and the interpretation rules **320**.

[0194] A fifth step of the processing of the content to produce the knowledge utilizing the confidence level includes integrating knowledge of the surviving equation package into a knowledge database. For example, integrating at least a portion of the reduced OCA combinations into a graphical database to produce updated knowledge. As another example, the portion of the reduced OCA combinations may be translated into rows and columns entries when utilizing a rows and columns database rather than a graphical database. When utilizing the rows and columns approach for the knowledge database, subsequent access to the knowledge database may utilize structured query language (SQL) queries.

[0195] As depicted in FIG. 8G, a specific example of the fifth step includes the IEI control module **308** recovering fact base information **600** from SS memory **96** to identify a portion of the knowledge database for potential modification utilizing the OCAs of the surviving interpretation SI 1 (i.e., compare a pattern of relationships between the OCAs of the surviving interpretation SI 1 from the interpreted information **344** to relationships of OCAs of the portion of the knowledge database including potentially new quality metric levels).

[0196] The fifth step further includes determining modifications (e.g., additions, subtractions, further clarifications required when information is complex, etc.) to the portion of the knowledge database based on the new quality metric levels. For instance, the IEI control module **308** causes adding the element “black” as a “describes” relationship of an existing bat OCA and adding the element “fruit” as an eats “does to” relationship to implement the modifications to the portion of the fact base information **600** to produce updated fact base information **608** for storage in the SS memory **96**.

[0197] FIG. 8H is a logic diagram of an embodiment of a method for processing content to produce knowledge for storage within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-8E, 8F, and also FIG. 8G. The method includes step **650** where a processing module of one or more processing modules of one or more computing devices of the computing system identifies words of an ingested phrase to produce tokenized

words. The identifying includes comparing words to known words of dictionary entries to produce identifiers of known words.

[0198] For each tokenized word, the method continues at step **651** where the processing module identifies one or more identigens that corresponds to the tokenized word, where each identigen describes one of an object, a characteristic, and an action (e.g., OCA). The identifying includes performing a lookup of identifiers of the one or more identigens associated with each tokenized word, where the different identifiers associated with each of the potential object, the characteristic, and the action associated with the tokenized word.

[0199] The method continues at step **652** where the processing module, for each permutation of sequential combinations of identigens, generates a plurality of equation elements to form a corresponding equation package, where each equation element describes a relationship between sequentially linked pairs of identigens, where each sequential linking pairs a preceding identigen to a next identigen. For example, for each permutation of identigens of each tokenized word, the processing module generates the equation package to include a plurality of equation elements, where each equation element describes the relationship (e.g., describes, acts on, is a, belongs to, did, did too, etc.) between sequentially adjacent identigens of a plurality of sequential combinations of identigens. Each equation element may be further associated with a quality metric to evaluate a favorability level of an interpretation in light of the sequence of identigens of the equation package.

[0200] The method continues at step **653** where the processing module selects a surviving equation package associated with most favorable interpretation. For example, the processing module applies rules (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens), to reduce the number of permutations of the sequential combinations of identigens to identify at least one equation package, where non-surviving equation packages are eliminated the compare unfavorably to pairing rules and/or are associated with an unfavorable quality metric levels to produce a non-surviving interpretation, where an overall quality metric level is assigned to each equation package based on quality metric levels of each pairing, such that a higher quality metric level indicates an equation package with a higher probability of favorability of correctness.

[0201] The method continues at step **654** where the processing module integrates knowledge of the surviving equation package into a knowledge database. For example, the processing module integrates at least a portion of the reduced OCA combinations into a graphical database to produce updated knowledge. The integrating may include recovering fact base information from storage of the knowledge database to identify a portion of the knowledge database for potential modifications utilizing the OCAs of the surviving equation package (i.e., compare a pattern of relationships between the OCAs of the surviving equation package to relationships of the OCAs of the portion of the knowledge database including potentially new quality metric levels). The integrating further includes determining modifications (e.g., additions, subtractions, further clarifications required when complex information is presented, etc.) to produce the updated knowledge database that is based on fit of acceptable quality metric levels, and imple-

menting the modifications to the portion of the fact base information to produce the updated fact base information for storage in the portion of the knowledge database.

[0202] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0203] FIGS. **8J** and **8K** are schematic block diagrams of another embodiment of a computing system that includes the content ingestion module **300** of FIG. **5E**, the element identification module **302** of FIG. **5E**, the interpretation module **304** of FIG. **5E**, the answer resolution module **306** of FIG. **5E**, and the SS memory **96** of FIG. **2**. Generally, an embodiment of this invention provides solutions where the computing system **10** supports for generating a query response to a query utilizing a knowledge database.

[0204] The generating of the query response to the query includes a series of steps. For example, a first step includes identifying words of an ingested query to produce tokenized words. As depicted in FIG. **8J**, a specific example of the first step includes the content ingestion module **300** comparing words of query info **138** to dictionary entries to produce formatted content **314** that includes identifiers of known words. For instance, the content ingestion module **300** produces identifiers for each word of the query “what black animal flies and eats fruit and insects?”

[0205] A second step of the generating of the query response to the query includes, for each tokenized word, identifying one or more identigens that correspond to the tokenized word, where each identigen describes one of an object, a characteristic, and an action (OCA). As depicted in FIG. **8J**, a specific example of the second step includes the element identification module **302** performing a look up of identifiers, utilizing an element list **332** and in accordance with element rules **318**, of the one or more identigens associated with each tokenized word of the formatted content **314** to produce identified element information **340**. A unique identifier is associated with each of the potential object, the characteristic, and the action associated with a particular tokenized word. For instance, the element identification module **302** produces a single identigen identifier for each of the black color, an animal, flies, eats, fruit, and insects.

[0206] A third step of the generating of the query response to the query includes, for each permutation of sequential combinations of identigens, generating a corresponding equation package (i.e., candidate interpretation). The equation package includes a sequential linking of pairs of identigens, where each sequential linking pairs a preceding identigen to a next identigen. An equation element describes a relationship between paired identigens (OCAs) such as describes, acts on, is a, belongs to, did, did to, etc.

[0207] As depicted in FIG. **8J**, a specific example of the third step includes the interpretation module **304**, for each permutation of identigens of each tokenized word of the identified element information **340**, generating the equation packages in accordance with interpretation rules **320** and a groupings list **334** to produce a series of equation elements that include pairings of identigens. For instance, the interpretation module **304** generates a first pairing to describe a black animal, a second pairing to describe an animal that flies, a third pairing to describe flies and eats, a fourth pairing to describe eats fruit, and a fifth pairing to describe eats fruit and insects.

[0208] A fourth step of the generating the query response to the query includes selecting a surviving equation package associated with a most favorable interpretation. As depicted in FIG. **8J**, a specific example of the fourth step includes the interpretation module **304** applying the interpretation rules **320** (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens) to reduce the number of permutations of the sequential combinations of identigens to produce interpreted information **344**. The interpreted information **344** includes identification of at least one equation package as a surviving interpretation SI **10**, where non-surviving equation packages, if any, are eliminated that compare unfavorably to pairing rules to produce a non-surviving interpretation.

[0209] A fifth step of the generating the query response to the query includes utilizing a knowledge database, generating a query response to the surviving equation package of the query, where the surviving equation package of the query is transformed to produce query knowledge for comparison to a portion of the knowledge database. An answer is extracted from the portion of the knowledge database to produce the query response.

[0210] As depicted in FIG. **8K**, a specific example of the fifth step includes the answer resolution module **306** interpreting the surviving interpretation SI **10** of the interpreted information **344** in accordance with answer rules **322** to produce query knowledge QK **10** (i.e., a graphical representation of knowledge when the knowledge database utilizes a graphical database). For example, the answer resolution module **306** accesses fact base information **600** from the SS memory **96** to identify the portion of the knowledge database associated with a favorable comparison of the query knowledge QK **10** (e.g., by comparing attributes of the query knowledge QK **10** to attributes of the fact base information **600**), and generates preliminary answers **354** that includes the answer to the query. For instance, the answer is “bat” when the associated OCAs of bat, such as black, eats fruit, eats insects, is an animal, and flies, aligns with OCAs of the query knowledge.

[0211] FIG. **8L** is a logic diagram of an embodiment of a method for generating a query response to a query utilizing knowledge within a knowledge database within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-8D**, **8J**, and also FIG. **8K**. The method includes step **655** where a processing module of one or more processing modules of one or more computing devices of the computing system identifies words of an ingested query to produce tokenized words. For

example, the processing module compares words to known words of dictionary entries to produce identifiers of known words.

[0212] For each tokenized word, the method continues at step 656 where the processing module identifies one or more identigens that correspond to the tokenized word, where each identigen describes one of an object, a characteristic, and an action. For example, the processing module performs a lookup of identifiers of the one or more identigens associated with each tokenized word, where different identifiers associated with each permutation of a potential object, characteristic, and action associated with the tokenized word.

[0213] For each permutation of sequential combinations of identigens, the method continues at step 657 where the processing module generates a plurality of equation elements to form a corresponding equation package, where each equation element describes a relationship between sequentially linked pairs of identigens. Each sequential linking pairs a preceding identigen to a next identigen. For example, for each permutation of identigens of each tokenized word, the processing module includes all other permutations of all other tokenized words to generate the equation packages. Each equation package includes a plurality of equation elements describing the relationships between sequentially adjacent identigens of a plurality of sequential combinations of identigens.

[0214] The method continues at step 658 where the processing module selects a surviving equation package associated with a most favorable interpretation. For example, the processing module applies rules (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens) to reduce the number of permutations of the sequential combinations of identigens to identify at least one equation package. Non-surviving equation packages are eliminated that compare unfavorably to pairing rules.

[0215] The method continues at step 659 where the processing module generates a query response to the surviving equation package, where the surviving equation package is transformed to produce query knowledge for locating the portion of a knowledge database that includes an answer to the query. As an example of generating the query response, the processing module interprets the surviving the equation package in accordance with answer rules to produce the query knowledge (e.g., a graphical representation of knowledge when the knowledge database utilizes a graphical database format).

[0216] The processing module accesses fact base information from the knowledge database to identify the portion of the knowledge database associated with a favorable comparison of the query knowledge (e.g., favorable comparison of attributes of the query knowledge to the portion of the knowledge database, aligning favorably comparing entigens without conflicting entigens). The processing module extracts an answer from the portion of the knowledge database to produce the query response.

[0217] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a

second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

[0218] FIGS. 9A, 9B, 9C, and 9D are schematic block diagrams of another embodiment of a computing system illustrating a method for interpreting meaning of content within the computing system. The computing system includes the content ingestion module 300 of FIG. 5E, the element identification module 302 of FIG. 5E, the interpretation module 304 of FIG. 5E, and a knowledge database 700. The knowledge database 700 may be implemented utilizing one or more of the memories of FIG. 2.

[0219] FIG. 9A illustrates example steps of a method for interpreting meaning of content where, in a first step the content ingestion module 300 obtains a phrase 702 that includes string of words. The obtaining includes receiving, looking up, retrieving, and generating. For example, the content ingestion module 300 receives the phrase 702 that includes the string of words “black bat eats fruit”.

[0220] Having obtained the phrase 702, in a second step the content ingestion module 300 generates a valid sequence of words 706 utilizing the phrase 702. The generating includes one or more of validating words of the string of words with valid words of dictionary, eliminating unneeded words, replacing undesired words with replacement words that are desired, and adding additional words that may provide enhanced clarity of understanding as is commonly used within a given language. For example, the content ingestion module 300 obtains a dictionary 704 from the knowledge database 700 and validates the words of the string of words with words of the dictionary 704 to produce the valid sequence of words 706. For instance, the content ingestion module 300 compares each word of the string of words “black bat eats fruit” to words of the dictionary 704 to validate each of the words producing the valid sequence of words 706 that includes “black, bat, eats, fruit.”

[0221] In a third step of the method for interpreting meaning of content, the element identification module 302 identifies a set of identigens for each word of the valid sequence of words 706 to produce a plurality of sets of identigens. A first set of identigens of the plurality of sets of identigens includes one or more different meanings of a first word of the valid sequence of words.

[0222] The identifying the set of identigens includes a series of sub-steps. A first sub-step includes accessing, for each word of the valid sequence of words 706, the knowledge database 700 to recover the set of identigens. For example, the element identification module 302 receives identigen information 708 from the knowledge database 700 that includes an identigen set #1 for the word “black”, where the identigen set #1 includes identigens iA1, iA2, and iA3. The identigens iA1, iA2, and iA3 represent three unique interpretations of the word black, including “dark-skin people”, “black color”, and “to make black.”

[0223] A second sub-step includes interpreting a query response from another computing device to produce the set of identigens, where the query response is received in response to an identigen query with regards to the word. For

example, the element identification module receives the plurality of sets of identigens from the other computing device.

[0224] FIG. 9B illustrates further example steps of the method for interpreting meaning of content where, in a fourth step the element identification module 302 identifies, for each identigen of the plurality of sets of identigens, a word type of a plurality of word types associated with phrase structure grammar rules to produce a plurality of sets of identigen-type associations 712. A first set of identigen-type associations of the plurality of sets of identigen-type associations includes one or more identigen-type associations of the first word of the valid sequence of words.

[0225] Word types include tens of various permutations of grammar based descriptions of words. Examples of word types includes noun, adjective, adverb, verb, etc. The phrase structure grammar rules apply to relationships between the parts of a natural language sentence. For example, a rule includes verbs following nouns. As another example, another rule includes adjectives preceding nouns.

[0226] The identifying, for each identigen of the plurality of sets of identigens, a word type of the plurality of word types associated with the phrase structure grammar rules to produce the plurality of sets of identigen-type associations 712 includes a series of sub-steps. A first sub-step includes accessing, for each identigen of the plurality of sets of identigens, the knowledge database 700 to recover the word type. For example, the element identification module 302 receives word type information 710 from the knowledge database 700 that includes a word type of “noun” for the identigen iA1, a word type of “adjective” for the identigen iA2, and a word type of “verb” for the identigen iA3, etc.

[0227] A second sub-step includes associating, for each identigen of the plurality of sets of identigens, the identigen with the corresponding word type to produce the plurality of sets of identigen-type associations 712. For example, the element identification module 302 associates “noun” with the identigen iA1, “adjective” with the identigen iA2, and “verb” with the identigen iA3, etc.

[0228] FIG. 9C illustrates further example steps of the method for interpreting meaning of content where, in a fifth step the interpretation module 304 interprets, utilizing the phrase structure grammar rules 714, the plurality of sets of identigen-type associations 712 to produce an entigen group 716. The entigen group 716 represents a most likely interpretation of the phrase 702. A first entigen of the entigen group 716 corresponds to an identigen of the first set of identigens having a most likely meaning of the one or more different meanings of the first word of the valid sequence of words.

[0229] The interpreting the plurality of sets of identigen-type associations 712 includes a variety of approaches. A first approach includes identifying valid identigen-type associations and a second approach includes eliminating invalid identigen-type associations.

[0230] When utilizing the first approach of identifying the valid identigen-type associations, the approach includes a series of sub-steps. A first sub-step includes identifying, for each set of identigen-type associations of the plurality of sets of identigen-type associations, a valid identigen-type association of a sequence of identigen-type associations complying with the phrase structure grammar rules 714. The sequence of identigen-type associations complying with the phrase structure grammar rules 714 includes a first identi-

gen-type association of a first set of identigen-type associations and a first identigen-type association of a second set of identigen-type associations. For example, the interpretation module 304 receives the phrase structure grammar rules 714 from the knowledge database 700 indicating that a sequence of the noun identigen iB2 and the verb identigen iC2 complies with the phrase structure grammar rules 714.

[0231] A second sub-step of the first approach includes identifying a sequence of identigens of the sequence of identigen-type associations complying with the phrase structure grammar rules to produce the entigen group 716. For example the interpretation module 304 receives the phrase structure grammar rules 714 from the knowledge database 700 indicating that a sequence of the adjective identigen iA2 followed by the noun identigen iB2 followed by the verb identigen iC2 followed by the noun identigen iD1 complies with the phrase structure grammar rules 714 and produces the entigen group 716.

[0232] When utilizing the second approach of identifying the invalid identigen-type associations, the approach includes a series of sub-steps. A first sub-step includes eliminating, for each set of identigen-type associations of the plurality of sets of identigen-type associations, an invalid identigen-type association of an invalid sequence of identigen-type associations based on non-compliance with the phrase structure grammar rules 714. The invalid sequence of identigen-type associations includes a first invalid identigen-type association of a first set of identigen-type associations and a first invalid identigen-type association of a second set of identigen-type associations. For example, the interpretation module 304 receives the phrase structure grammar rules 714 from the knowledge database 700 indicating that a sequence of the noun identigen iB1 and the verb identigen iC2 does not comply with the phrase structure grammar rules 714 and that a sequence of the noun identigen iB2 and the noun identigen iC1 does not comply with the phrase structure grammar rules 714.

[0233] A second sub-step of the second approach includes identifying, for each set of identigen-type associations of the plurality of sets of identigen-type associations, a remaining identigen-type association of another sequence of identigen-type associations complying with the phrase structure grammar rules. The sequence of identigen-type associations complying with the phrase structure grammar rules includes a first valid identigen-type association of the first set of identigen-type associations and a first valid identigen-type association of the second set of identigen-type associations. For example, the interpretation module 304 receives the phrase structure grammar rules 714 from the knowledge database 700 indicating that the sequence of the noun identigen iB2 and the verb identigen iC2 complies with the phrase structure grammar rules 714.

[0234] A third sub-step includes identifying a sequence of identigens of the other sequence of identigen-type associations complying with the phrase structure grammar rules to produce the entigen group 716. For example the interpretation module 304 receives the phrase structure grammar rules 714 from the knowledge database 700 indicating that the sequence of the adjective identigen iA2 followed by the noun identigen iB2 followed by the verb identigen iC2 followed by the noun identigen iD1 complies with the phrase structure grammar rules 714 and produces the entigen group 716.

[0235] FIG. 9D illustrates further example steps of the method for interpreting meaning of content where, in a sixth step the interpretation module 304 outputs the entigen group 716 to a receiving entity (e.g., to another computing device). Alternatively, or in addition to, the interpretation module 304 facilitates storage of the entigen group 716 in the knowledge database 700. For example, the interpretation module 304 sends the entigen group 716 to the knowledge database 700 for storage, where the identigen-type associations for each identigen are stored with the identigens, and where indications are stored of the relationships between the valid sequence of identigens. For instance, the adjective black operates on the entigen representing noun bat, the entigens representing the noun bat operates on the entigen representing the verb eats, and the entigens representing the verb eats operates on the entigen representing the noun fruit.

[0236] The method described above with reference to FIG. 9A-9D and in conjunction with the various modules can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a non-transitory computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

[0237] FIG. 10A is a schematic block diagram of another embodiment of a computing system that includes grammar rich content sources 750, the user device 12-1 of FIG. 1, and the artificial intelligence (AI) server 20-1 of FIG. 1. The grammar rich content sources 750 includes the content sources 16-1 through 16-N of FIG. 1. In particular, content sources associated with the grammar rich content sources 750 provides any type of content where at least a portion of the content includes examples of text in accordance with parts of speech rules. The AI server 20-1 includes the processing module 50-1 of FIG. 2 and the solid state (SS) memory 96 of FIG. 2. The processing module 50-1 includes the identigen entigen intelligence (IEI) module 122 of FIG. 4A. Generally, an embodiment of this invention presents solutions where the computing system functions to interpret meaning.

[0238] In an example of operation of the interpreting of the meaning the IEI module 122 receives a query request 136 that includes a sentence and a request to determine knowledge of the sentence. The IEI module 122 matches words of the sentence to entry words of a word dictionary to identify valid sequential words of the sentence. The dictionary is extracted from fact base information 600 retrieved from the SS memory 96. From time to time, the IEI module 122 IEI processes grammar rich content responses 754 from the grammar rich content sources 750 in response to grammar rich content requests 752 to produce knowledge stored as the fact base information 600 in the SS memory 96.

[0239] Having identified the valid sequential words, the IEI module 122 identifies parts of the sentence in accordance with phrase structure grammar rules (e.g., separated by a separator like and, or). For example, the IEI module 122

identifies a first and second part where each part includes 3 to 5 words and where the first and second part are separated by the word “and”.

[0240] Having identified the parts of the sentence, the IEI module 122 identify grammar structures in accordance with the phrase structure grammar rules (e.g., noun phrase, verb phrase, etc.). For example, the IEI module 122 compares words of a first part to a pattern that represents a particular phrase structure grammar rule and indicates the grammar structure when the comparison is favorable.

[0241] Having identified the grammar structures, the IEI module 122 converts the grammar structures into a standard form in accordance with the phrase structure grammar rules (e.g., ordering of words of phrases). For example, the IEI module 122 obtains the phrase structure grammar rules as fact base information 600 from SS memory 96 and rearranges the identified grammar structures in accordance with the phrase structure grammar rules.

[0242] Having produced the standard form, the IEI module 122 retrieves, from the SS memory 96 as fact base information 600, a set of identigens for each word of the standard form. For example, the IEI module 122 indexes into storage of sets of identigens utilizing the word to recover the set of identigens for the word.

[0243] Having retrieved the sets of identigens, the IEI module 122 eliminates identigens from each set of identigens that do not comply with grammar and/or identigen sequencing rules to produce an entigen set. The IEI module 122 issues a query response 140 to the user device 12-1 that includes the entigen set.

[0244] FIG. 10B is a data flow diagram illustrating an example of a method for interpreting meaning that includes a series of steps. A first step includes receiving a word string: “the bat is black and eat its fruit”. A second step includes identifying valid words of the word string. For example, stored words of a dictionary of word identigen information 760 is compared to each of the words to identify valid words: the, bat, is, black, and, eats, fruit.

[0245] A third step includes identifying the parts of the word string by searching for parts (e.g., parts versus separators) indicated by phrase structure grammar rule 762. For example, the separator “and” is identified which partitions the word string into a first part that includes words “the bat is black” and a second part that includes words “eats fruit”.

[0246] A fourth step includes applying grammar structures of the phrase structure grammar rules 762 to each part of the word string to identify standard form phrases. For example, the word string is identified to include a noun phrase “the bat”, another noun phrase “is black”, the separator “and”, and a verb phrase “eat fruit”.

[0247] A fifth step includes applying parts of speech rules of the phrase structure grammar rules 762 to arrange the identified standard form phrases into a standard form sequence. For example, the phrases are arranged to create a standard form that includes “the black bat eats fruit” in accordance with the parts of speech rules.

[0248] A sixth step includes retrieving, for each word of the standard form, a set of identigens from the word identigen information 760 and eliminating identigens from each set of identigens that do not comply with the grammar and/or identigen sequencing rules of the phrase structure grammar rules 762 to produce one identigen for each set of identigens.

[0249] A seventh step includes identifying the remaining one identigen for each set of identigens as a sequence of

entigens of an antigen group. The antigen group represents a most likely meaning of the word string.

[0250] FIG. 10C is a logic diagram of an embodiment of a method for interpreting meaning within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-8L, 10A-10B, and also FIG. 10C. The method includes step 780 where a processing module of one or more processing modules of one or more computing devices of the computing system matches words of a sentence to dictionary entries of valid words to produce a valid sequence of words. For example, the processing module accesses the dictionary utilizing each word to retrieve valid words to produce the valid sequential words.

[0251] The method continues at step 782 where the processing module identifies sentence parts. For example, the processing module identifies separators to delineate the sentence parts. The method continues at step 784 where the processing module identifies grammar structures of the sentence parts. For example, for each sentence parts, the processing module matches words of the part to patterns of candidate grammar structure types to produce the grammar structures.

[0252] The method continues at step 786 where the processing module determines standard form sentence utilizing the grammar structures. For example, the processing module orders words of phrases in accordance with phrase structure grammar rules.

[0253] The method continues at step 788 where the processing module retrieves, for each word of the standard form sentence, a set of identigens. For example, the processing module accesses a knowledge database utilizing the word to retrieve a set of identigens.

[0254] The method continues at step 790 where the processing module selects one antigen from each set of identigens to produce an antigen group. For example, the processing module eliminates identigens that don't comply with the grammar and/or antigen sequencing rules to produce the antigen group for the sentence.

[0255] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

[0256] FIG. 11A is a schematic block diagram of another embodiment of a computing system that includes domain content sources 800, the user device 12-1 of FIG. 1, and the artificial intelligence (AI) server 20-1 of FIG. 1. The domain content sources 800 includes the content sources 16-1 through 16-N of FIG. 1. In particular, content sources associated with the domain content sources 800 provides any type of content where at least a portion of the content includes content associated by domain. The AI server 20-1

includes the processing module 50-1 of FIG. 2 and the solid state (SS) memory 96 of FIG. 2. The processing module 50-1 includes the antigen antigen intelligence (IEI) module 122 of FIG. 4A. Generally, an embodiment of this invention presents solutions where the computing system functions to interpret meaning.

[0257] In an example of operation of the interpreting of the meaning the IEI module 122 receives a query request 136 that includes a word string and a request to interpret knowledge of the word string. The IEI module 122 matches words of the word string to entry words of a word dictionary to identify valid sequential words of the word string. For example, the IEI module 122 extracts dictionary entries from fact base information 600 retrieved from the SS memory 96. From time to time, the IEI module 122 IEI processes domain content responses 804 from the domain content sources 800 in response to domain content requests 802 to produce knowledge stored as the fact base information 600 in the SS memory 96.

[0258] Having identified the valid sequential words of the word string, the IEI module 122, for each word, retrieves, from the SS memory 96 as fact base info 600, a set of identigens and an associated domain to produce a plurality of sets of identigens and domains. The IEI module 122 applies domain rules to the plurality of sets of identigens in accordance with the domains to eliminate identigens that do not comply with the rules and to keep identigens that share common domains to produce retained identigens.

[0259] Having retained one remaining valid antigen for each set of identigens, the IEI module 122 outputs retained identigens as a query response 140. The retained identigens represents a most likely meaning of the string of words.

[0260] FIG. 11B is a data flow diagram illustrating an example of a method for interpreting meaning that includes a series of steps. A first step includes receiving a word string for knowledge interpretation. For example, the word string includes "the bat hit a ball". A second step includes comparing words of the word string to valid words from a dictionary to produce the valid words: the, bat, hit, a, and ball.

[0261] A third step includes retrieving, for each word, a set of identigens where each antigen is associated with a domain. For example, a set of identigens for the word bat includes a first antigen associated with a sports domain of a baseball bat, a second antigen associated with an animal domain of an animal, and a third antigen associated with an action domain of to hit.

[0262] A fourth step includes identifying identigens associated with a reoccurring, domain amongst the sets of identigens as most likely meaning identigens. For example, the sports domain is identified as the most common domain and the identigens associated with the baseball bat, striking, and a sports ball are identified as a most likely permutation of identigens amongst the sets of identigens.

[0263] FIG. 11C is a logic diagram of an embodiment of a method for interpreting meaning within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-8L, 11A-11B, and also FIG. 11C. The method includes step 830 where a processing module of one or more processing modules of one or more computing devices of the computing system matches words of a string of words to dictionary entries of valid words to produce a valid sequence of words. For

example, the processing module accesses the dictionary utilizing each word to retrieve valid words of the valid sequence of words.

[0264] For each word of the valid sequence of words, the method continues at step **832** where the processing module retrieves a set of identigens with a corresponding set of domains to produce a plurality of sets of identigens and domains. For example, the processing module accesses a knowledge database using the word to retrieve the set of identigens, where each identigen is paired with a domain identifier.

[0265] The method continues at step **834** where the processing module applies domain rules to the plurality of sets of identigens and domains to retained identigens of a common domain to produce retained identigens. For example, the processing module identifies identigens of a most common domain, a next most common domain, etc. and links identigens of the different sets of identigens by a level of domain commonality.

[0266] The method continues at step **836** for the processing module outputs the retained identigens and an indication of a level of domain compliance (e.g., the level of domain commonality). For example, the processing module generates the level of domain compliance based on the level of domain commonality. For instance, the processing module generates the domain compliance to include a number of identigens sharing a particular common domain.

[0267] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0268] FIG. **12A** is a schematic block diagram of another embodiment of a computing system that includes adjacency content sources **850**, the user device **12-1** of FIG. **1**, and the artificial intelligence (AI) server **20-1** of FIG. **1**. The adjacency content sources **850** includes the content sources **16-1** through **16-N** of FIG. **1**. In particular, content sources associated with the adjacency content sources **850** provides any type of content where at least a portion of the content includes repeat word utilization. The AI server **20-1** includes the processing module **50-1** of FIG. **2** and the solid state (SS) memory **96** of FIG. **2**. The processing module **50-1** includes the identigen entigen intelligence (IEI) module **122** of FIG. **4A**. Generally, an embodiment of this invention presents solutions where the computing system functions to interpret meaning.

[0269] In an example of operation of the interpreting of the meaning, the IEI module **122** receives a query request **136** that includes a word string and a request to interpret knowledge of the word string. The IEI module **122** matches words of the word string to entry words of a word dictionary to identify valid sequential words of the word string. The dictionary is extracted from fact base information **600**

retrieved from the SS memory **96**. From time to time, the IEI module **122** IEI processes adjacency content responses **856** from the adjacency content sources **850** in response to adjacency content requests **854** to produce knowledge stored as the fact base information **600** in the SS memory **96**.

[0270] Having identified the valid sequential words of the word string, the IEI module **122**, for each word, retrieves, from the SS memory **96** as fact base info **600**, a set of identigens to produce a plurality of sets of identigens. The IEI module **122**, for each adjacent pairing of identigens, retrieves, from the SS memory **96** as fact base info **600**, an adjacency score (e.g., number of words separating a common word, number of repeat utilizations of the word within a range of words, etc.).

[0271] The IEI module **122**, for each adjacent pairing of identigens, eliminates identigens associated with adjacency scores below a low threshold level. For example, the IEI module **122** eliminates identigens that are not frequently repeated.

[0272] The IEI module **122**, for each remaining permutation of sequential identigens of the plurality of sets of identigens, retrieves, from the SS memory **96** as fact base info **600**, updated adjacency scores for pairs of adjacent identigens of the permutation. For example, the IEI module **122** retrieves adjacency scores for words across an entire permutation of identigens.

[0273] The IEI module **122** rank orders a remaining permutation of sequential identigens by total scores, where a first total score includes pairwise undated adjacency scores for a first remaining permutation of sequential identigens.

[0274] The IEI module **122** outputs a highest ranked ordered permutation of sequential identigens as a query response **140**. The highest ranked ordered permutation of sequential identigens represents a potential most likely meaning of the string of words.

[0275] FIG. **12B** is a data flow diagram illustrating an example of a method for interpreting meaning that includes a series of steps. A first step includes receiving a word string for knowledge interpretation. For example, the word string includes "the bat hit a ball". A second step includes identifying valid words of the word string by comparing words of the word string to stored words of the dictionary. For example, the valid words are identified to include the, that, hit, a, and ball.

[0276] A third step includes, for each word of the valid words, retrieving a set of identigens from a knowledge database and retrieving adjacency scores between permutations of adjacent identigens of adjacent sets of identigens. For example, a first identigen for the word bat represents baseball bat and is associated with an adjacency score of 90 (e.g., a high score representing a more likely combination) to a first identigen of the word hit meaning to strike. The first identigen for the word that is also associated with an identigen adjacency score of 42 and a second identigen of the word hit meaning struck.

[0277] The third step further includes eliminating permutations of identigens associated with low adjacency scores. For example, the third identigen for the word bat that means to hit is eliminated along with the second identigen of the word hit for struck as the adjacency scores associated with the third identigen of the word bat are low as compared to the other adjacency scores (e.g., 5 is the lowest).

[0278] In a similar fashion, the fourth step includes eliminating further identigens of remaining sets of identigens

based on identifying identigens associated with adjacency scores that are below a low threshold level. For example, fourth and fifth identigens for the word ball that mean ball of a foot and forming a round shape are eliminated when the adjacency scores are the lowest compared to the other identigens for the word ball (e.g., 5 and 40 are lower than 70, 80, and 90 for the other identigens).

[0279] A fifth step includes rank ordering permutations of identigens across the sets of identigens based on entigen group permutation scores based on aggregates of adjacency scores for remaining identigens. For example, a first permutation includes the baseball bat identigen for the word bat the strike identigen for the word hit and the sports ball identigen for the word ball when a sum of associated adjacency scores is 180 (e.g., 90+90). Still further permutations are identified to include less likely interpretations of the word string for permutations with lower summarized adjacency scores.

[0280] FIG. 12C is a logic diagram of an embodiment of a method for interpreting meaning within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-8L, 12A-12B, and also FIG. 12C. The method includes step 900 where a processing module of one or more processing modules of one or more computing devices of the computing system matches words of a string of words to dictionary entries of valid words to produce a valid sequence of words. For example, the processing module accesses the dictionary utilizing each word to retrieve valid words of the valid string of words.

[0281] For each word of the valid sequence of words, the method continues at step 902 where the processing module retrieves a set of identigens to produce a plurality of sets of identigens. For example, the processing module accesses a knowledge database using the word to retrieve the set of identigens.

[0282] The method continues at step 904 where the processing module eliminates identigens associated with adjacency scores of pairwise identigens that are below a low threshold level. For example, for each adjacent pairing of identigens, the processing module retrieves an adjacency score, compares the adjacency score to the low threshold level, and identifies identigens associated with an adjacency score below the low threshold number based on the comparison.

[0283] The method continues at step 906 where the processing module rank orders remaining permutations of sequential identigens based on updated adjacency scores for pairs of identigens when used in a permutation. For example, for each adjacent pairing of identigens, the processing module retrieves an updated adjacency score for each pairing for the full permutation of identigens, sums the scores of the permutations, and rank orders by total score.

[0284] The method continues at step 908 where the processing module outputs highest ranked ordered permutations of sequential identigens that represent potential most likely meanings of the string of words. For example, the processing module issues a response to a requesting entity that includes the rankings and permutations of identigens.

[0285] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g.,

a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

[0286] FIG. 13A is a schematic block diagram of another embodiment of a computing system that includes word position content sources 930, the user device 12-1 of FIG. 1, and the artificial intelligence (AI) server 20-1 of FIG. 1. The word position content sources 930 includes the content sources 16-1 through 16-N of FIG. 1. In particular, content sources associated with the word position content sources 930 provides any type of content where at least a portion of the content includes repeated utilization of a common word. The AI server 20-1 includes the processing module 50-1 of FIG. 2 and the solid state (SS) memory 96 of FIG. 2. The processing module 50-1 includes the identigen entigen intelligence (IEI) module 122 of FIG. 4A. Generally, an embodiment of this invention presents solutions where the computing system functions to interpret meaning.

[0287] In an example of operation of the interpreting of the meaning, the IEI module 122 receives a query request 136 that includes a word string and a request to interpret knowledge of the word string. The IEI module 122 matches words of the word string to entry words of a word dictionary to identify valid sequential words of the word string. The dictionary is extracted from fact base information 600 retrieved from the SS memory 96. From time to time, the IEI module 122 IEI processes word position content responses 936 from the word position content sources 930 in response to word position content requests 934 to produce knowledge stored as the fact base information 600 in the SS memory 96.

[0288] The IEI module 122, for each word, determines proximity to identical words of the word string (e.g., how many words away, frequency of use, etc.). The IEI module 122, for each word, retrieves, from the SS memory 96 as fact base info 600, set of identigens to produce a plurality of sets of identigens.

[0289] The IEI module 122, for a first portion of the word string, selects, for each set of identigens, one identigen based on identigen adjacency rules (e.g., sequentially advance through pairs of identigens, analyze as a whole). For example, the processing module selects a common identigen amongst two or more sets of identigens when a frequency of use is greater than a minimum threshold level across a range of words.

[0290] The IEI module 122 selects, for another portion of the word string, for each set of identigens, one identigen based on selected identigens for words of the first portion that are within a favorable proximity of the same words of the other portions to produce a selected permutation of sequential identigens for the word string. The IEI module 122 outputs the selected permutation of sequential identigens as a query response 140, where the selected permutation of sequential identigens represents a most likely meanings of the word string.

[0291] FIG. 13B is a data flow diagram illustrating an example of a method for interpreting meaning that includes

a series of steps. A first step includes receiving a word string for knowledge interpretation. For example, the word string is a received to include “the bat hit a ball” and “boycott the ball”.

[0292] A second step includes comparing words of the word strings to stored words of a dictionary to identify valid words for one or more portions of the word string. For example, the valid words are identified to include, for a first portion, the, bat, hit, a, and for a second portion, ball, boy, caught, the, and ball.

[0293] A third step includes determining frequency of occurrence of a given word that repeats within the word string. For example, the word ball is identified to occur twice within a range of four words. Such close proximity may indicate that the word has the same meaning in both occurrences.

[0294] A fourth step includes retrieving, for each word of the word string, a set of identigens from a knowledge database. The fourth step further includes selecting identigens for a first portion of the word string based on identigen sequencing rules. For example, an identigen meaning baseball bat of the word bat is selected with an identigen meaning to strike for the word hit and an identigen meaning sports ball for the word ball in accordance with the identigen sequencing rules.

[0295] A fifth step includes selecting identigens for a second portion of the overall word string based on the selected identigens of the first portion, where an entigen of the second portion is selected to be identical to the identigens selected in the first portion for common words. For example, the identigen for the sports ball is selected for the second portion since the identigen for the sports ball was selected in the first portion and the frequency of occurrence and word separation (e.g., absolute and/or average number of words between sequential occurrences) compare favorably to an associated proximity threshold level. Having selected the identigens for the common words of the portions, remaining identigens are selected in accordance with the identigen sequencing rules to produce selected identigens for the word string.

[0296] A sixth step includes outputting the selected identigens as an entigen group representing a most likely meaning of the word string. For example, the entigen group includes entigens for baseball bat, strike, ball, child, intercept, and sports ball.

[0297] FIG. 13C is a logic diagram of an embodiment of a method for interpreting meaning within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. 1-8L, 13A-13B, and also FIG. 13C. The method includes step 970 where a processing module of one or more processing modules of one or more computing devices of the computing system matches words of a string of words to dictionary entries of valid words to produce a valid sequence of words. For example, the processing module accesses the dictionary utilizing each word to retrieve valid words of the valid sequence of words.

[0298] The method continues at step 972 where the processing module determines proximity of identical words of the string of words. For example, the processing module counts how many words away and a frequency of occurrence of the word.

[0299] The method continues at step 974, where, for each word of the valid sequence of words, the processing module retrieves a set of identigens to produce a plurality of sets of identigens. For example, the processing module accesses a knowledge database using the word to retrieve the set of identigens.

[0300] The method continues at step 976 where the processing module selects, for each set of identigens of a first portion of the string of words, one identigen that represents a most likely meaning of the word corresponding to the set of identigens. For example, the processing module selects based on identigen adjacency rules (e.g., sequentially advance through pairs of identigens, and analyze the permutation of identigens is a whole).

[0301] The method continues at step 978 where the processing module selects, for an incremental occurrence of a word that is within a favorable proximity of an initial occurrence of the word in the first portion, the identigen that was selected for the initial occurrence of the word. For example, the processing module selects a same identigen for a same word when the word is less than a maximum word distance away.

[0302] The method continues at step 980 where the processing module outputs a permutation of selected identigens for each of the plurality of identigens. For example, the processing module completes selections based on identigen sequencing rules and/or previously selected identigens for same words. In an embodiment, a common word is chained from each occurrence to another occurrence where a distance in terms of number of words between each occurrence is less than a maximum distance number.

[0303] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system 10 of FIG. 1 or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system 10, cause the one or more computing devices to perform any or all of the method steps described above.

[0304] It is noted that terminologies as may be used herein such as bit stream, stream, signal sequence, etc. (or their equivalents) have been used interchangeably to describe digital information whose content corresponds to any of a number of desired types (e.g., data, video, speech, text, graphics, audio, etc. any of which may generally be referred to as ‘data’).

[0305] As may be used herein, the terms “substantially” and “approximately” provides an industry-accepted tolerance for its corresponding term and/or relativity between items. For some industries, an industry-accepted tolerance is less than one percent and, for other industries, the industry-accepted tolerance is 10 percent or more. Other examples of industry-accepted tolerance range from less than one percent to fifty percent. Industry-accepted tolerances correspond to, but are not limited to, component values, integrated circuit process variations, temperature variations, rise and fall times, thermal noise, dimensions, signaling errors, dropped

packets, temperatures, pressures, material compositions, and/or performance metrics. Within an industry, tolerance variances of accepted tolerances may be more or less than a percentage level (e.g., dimension tolerance of less than $\pm 1\%$). Some relativity between items may range from a difference of less than a percentage level to a few percent. Other relativity between items may range from a difference of a few percent to magnitude of differences.

[0306] As may also be used herein, the term(s) “configured to”, “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via an intervening item (e.g., an item includes, but is not limited to, a component, an element, a circuit, and/or a module) where, for an example of indirect coupling, the intervening item does not modify the information of a signal but may adjust its current level, voltage level, and/or power level. As may further be used herein, inferred coupling (i.e., where one element is coupled to another element by inference) includes direct and indirect coupling between two items in the same manner as “coupled to”.

[0307] As may even further be used herein, the term “configured to”, “operable to”, “coupled to”, or “operably coupled to” indicates that an item includes one or more of power connections, input(s), output(s), etc., to perform, when activated, one or more its corresponding functions and may further include inferred coupling to one or more other items. As may still further be used herein, the term “associated with”, includes direct and/or indirect coupling of separate items and/or one item being embedded within another item.

[0308] As may be used herein, the term “compares favorably”, indicates that a comparison between two or more items, signals, etc., provides a desired relationship. For example, when the desired relationship is that signal 1 has a greater magnitude than signal 2, a favorable comparison may be achieved when the magnitude of signal 1 is greater than that of signal 2 or when the magnitude of signal 2 is less than that of signal 1. As may be used herein, the term “compares unfavorably”, indicates that a comparison between two or more items, signals, etc., fails to provide the desired relationship.

[0309] As may be used herein, one or more claims may include, in a specific form of this generic form, the phrase “at least one of a, b, and c” or of this generic form “at least one of a, b, or c”, with more or less elements than “a”, “b”, and “c”. In either phrasing, the phrases are to be interpreted identically. In particular, “at least one of a, b, and c” is equivalent to “at least one of a, b, or c” and shall mean a, b, and/or c. As an example, it means: “a” only, “b” only, “c” only, “a” and “b”, “a” and “c”, “b” and “c”, and/or “a”, “b”, and “c”.

[0310] As may also be used herein, the terms “processing module”, “processing circuit”, “processor”, “processing circuitry”, and/or “processing unit” may be a single processing device or a plurality of processing devices. Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions. The processing module, module, processing circuit, processing circuitry, and/or processing

unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, processing circuitry, and/or processing unit. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, processing circuitry, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network). Further note that if the processing module, module, processing circuit, processing circuitry and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, processing circuitry and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture.

[0311] One or more embodiments have been described above with the aid of method steps illustrating the performance of specified functions and relationships thereof. The boundaries and sequence of these functional building blocks and method steps have been arbitrarily defined herein for convenience of description. Alternate boundaries and sequences can be defined so long as the specified functions and relationships are appropriately performed. Any such alternate boundaries or sequences are thus within the scope and spirit of the claims. Further, the boundaries of these functional building blocks have been arbitrarily defined for convenience of description. Alternate boundaries could be defined as long as the certain significant functions are appropriately performed. Similarly, flow diagram blocks may also have been arbitrarily defined herein to illustrate certain significant functionality.

[0312] To the extent used, the flow diagram block boundaries and sequence could have been defined otherwise and still perform the certain significant functionality. Such alternate definitions of both functional building blocks and flow diagram blocks and sequences are thus within the scope and spirit of the claims. One of average skill in the art will also recognize that the functional building blocks, and other illustrative blocks, modules and components herein, can be implemented as illustrated or by discrete components, application specific integrated circuits, processors executing appropriate software and the like or any combination thereof.

[0313] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with one or more other routines. In addition, a flow diagram may include

an “end” and/or “continue” indication. The “end” and/or “continue” indications reflect that the steps presented can end as described and shown or optionally be incorporated in or otherwise used in conjunction with one or more other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0314] The one or more embodiments are used herein to illustrate one or more aspects, one or more features, one or more concepts, and/or one or more examples. A physical embodiment of an apparatus, an article of manufacture, a machine, and/or of a process may include one or more of the aspects, features, concepts, examples, etc. described with reference to one or more of the embodiments discussed herein. Further, from figure to figure, the embodiments may incorporate the same or similarly named functions, steps, modules, etc. that may use the same or different reference numbers and, as such, the functions, steps, modules, etc. may be the same or similar functions, steps, modules, etc. or different ones.

[0315] Unless specifically stated to the contra, signals to, from, and/or between elements in a figure of any of the figures presented herein may be analog or digital, continuous time or discrete time, and single-ended or differential. For instance, if a signal path is shown as a single-ended path, it also represents a differential signal path. Similarly, if a signal path is shown as a differential path, it also represents a single-ended signal path. While one or more particular architectures are described herein, other architectures can likewise be implemented that use one or more data buses not expressly shown, direct connectivity between elements, and/or indirect coupling between other elements as recognized by one of average skill in the art.

[0316] The term “module” is used in the description of one or more of the embodiments. A module implements one or more functions via a device such as a processor or other processing device or other hardware that may include or operate in association with a memory that stores operational instructions. A module may operate independently and/or in conjunction with software and/or firmware. As also used herein, a module may contain one or more sub-modules, each of which may be one or more modules.

[0317] As may further be used herein, a computer readable memory includes one or more memory elements. A memory element may be a separate memory device, multiple memory devices, or a set of memory locations within a memory device. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, a quantum register or other quantum memory and/or any other device that stores data in a non-transitory manner. Furthermore, the memory device may be in a form of a solid-state memory, a hard drive memory or other disk storage, cloud memory, thumb drive, server memory, computing device memory, and/or other non-transitory medium for storing data. The storage of data includes temporary storage (i.e., data is lost when power is removed from the memory element) and/or persistent stor-

age (i.e., data is retained when power is removed from the memory element). As used herein, a transitory medium shall mean one or more of: (a) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for temporary storage or persistent storage; (b) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for temporary storage or persistent storage; (c) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for processing the data by the other computing device; and (d) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for processing the data by the other element of the computing device. As may be used herein, a non-transitory computer readable memory is substantially equivalent to a computer readable memory. A non-transitory computer readable memory can also be referred to as a non-transitory computer readable storage medium.

[0318] While particular combinations of various functions and features of the one or more embodiments have been expressly described herein, other combinations of these features and functions are likewise possible. The present disclosure is not limited by the particular examples disclosed herein and expressly incorporates these other combinations.

What is claimed is:

1. A method for execution by a computing device, the method comprises:

determining a set of identigens for each word of a multitude of words of a phrase to produce a multitude of sets of identigens, wherein a set of identigens of the multitude of sets of identigens represents one or more different meanings of a word of the multitude of words, wherein each identigen of the multitude of sets of identigens includes a meaning identifier for a particular different meaning of the one or more different meanings, an instance identifier that includes a particular spelling in a particular language for the word, and a time reference, wherein each meaning identifier associated with a particular set of identigens of the multitude of sets of identigens represents a different meaning of one or more different meanings of a corresponding word of the multitude of words, wherein each time reference provides time information when a corresponding different meaning of the one or more different meanings applies, wherein a first set of identigens of the multitude of sets of identigens includes one or more different meanings of a first word of the multitude of words, wherein the determining the set of identigens for each word of the multitude of words of the phrase to produce the multitude of sets of identigens further includes:

extracting a set of at least two identigens for each word of at least ten words of the multitude of words of the phrase from a knowledge database to produce the multitude of sets of identigens;

identifying, based on identigen adjacency scores of the knowledge database, levels of pairing of all combinations of sequentially adjacent identigens between each adjacent set of identigens of the multitude of sets of identigens;

interpreting, based on the identigen adjacency scores of the sequentially adjacent identigens between each adjacent set of identigens, every permutation of sequentially adjacent

identigens that links across every set of identigens of the multitude of sets of identigens to produce permutations of entigen groups such that a first entigen group of the permutations of entigen groups corresponds to a first identigen of the first set of identigens which links to a first identigen of a second set of identigens which in turn links to a first identigen of a third set of identigens;

determining a permutation score for each entigen group of the permutations of entigen groups based on the identigen adjacency scores of the sequentially adjacent identigens between each adjacent set of identigens; and

selecting an entigen group of the permutations of entigen groups based on the permutation score for each entigen group to produce a selected entigen group comprising a multitude of entigens, wherein the selected entigen group represents a most likely meaning interpretation of the phrase, wherein a first entigen of the selected entigen group corresponds to the first identigen of the first set of identigens, wherein each remaining entigen of the selected entigen group corresponds to a selected identigen of a remaining set of identigens having a selected meaning of one or more different meanings of a remaining word of the multitude of words, wherein each entigen of the entigen group represents a corresponding single conceivable and perceivable thing in space and time that is independent of language, wherein the knowledge database includes a multitude of groups of entigens associated with a multitude of topics, wherein the multitude of topics includes a topic of the phrase, wherein each of groups of entigens includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens.

2. The method of claim 1 further comprises:
facilitating storage of the selected entigen group in the knowledge database; and
outputting the selected entigen group to a receiving entity.

3. The method of claim 1, wherein the determining the set of identigens for each word of the multitude of words to produce the multitude of sets of identigens comprises one or more of:

accessing, for each word of the at least ten words of the multitude of words, the knowledge database to recover the set of the at least two identigens; and

interpreting a query response from another computing device to produce the set of the at least two identigens, wherein the query response is received in response to an identigen query with regards to the word, and wherein the other computing device generates the query response based on accessing the knowledge database.

4. The method of claim 1, wherein the identifying the levels of pairing of all the combinations of sequentially adjacent identigens between each adjacent set of identigens of the multitude of sets of identigens comprises:

accessing, based on identifiers of a pair of sequentially adjacent identigens, the knowledge database to recover an identigen adjacency score for the pair of sequentially adjacent identigens; and

establishing a level of pairing for the pair of sequentially adjacent identigens based on the identigen adjacency score for the pair of sequentially adjacent identigens.

5. The method of claim 1, wherein the determining the permutation score for each entigen group of the permutations of entigen groups comprises:

establishing a first permutation score of the first entigen group as an aggregate of the identigen adjacency scores of the sequentially adjacent identigens between each adjacent set of identigens corresponding to the first entigen group; and

nulling out the first permutation score when at least one of the identigen adjacency scores indicates that corresponding sequentially adjacent identigens do not affirmatively pair.

6. The method of claim 1, wherein the selecting the entigen group of the permutations of entigen groups based on the permutation score for each entigen group to produce the selected entigen group comprises:

identifying a particular permutation of the permutations of entigen groups associated with a greatest permutation score to produce an identified permutation; and
establishing an entigen group associated with the particular permutation as the selected entigen group.

7. A computing device of a computing system, the computing device comprises:

an interface;

a local memory; and

a processor operably coupled to the interface and the local memory, wherein the processor executes operational instructions stored in the local memory to cause the processor to function to:

determine a set of identigens for each word of a multitude of words of a phrase to produce a multitude of sets of identigens, wherein a set of identigens of the multitude of sets of identigens represents one or more different meanings of a word of the multitude of words, wherein each identigen of the multitude of sets of identigens includes a meaning identifier for a particular different meaning of the one or more different meanings, an instance identifier that includes a particular spelling in a particular language for the word, and a time reference, wherein each meaning identifier associated with a particular set of identigens of the multitude of sets of identigens represents a different meaning of one or more different meanings of a corresponding word of the multitude of words, wherein each time reference provides time information when a corresponding different meaning of the one or more different meanings applies, wherein a first set of identigens of the multitude of sets of identigens includes one or more different meanings of a first word of the multitude of words, wherein the processor determines the set of identigens for each word of the multitude of words of the phrase to produce the multitude of sets of identigens further by

extracting a set of at least two identigens for each word of at least ten words of the multitude of words of the phrase from a knowledge database to produce the multitude of sets of identigens;

identify, based on identigen adjacency scores of the knowledge database, levels of pairing of all combinations of sequentially adjacent identigens between each adjacent set of identigens of the multitude of sets of identigens;

interpret, based on the identigen adjacency scores of the sequentially adjacent identigens between each adjacent set of identigens, every permutation of sequentially adjacent identigens that links across every set of identigens of the multitude of sets of identigens to produce permutations of entigen groups such that a first entigen group of the permutations of entigen groups corre-

sponds to a first identigen of the first set of identigens which links to a first identigen of a second set of identigens which in turn links to a first identigen of a third set of identigens;

determine a permutation score for each entigen group of the permutations of entigen groups based on the identigen adjacency scores of the sequentially adjacent identigens between each adjacent set of identigens; and select an entigen group of the permutations of entigen groups based on the permutation score for each entigen group to produce a selected entigen group comprising a multitude of entigens, wherein the selected entigen group represents a most likely meaning interpretation of the phrase, wherein a first entigen of the selected entigen group corresponds to the first identigen of the first set of identigens, wherein each remaining entigen of the selected entigen group corresponds to a selected identigen of a remaining set of identigens having a selected meaning of one or more different meanings of a remaining word of the multitude of words, wherein each entigen of the entigen group represents a corresponding single conceivable and perceivable thing in space and time that is independent of language, wherein the knowledge database includes a multitude of groups of entigens associated with a multitude of topics, wherein the multitude of topics includes a topic of the phrase, wherein each of groups of entigens includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens.

8. The computing device of claim 7, wherein the processor further functions to:

facilitate storage, via the interface, of the selected entigen group in the knowledge database; and

output, via the interface, the selected entigen group to a receiving entity.

9. The computing device of claim 7, wherein the processor functions to determine the set of identigens for each word of the multitude of words to produce the multitude of sets of identigens by one or more of:

accessing, via the interface for each word of the at least ten words of the multitude of words, the knowledge database to recover the set of the at least two identigens; and

interpreting a query response from another computing device to produce the set of the at least two identigens, wherein the query response is received in response to an identigen query with regards to the word, and wherein the other computing device generates the query response based on accessing the knowledge database.

10. The computing device of claim 7, wherein the processor functions to identify the levels of pairing of all the combinations of sequentially adjacent identigens between each adjacent set of identigens of the multitude of sets of identigens by:

accessing, via the interface and based on identifiers of a pair of sequentially adjacent identigens, the knowledge database to recover an identigen adjacency score for the pair of sequentially adjacent identigens; and

establishing a level of pairing for the pair of sequentially adjacent identigens based on the identigen adjacency score for the pair of sequentially adjacent identigens.

11. The computing device of claim 7, wherein the processor functions to determine the permutation score for each entigen group of the permutations of entigen groups by:

establishing a first permutation score of the first entigen group as an aggregate of the identigen adjacency scores of the sequentially adjacent identigens between each adjacent set of identigens corresponding to the first entigen group; and

nulling out the first permutation score when at least one of the identigen adjacency scores indicates that corresponding sequentially adjacent identigens do not affirmatively pair.

12. The computing device of claim 7, wherein the processor functions to select the entigen group of the permutations of entigen groups based on the permutation score for each entigen group to produce the selected entigen group by: identifying a particular permutation of the permutations of entigen groups associated with a greatest permutation score to produce an identified permutation; and establishing an entigen group associated with the particular permutation as the selected entigen group.

13. A non-transitory computer readable memory comprises:

a first memory element that stores operational instructions that, when executed by a processor, causes the processor to:

determine a set of identigens for each word of a multitude of words of a phrase to produce a multitude of sets of identigens, wherein a set of identigens of the multitude of sets of identigens represents one or more different meanings of a word of the multitude of words, wherein each identigen of the multitude of sets of identigens includes a meaning identifier for a particular different meaning of the one or more different meanings, an instance identifier that includes a particular spelling in a particular language for the word, and a time reference, wherein each meaning identifier associated with a particular set of identigens of the multitude of sets of identigens represents a different meaning of one or more different meanings of a corresponding word of the multitude of words, wherein each time reference provides time information when a corresponding different meaning of the one or more different meanings applies, wherein a first set of identigens of the multitude of sets of identigens includes one or more different meanings of a first word of the multitude of words, wherein the processor determines the set of identigens for each word of the multitude of words of the phrase to produce the multitude of sets of identigens further by:

extracting a set of at least two identigens for each word of at least ten words of the multitude of words of the phrase from a knowledge database to produce the multitude of sets of identigens;

a second memory element that stores operational instructions that, when executed by the processor, causes the processor to:

identify, based on identigen adjacency scores of the knowledge database, levels of pairing of all combinations of sequentially adjacent identigens between each adjacent set of identigens of the multitude of sets of identigens; and

interpret, based on the identigen adjacency scores of the sequentially adjacent identigens between each adjacent set of identigens, every permutation of sequentially adjacent identigens that links across every set of identigens of the multitude of sets of identigens to produce permutations of entigen groups such that a first entigen group of the permutations of entigen groups corresponds to a first identigen of the first set of identigens

which links to a first identigen of a second set of identigens which in turn links to a first identigen of a third set of identigens;

a third memory element that stores operational instructions that, when executed by the processor, causes the processor to:

determine a permutation score for each entigen group of the permutations of entigen groups based on the identigen adjacency scores of the sequentially adjacent identigens between each adjacent set of identigens; and a fourth memory element that stores operational instructions that, when executed by the processor, causes the processor to:

select an entigen group of the permutations of entigen groups based on the permutation score for each entigen group to produce a selected entigen group comprising a multitude of entigens, wherein the selected entigen group represents a most likely meaning interpretation of the phrase, wherein a first entigen of the selected entigen group corresponds to the first identigen of the first set of identigens, wherein each remaining entigen of the selected entigen group corresponds to a selected identigen of a remaining set of identigens having a selected meaning of one or more different meanings of a remaining word of the multitude of words, wherein each entigen of the entigen group represents a corresponding single conceivable and perceivable thing in space and time that is independent of language, wherein the knowledge database includes a multitude of groups of entigens associated with a multitude of topics, wherein the multitude of topics includes a topic of the phrase, wherein each of groups of entigens includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens.

14. The non-transitory computer readable memory of claim 13 further comprises:

a fifth memory element that stores operational instructions that, when executed by the processor, causes the processor to:

facilitate storage of the selected entigen group in the knowledge database; and

output the selected entigen group to a receiving entity.

15. The non-transitory computer readable memory of claim 13, wherein the processor functions to execute the operational instructions stored by the first memory element to cause the processor to determine the set of identigens for each word of the multitude of words to produce the multitude of sets of identigens by one or more of:

accessing, for each word of the at least ten words of the multitude of words, the knowledge database to recover the set of the at least two identigens; and

interpreting a query response from another computing device to produce the set of the at least two identigens, wherein the query response is received in response to an identigen query with regards to the word, and wherein the other computing device generates the query response based on accessing the knowledge database.

16. The non-transitory computer readable memory of claim 13, wherein the processor functions to execute the operational instructions stored by the second memory element to cause the processor to identify the levels of pairing of all the combinations of sequentially adjacent identigens between each adjacent set of identigens of the multitude of sets of identigens by:

accessing, based on identifiers of a pair of sequentially adjacent identigens, the knowledge database to recover an identigen adjacency score for the pair of sequentially adjacent identigens; and

establishing a level of pairing for the pair of sequentially adjacent identigens based on the identigen adjacency score for the pair of sequentially adjacent identigens.

17. The non-transitory computer readable memory of claim 13, wherein the processor functions to execute the operational instructions stored by the third memory element to cause the processor to determine the permutation score for each entigen group of the permutations of entigen groups by: establishing a first permutation score of the first entigen group as an aggregate of the identigen adjacency scores of the sequentially adjacent identigens between each adjacent set of identigens corresponding to the first entigen group; and

nulling out the first permutation score when at least one of the identigen adjacency scores indicates that corresponding sequentially adjacent identigens do not affirmatively pair.

18. The non-transitory computer readable memory of claim 13, wherein the processor functions to execute the operational instructions stored by the fourth memory element to cause the processor to select the entigen group of the permutations of entigen groups based on the permutation score for each entigen group to produce the selected entigen group by:

identifying a particular permutation of the permutations of entigen groups associated with a greatest permutation score to produce an identified permutation; and establishing an entigen group associated with the particular permutation as the selected entigen group.

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